

HOUSING CONSTRAINTS AND INTERGENERATIONAL MOBILITY*

Ailin Zhang[†]

September 30, 2026

PRELIMINARY DRAFT — COMMENTS WELCOME

ABSTRACT

This paper studies how housing supply constraints shape intergenerational educational mobility through house-price capitalization. Using the 2015 Chinese Census and 2016 China Migrants Dynamic Survey, I document asymmetric effects of housing-price appreciation: a 10% increase raises upward mobility among children in low-educated native homeowner households by 1.9 percentage points, reduces net migration of low-educated parents by 2.3 points, and increases parent–child separation among low-educated migrant families by 3.5 points. I develop a quantitative spatial model linking housing wealth, migration, family arrangements, education investment, and school access. Counterfactuals show that more elastic housing supply mainly redistributes the gains from urban growth, while equalising migrant children’s school access raises aggregate upward mobility by 0.18 percentage points.

*I am deeply indebted to my advisors Gabriel Ahlfeldt and Christian Hilber for their guidance and support. I thank Sara Bagagli, Felipe Carozzi, Michael Gmeiner, Hans Koster, Amrita Kulka, Henry Overman, Giorgio Pietrabissa, Olmo Silva, Daniel Sturm, Yanos Zylberberg, and other seminar participants at LSE, 13th Warwick Economics PhD Conference, and Reading Workshop in Urban Economics and Economic Geography for helpful discussions.

[†]London School of Economics and Political Science. Email: a.zhang12@lse.ac.uk.

1 INTRODUCTION

Intergenerational mobility varies sharply across space. In China, children from low-skilled local families in cities such as Beijing and Shanghai appear more likely to move up the education ladder, as shown in Fig. 1. These cities also experienced some of the largest increases in housing prices during 2003–2012. This spatial pattern is difficult to interpret. Is the positive relationship between housing-price growth and upward mobility causal? Through what mechanisms does housing price appreciation make some children more, and others less, able to move upward relative to their parents? And do housing constraints improve intergenerational mobility on average, or mainly redistribute opportunity across households and locations?

This paper studies the distributional incidence and aggregate implications of housing constraints for intergenerational educational mobility. I argue that housing constraints reshape the intergenerational incidence of urban growth through two key mechanisms that arise when local demand growth is capitalized into housing prices. Price appreciation may expand the resources available to households holding housing assets, while higher housing costs and school-access barriers restrict low-skilled migrant families’ access to productive cities both by discouraging migration and by separating parents from their children. Whether these opposing effects translate into higher or lower aggregate mobility is a quantitative question. I address it by combining reduced-form evidence on these wealth and access margins with a quantitative spatial model linking housing prices, location choices, family arrangements, private education investment, and access to public education.

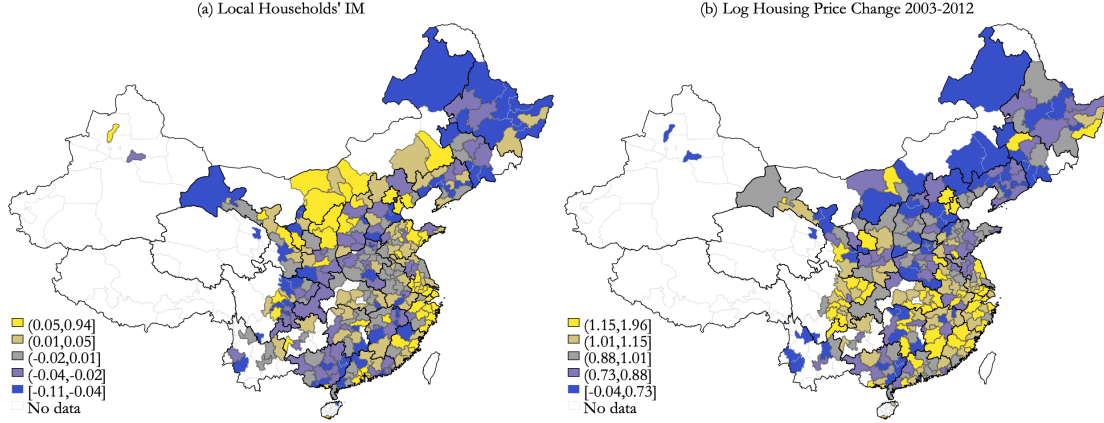


Figure 1: House price change and intergenerational education mobility.

Panel (a) shows intergenerational education mobility (IM) among local households, and Panel (b) depicts the change in log housing prices during 2003–2012 across Chinese prefectural cities. City-level IM is measured by the coefficients on city fixed effects from individual-level regressions of upward mobility, controlling for individual and family characteristics. The sample includes natives (non-migrants) born between 1986–1993 whose best-educated parent has ≤ 9 years of schooling. Upward mobility is defined as attaining junior college or above (≥ 15 years of schooling). Data sources: 2015 Census and CSYRE. White regions indicate missing data.

Understanding these forces matters because intergenerational mobility is both declining and spatially unequal in China. Recent evidence points to growing persistence across generations: [Fan et al. \(2021\)](#), for example, find that the intergenerational elasticity of income increased from 0.390 for the 1970–1980 birth cohorts to 0.442 for the 1981–1988 cohorts.¹ Mobility is also closely related to inequality: areas with greater inequality tend to exhibit stronger intergenerational persistence ([Krueger, 2012](#); [Corak, 2013](#); [Chetty et al., 2014](#); [Fan et al., 2021](#)). Identifying the spatial forces that shape mobility is therefore important for understanding how economic opportunities are transmitted across generations.

An emerging literature shows that intergenerational mobility varies systematically across places. In the United States, [Chetty et al. \(2014\)](#) show that areas with higher mobility tend to have less residential segregation, lower income inequality, better schools, greater social capital, and more stable family structures. Beyond the United States, [Alesina et al. \(2021\)](#) document persistent spatial differences in educational mobility across Africa, while [Fletcher and Han \(2019\)](#) show that state-level education policies are associated with mobility differences in the United States. A separate literature studies how housing supply constraints shape spatial mobility and aggregate efficiency. Binding land-use regulations in productive cities raise housing prices, restrict population growth, and generate spatial misallocation ([Hsieh and Moretti, 2019](#)); in China, the inland-favoring allocation of construction land quotas similarly redirected land supply away from more productive eastern cities and reduced aggregate welfare ([Fang et al., 2021](#)). Taken together, these literatures show that place matters for intergenerational mobility and that housing constraints shape access to productive locations. Yet little is known about how these constraints affect the intergenerational distribution of the gains from urban growth.

This paper brings these two literatures together by studying how housing constraints shape upward educational mobility across families with different exposure to housing-price appreciation. Throughout the paper, I focus on children from low-skilled families and define upward mobility as attaining a higher level of education than their parents. This household-level measure captures the intergenerational transmission of human capital and allows me to study how the incidence of housing-price appreciation depends on parents’ housing tenure, migration, and family-arrangement decisions.

The relationship between housing constraints and upward mobility is theoretically ambiguous because housing price appreciation changes both household resources and access to high-opportunity cities. This distinction mirrors a broader view of housing as both capital and a bundle of location-specific opportunities ([Ioannides and Ngai, 2025](#)). On the one hand, higher house prices increase the resources of households that already own housing. Incumbent homeowners may increase education investment in their children through higher housing wealth, collateral value, or lower precautionary saving motives ([Lovenheim, 2011](#); [Gan, 2010](#); [Peng et al., 2019](#); [Fan and Yavas, 2020](#); [Painter et al., 2022](#)).² Renters, by contrast, do not receive this capital gain and may face tighter budgets for housing

¹Intergenerational mobility is commonly measured in terms of income or educational attainment. This paper focuses on educational mobility because education closely predicts lifetime economic outcomes, is observable earlier in life, and provides a direct measure of intergenerational human-capital mobility. See [Fan et al. \(2015\)](#); [Gruijters et al. \(2019\)](#); [Fan et al. \(2021\)](#); [Xie et al. \(2022\)](#) for evidence on declining mobility in China. Using data from [Xie et al. \(2022\)](#), Fig. 6 in the Appendix plots the decline in intergenerational educational mobility over recent decades.

²The housing-wealth channel may operate differently in China than in countries with more developed mortgage-refinancing markets. Individual mortgage refinancing is less central, while collateral and precautionary-saving channels may be more relevant. At the same time, housing appreciation could reduce education investment if non-wage wealth lowers the perceived return to schooling or if real-estate booms raise the opportunity cost of education ([Charles et al., 2015](#); [Li and Xiao, 2020](#)). The reduced-form evidence therefore tests which force dominates in this setting.

and education (Newman and Holupka, 2014; Angst et al., 2023).

On the other hand, higher housing costs can raise the cost of accessing productive cities for low-skilled migrant families. Supply-constrained cities tend to offer better labour-market opportunities, amenities, and public services (Glaeser and Gyourko, 2005; Gyourko et al., 2013), but high housing costs can disproportionately deter low-skilled parents from migrating to these destinations. For families that do migrate, housing costs can discourage parent–child co-residence, while school-access barriers reduce the educational value of bringing children to the destination (Imbert et al., 2026). Even migrant children who accompany their parents may have incomplete access to local public schools, limiting their ability to benefit from local education resources (Lee, 2011; Li et al., 2015; Feng and He, 2022). Public education finance may further amplify this segregation effect: if high-price cities attract skilled adults while excluding disadvantaged children, local tax resources per eligible child may rise, widening differences in public education access across places (Zheng and Graham, 2022). Taken together, these mechanisms imply sharply different incidence across households. Housing-price appreciation may benefit initial asset holders while raising the spatial and family costs of access for migrant families.

In terms of structure, I first describe the institutional origins of housing-wealth differences and document two motivating facts on migrant sorting and access to public education. First, expensive cities attract more migrant workers, yet migrant parents are less likely to co-reside with their children in these destinations. Second, high-price cities concentrate greater public education resources, while migrant children are less likely to enrol in local public schools. These cross-sectional patterns suggest that access to the opportunities available in expensive cities is unevenly distributed between local and migrant families, but they do not identify the causal role of housing prices, which may correlate with local labour-market opportunities and amenities.

I then estimate the incidence of supply-amplified housing-price appreciation. I instrument housing-price growth with interactions between shift-share labour-demand shocks and two predetermined physical supply constraints: the share of developable land already developed and mean terrain slope. A 10% increase in housing prices raises upward mobility among children observed in low-educated native homeowner households by 1.9 percentage points, while the corresponding effects for migrant children are not statistically significant. The same increase reduces the net migration of low-educated parents by 2.3 percentage points and raises parent–child separation among low-educated migrant families by 3.5 percentage points. I therefore interpret the reduced form as documenting asymmetric incidence patterns: a positive mobility response concentrated among homeowner households, consistent with a housing-resource channel, and migration and family-arrangement responses among migrant families, consistent with a housing-cost and access channel.

The reduced-form evidence documents who gains and how families adjust when local demand growth is capitalized into housing prices, but these conditional responses cannot be directly aggregated into the effect of a nationwide policy change. Housing prices affect which parents migrate, whether children move with them, and where children are educated. Migrant and co-residing families are therefore selected endogenously, while migration changes local populations, housing markets, and public education resources. A further limitation is the “missing-intercept” problem in reduced-form counterfactual analysis (Wolf, 2023). Because the estimates are identified from cross-city differences in exposure, they recover relative responses across locations but not the common component of a counterfactual that affects all cities. I therefore develop a quantitative spatial model to aggregate these household responses and evaluate nationwide counterfactuals while allowing migration, family

arrangements, child schooling locations, housing markets, and local public education resources to adjust jointly.

The model features heterogeneous households that differ by education, origin, and initial housing endowment. Parents choose whether to co-reside with their children and where to live, jointly determining children’s locations and access to local education resources. Housing prices operate through two structural margins motivated by the empirical patterns: appreciation changes the resources of initial asset holders, while housing costs affect migration and parent–child co-residence. School-access barriers further reduce the educational return to bringing children to high-opportunity destinations.

Three features of the Chinese institutional setting discipline these margins. First, the privatization of state-owned housing beginning in 1994 allowed incumbent urban tenants to acquire housing at subsidized prices, creating persistent differences in initial housing endowments between incumbent residents and rural–urban migrants (Wang, 2011; Huang and Yi, 2011). Second, public education is financed largely at the local level, so migration and local economic activity affect the resources available for education (Jia et al., 2014; Lü, 2015; Huang et al., 2021; Liu et al., 2017; Rozelle and Hell, 2020). Third, access to these locally financed resources is mediated by the *hukou* system and destination-specific eligibility requirements. In the model, these institutions map respectively into initial housing endowments, local education budgets, and a school-access wedge governing migrant children’s effective access to local public education. Together, they determine who owns the asset before urban growth occurs, where public educational resources are located, and who can access them.

I bring the model to the data by estimating the parameters governing location and family-arrangement choices, recovering unobserved local fundamentals. The counterfactuals show that housing constraints have much larger effects on the distribution of the intergenerational gains from urban growth than on national average mobility. For a given urban growth shock, making housing supply more elastic weakens price capitalization and changes migration and parent–child co-residence, while generating only a negligible increase in aggregate upward mobility. Equalising local school access between migrant and incumbent children raises aggregate upward mobility by 0.18 percentage points, largely through endogenous migration and co-residence. Finally, removing the housing-wealth gains created by public-housing privatisation lowers education investment and mobility among the descendants of initial asset holders, showing that the distribution of housing assets can have persistent intergenerational consequences.

Related literature

This paper first contributes to the literature on the geography of intergenerational mobility and human capital accumulation (Chetty et al., 2014; Fletcher and Han, 2019; Fogli and Guerrieri, 2019; Sieg et al., 2020; Alesina et al., 2021). This literature shows that upward mobility varies systematically across places and is shaped by local inequality, schools, social capital, and neighbourhood conditions. Recent structural work further studies how broad economic forces, such as the shift from manufacturing to services, reshape the spatial distribution of inequality and mobility (Takeda, 2022). I add to this literature by studying how housing supply constraints shape the intergenerational incidence of urban growth through both housing-wealth capitalization and unequal access to productive cities. More broadly, Ioannides and Ngai (2025) organize the housing–inequality nexus around housing as consumption, capital, and location. I study how housing supply constraints jointly shape these margins through housing costs, housing wealth, and access to productive cities. Most closely, Zheng and Graham (2022) document a positive relationship between housing prices and intergenerational

mobility across U.S. school districts, where public school funding depends heavily on local property tax revenues. I differ by studying a setting in which housing prices affect mobility through household asset exposure and migrant families’ access to high-opportunity cities. I combine reduced-form evidence on the unequal incidence of supply-amplified appreciation with a quantitative spatial model incorporating initial housing endowments, family arrangements, and school-access barriers.

Second, this paper contributes to the literature on housing supply constraints and spatial mobility (Ganong and Shoag, 2017; Glaeser and Gyourko, 2018; Hsieh and Moretti, 2019; Chen et al., 2019; Fang et al., 2021; Favilukis et al., 2023; Duranton and Puga, 2023). Existing work shows that constrained housing supply raises prices, limits migration to productive places, and generates spatial misallocation. I extend this literature by showing that housing constraints also shape how the gains from productive cities are transmitted across generations. By affecting where parents live, whether children move with them, and how much households and local governments invest in education, housing constraints can translate intragenerational spatial frictions into intergenerational differences in opportunity. A recent paper, Binder et al. (2026), and this paper examine complementary dimensions of the intergenerational effects of housing supply constraints in different institutional contexts. Binder et al. (2026) study the transmission of housing capital across generations in the United States, primarily through differences in homeownership, while I study the human-capital dimension in China, examining how housing supply conditions shape children’s educational mobility through parental housing wealth, migration, family arrangements, and access to local public education.

Third, this paper contributes to structural models of spatial equilibrium and location choice (Ahlfeldt et al., 2015; Monte et al., 2018; Bryan and Morten, 2019; Tombe and Zhu, 2019; Imbert et al., 2026). This literature has incorporated heterogeneous workers, non-homothetic preferences, and rich migration margins into urban and spatial models (Diamond, 2016; Couture et al., 2019; Diamond and Gaubert, 2022; Tsivanidis, 2019). I extend this work by introducing parent–child family arrangements and child schooling locations into a model of housing constraints and human-capital accumulation. This feature is particularly important in China, where many migrant parents move for work while leaving dependent children behind in their places of origin. Closest to my paper, Imbert et al. (2026) model the endogenous decision to migrate with or without children and study its implications for location choices and remittances. I instead focus on how housing prices, initial housing assets, and school-access barriers jointly shape intergenerational educational mobility.

The rest of the paper proceeds as follows. Section 2 describes the institutional background and motivating facts. Section 3 describes the data. Section 4 presents reduced-form evidence on the incidence of supply-amplified housing-price appreciation. Section 5 develops the quantitative spatial model, which is brought to the data in Section 6. Section 7 uses the calibrated model to conduct counterfactuals. Section 8 concludes.

2 CONTEXT AND MOTIVATING FACTS

2.1 URBAN HOUSING SUPPLY CONSTRAINTS

China’s escalating housing prices over the past decades have created severe affordability challenges, especially in mega-cities such as Beijing and Shanghai. Between 2003 and 2013, annual housing prices in 35 large and medium-sized Chinese cities rose by 17%, far outpacing GDP growth of 10% (Tian et al., 2020). Although the primary drivers remain debated, existing work points to the interaction

between strong urban demand growth and constrained housing supply as an important force behind this price appreciation (Dong, 2016; Lai and Van Order, 2020; Fang and Huang, 2022). Housing supply constraints determine how strongly local demand growth is capitalized into housing prices rather than absorbed through housing construction and population adjustment. They include physical constraints, such as topography, water bodies, and the scarcity of developable land, as well as regulatory constraints, such as construction-land quotas, FAR limits, and height restrictions.³

In this paper, I focus on two predetermined physical constraints: the scarcity of developable land and terrain slope. Regulatory restrictions may similarly make housing supply less elastic, but their effective stringency is difficult to measure and may respond endogenously to local housing-market conditions, complicating causal identification.

2.2 PRIVATISATION OF PUBLIC HOUSING

Before China’s housing reforms, urban housing was largely provided through a welfare system in which state-owned housing units were allocated to employees of state-owned enterprises and public-sector work units at very low rents. Although private housing supply gradually expanded after economic reform, around 40% of urban households still lived in state-owned housing in 1993 (Wang, 2011). This system created severe underinvestment, housing shortages, and crowding, placing increasing fiscal pressure on local governments and work units (Fu, 2016).

The 1994 *Decision on Deepening the Reform of the Urban Housing System* allowed incumbent urban tenants to purchase formerly public housing from state employers at heavily discounted prices (State Council, 1994). As in privatisation episodes in Eastern Europe and the Russian Federation, the reform sharply expanded private urban homeownership. Using CHNS data, Wang (2011) show that the urban homeownership rate rose from around 55% before the reform to above 80% in the 2000s. In 1998, most state-owned enterprises were prohibited from providing new welfare housing, and new residential units were required to be transacted through the market, marking the end of the urban welfare housing system. This institutional background matters for two reasons. First, privatised housing converted subsidised occupancy rights into marketable assets, creating scope for housing-wealth effects on household consumption and education investment. Consistent with this channel, Li and Shin (2013) show that parents with allocated public housing are in a stronger financial position and provide more financial support to their children. Second, the reform contributed to an unequal initial distribution of urban housing wealth. Incumbent urban tenants acquired subsidised housing assets, whereas rural-urban migrants generally lacked comparable access despite often holding collectively owned residential land in their rural origins (*zhai ji di*) (Huang and Yi, 2011). In the model, the privatisation windfall enters as one component of owner households’ initial housing endowments; I examine its intergenerational consequences in the counterfactual analysis.

2.3 FISCAL DECENTRALIZATION, HUKOU, AND PUBLIC EDUCATION SPENDING

Public education in China is financed largely at the local level, creating substantial spatial variation in education resources. The 1994 Tax Sharing System recentralized tax revenues while leaving local

³For related evidence on physical constraints, see Hilber and Mayer (2009); Saiz (2010); Wang et al. (2012); for regulatory constraints, see Glaeser et al. (2005); Saks (2008); Hilber and Vermeulen (2016); Hsieh and Moretti (2019); Tan et al. (2020); Fang and Huang (2022).

governments responsible for major expenditure responsibilities, including education, generating a vertical fiscal imbalance (Jia et al., 2014). During this period, town-level governments with limited fiscal capacity bore much of the cost of rural schooling, while intergovernmental transfers were insufficient to close gaps between local fiscal capacity and expenditure needs, widening regional and urban–rural disparities in compulsory education (Lü, 2015; Huang et al., 2021). Subsequent reforms shifted education finance toward higher levels of government, including the 2001 transition from township-based to county-based education finance and the 2006 reform eliminating compulsory-school fees. Nevertheless, inequality in compulsory education finance persisted and remained closely tied to spatial differences in economic development (Wang, 2014; Liu et al., 2017). This decentralized system implies that cities attracting more skilled adults and higher tax bases can provide greater public education resources.

Access to locally financed public education is also shaped by the *hukou* system and destination-specific enrolment requirements. Relative to local children, migrant children without local *hukou* may face additional residence, employment, social-insurance, documentation, and school-capacity requirements when applying to public schools. In some cities, proof of stable residence or housing within a school catchment also affects enrolment eligibility or priority (Zhang et al., 2020; Peng et al., 2021). These requirements create an education-access wedge between local and migrant children. The wedge operates both after and before migration: children who accompany their parents may have more limited access to local public schools, while parents anticipating these barriers may be less willing to bring their children to the destination city in the first place. As a result, the benefits of greater public education spending in high-resource cities may not be equally accessible to migrant families.

2.4 MOTIVATING FACTS

Fact 1: Expensive cities attract migrant labour, but fewer parents bring their children

Using individual-level data from the 2015 census and city-level housing-price data from CSYRE, Fig. 2 contrasts migrant workers’ spatial sorting with migrant parents’ decisions to bring their children. Panel (a) relates the log number of migrant workers who arrived in each city during 2005–2015 to log housing prices in 2005. The relationships are reported separately for high-skilled migrants, defined as those who entered senior-secondary education or above, and low-skilled migrants, defined as those with junior-secondary education or below. Consistent with expensive cities offering better labour-market opportunities and amenities, migrants in both skill groups sort disproportionately towards initially high-price cities. Although low-skilled workers constitute the majority of migrants, their share is lower in more expensive destinations, indicating stronger skill selection into high-price cities.

Panel (b) turns to the family-migration margin and relates initial housing prices to the share of migrant mothers with living children who co-reside with at least one child. Because the census reports fertility information only for women and does not record the ages of non-co-residing children, the panel focuses on female migrants aged 15–38, whose children are more likely to be dependent and school-aged. The co-residence rate declines with initial housing prices, particularly among low-skilled mothers. Taken together, the two panels show that expensive cities attract migrant labour, but migrant parents—especially the low-skilled—are less likely to bring their children. Although these relationships are descriptive rather than causal, they motivate an access channel through which high housing costs restrict disadvantaged migrant families’ ability to enter expensive cities together with their children.

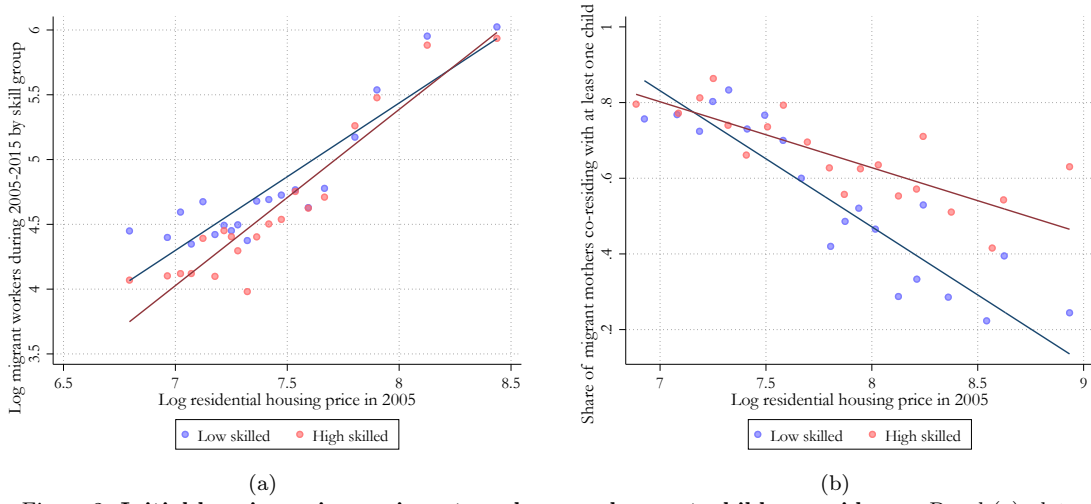


Figure 2: **Initial housing prices, migrant workers, and parent-child co-residence.** Panel (a) plots the log number of migrant workers arriving in prefecture-level cities during 2005–2015 against log housing prices in 2005, separately for high- and low-skilled migrants. Panel (b) plots the share of female migrants with living children whose children co-reside with them against initial housing prices, again by skill group. High-skilled migrants have entered senior-secondary education or above; low-skilled migrants have junior-secondary education or below. Migrant workers are individuals aged 16–64 who live outside their place of *hukou* registration, have been away for fewer than ten years, and were employed in the week preceding the census. Panel (b) restricts the sample to female migrants aged 15–38 with living children. Migration and family data are from the 2015 Population Census; housing-price data are from CSYRE.

Fact 2: Expensive cities concentrate public education resources but limit migrant access

Using city-level fiscal data and the 2011 China Migrants Dynamic Survey (CMDS), Fig. 3 documents a mismatch between the spatial distribution of public education resources and migrant children’s access to them. Panel (a) relates log local government education expenditure per capita to log housing prices in 2005. Public education spending is higher in more expensive cities, consistent with evidence from the United States that more expensive school districts tend to have greater school funding and higher upward mobility (Zheng and Graham, 2022). Although the city-level fiscal data do not distinguish expenditure on primary, secondary, and adult education, the pattern indicates that public resources relevant for human-capital formation are concentrated in high-price cities.

Panel (b) relates initial housing prices to the share of migrant children enrolled in local public schools. Because public schools in urban China are typically better resourced and much less expensive than migrant or private schools, this enrolment share provides a useful proxy for effective access, as eligible migrant families are unlikely to choose alternative schools. The enrolment share declines in more expensive cities, consistent with more binding *hukou*-related access barriers in high-price destinations. The figure highlights an education-access margin: high-price cities provide greater public education resources, while migrant children—particularly those from disadvantaged families—face more limited access to them.

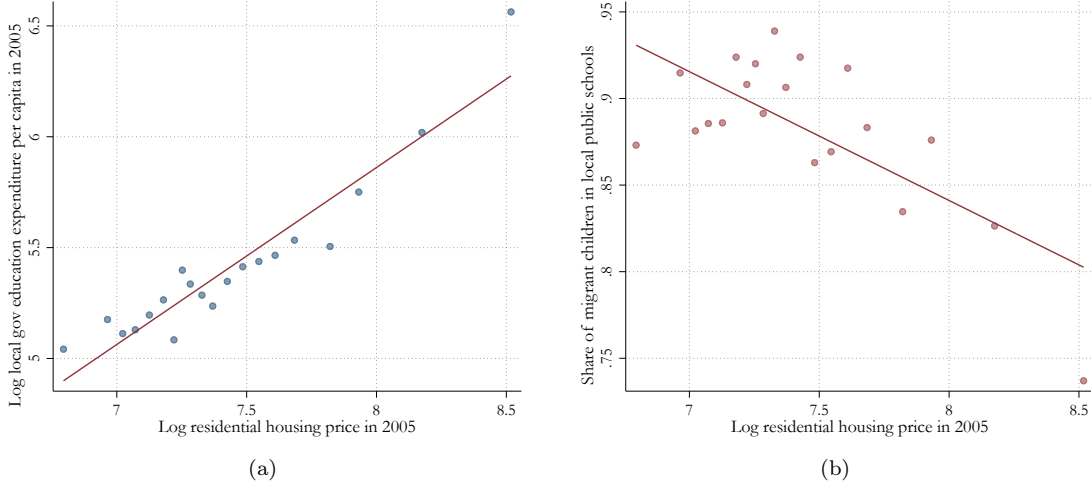


Figure 3: **Housing prices, public education resources, and migrant children's access.** Panel (a) plots log local government education expenditure per capita against log housing prices in 2005 across prefecture-level cities. Panel (b) plots the share of migrant children enrolled in local public schools against initial housing prices. The enrolment share is constructed from the 2011 China Migrants Dynamic Survey (CMDS), restricting the sample to school-aged migrant children living in the destination city. Education expenditure and population data are from the CSY; housing-price data are from CSYRE.

Taken together, the institutional setting and motivating facts point to two sources of asymmetric exposure to the capitalization of urban growth into housing prices. Initial housing-wealth differences allow incumbent homeowners to benefit from price appreciation, while higher housing costs and unequal access to local public education raise the cost of spatial and family access for migrant households. These patterns motivate the causal analysis that follows.

3 DATA

This section describes the main data sources used in the reduced-form analysis and the quantitative spatial model. The analysis combines three types of data: household data on location choices and upward mobility, housing and land-market data on prices and supply constraints, and city-level controls. The quantitative model analysis focus on 270 cities with complete data inputs.

Household data and upward-mobility measure

The main household data come from microdata extracts of the 2005 and 2015 National 1% Population Sample Surveys, covering approximately 2‰ and 1‰ of the national population, respectively. I refer to these surveys below as the Census data. For the quantitative analysis, I use the 2005 Census to construct baseline adult populations and wages, restricting the adult sample to ages 19–64. The Census data record rich information including current residence, hukou origin, education, employment, fertility histories, household relationships, and housing characteristics. Their nationwide coverage permits substantially broader spatial analysis than household surveys such as CHIP, CGSS, and CHFS. The data nevertheless have two important limitations. First, wage information is available only in the 2005 extract, and current household housing wealth is not observed. The model therefore uses the 2005 Census to construct wages, while the reduced-form analysis measures homeowners' exposure

using city-level house-price growth, instrumented by interactions between local labour-demand shocks and predetermined housing-supply constraints. Second, parent–child links can be identified only for family members observed in the same household. I therefore complement the 2015 Census with the 2016 China Migrants Dynamic Survey (CMDS) when analysing the effects on migrant households, which records migrants’ children whether they co-reside at destination or remain in the hukou place. CMDS does not, however, identify the precise city in which a separated child resides.

In my reduced-form analysis using data from the 2015 Census and the 2016 wave of CMDS, a child whose best-educated parent has ≤ 9 years of schooling (junior high school or below) is defined as achieving intergenerational upward mobility if he or she attains junior college or above (≥ 15 years of schooling).⁴ In the 2015 Census sample, junior college or above corresponds to roughly the top 27% of the education distribution for the 1986–1993 birth cohorts, while the best-educated parent has junior high school or below in about 80% of linked families. The native-homeowner Census sample and the co-residing and separated migrant-child CMDS samples are all restricted to these cohorts. They were aged 10–17 at the start of the house-price shock window in 2003 and aged 22–30 when their educational attainment was observed in the available data. The cohort restriction therefore ensures that exposure began before college entry and that postsecondary attainment can be measured in the outcome data, while parental education predates the shock.

Housing price

Data on housing prices in 286 prefecture-level cities from 2003–2012 are obtained from the *China Statistical Yearbook for Regional Economy* (CSYRE), dividing the total sales value of commercial housing by the corresponding floor space sold, as in [Liang et al. \(2016\)](#) and [Jia et al. \(2021\)](#). This measure has several limitations. First, it is only a proxy for residential housing prices because commercial-housing sales also include non-residential properties. Nevertheless, [Wrenn et al. \(2019\)](#) shows that more than 90% of the variation in this measure between 2000 and 2005 is attributable to changes in residential housing prices. Second, the simple average does not account for changes in the quality of newly sold units or depreciation of the existing housing stock. Third, the measure may be subject to government manipulation, although this is less concerning if such distortions are similar across cities. Despite these limitations, no alternative series provides comparable coverage back to 2003. Following [Chen and Wen \(2017\)](#), I validate the CSYRE measure against one of the most authoritative housing-price series for China, constructed by [Fang et al. \(2016\)](#) and covering 120 cities. The correlation between the two log price measures over the overlapping sample period is 0.85.⁵

Physical constraints

The first physical constraint is terrain slope constructed from the China Historical Geographic Information System (CHGIS) V5 digital elevation model (DEM). The archived CHGIS raster is the 2015 CHGIS V5 DEM, revised for QGIS in 2019, and is based on the U.S. Geological Survey’s GTOPO30 global DEM.⁶ The raster has 30-arc-second cells (approximately 1 km at the equator) and is supplied on a 1-km projected grid. I calculate slope in degrees using the Horn 3×3 finite-difference algorithm with a 1-km horizontal cell size. For each prefecture-level city, I then average the slope

⁴Similar measures are commonly adopted in the related literature; see, for example, [Card et al. \(2022\)](#).

⁵I thank Hanming Fang for kindly sharing the housing-price data used in [Fang et al. \(2016\)](#).

⁶The archived data documentation identifies the source as “CHGIS V5 DEM (2015). Based on GTOPO-30 from the U.S. Geological Survey.” The CHGIS V5 distribution is available through the Harvard Dataverse at https://dataverse.harvard.edu/dataverse/chgis_v5; the underlying GTOPO30 documentation is available at <https://1ta.cr.usgs.gov/GTOPO30>.

values for raster-cell centres lying within a 10-km buffer around the city centre and inside the city’s administrative boundary. This mean slope measure captures the topographic component of physical housing-supply constraints, consistent with the role of terrain in limiting urban expansion documented in existing literature (Burchfield et al., 2006; Saiz, 2010; Hilber and Vermeulen, 2016).

The second measure of physical constraints, the share of developable land that is developed (% developed), is calculated using the 2000 1km Multi-period Remote Sensing Monitoring Dataset of Land Use/Land Cover in China (CNLUCC) constructed by Xu et al. (2018) from the Data Center of Resources and Environmental Sciences, Chinese Academy of Sciences.⁷ First developed in 1980 and updated at a 5-year interval, CNLUCC uses remote sensing information mostly from Landsat-MSS/TM/ETM, categorizing land into 6 first-level cover types and 25 second-level cover types.⁸ Land use ‘urban built-up’ is defined as ‘developed’ in this paper; ‘developable’ lands further include land uses: ‘woodland’, ‘meadow’, ‘rural residence’ and ‘bare or otherwise unused land’.⁹ To evaluate the land scarcity in urban area, for each city I identify the city center using Gaode Map API and calculate the ratio of developed land area to developable land area in the 10km surrounding buffer.

Other data sources

Urban employment data in each of 19 industries along with other control variables are extracted from annual City Statistic Yearbook (CSY).

4 REDUCED-FORM ANALYSIS

4.1 IDENTIFICATION STRATEGY

The reduced-form analysis estimates the incidence of supply-amplified housing-price appreciation across homeowner and migrant families. I follow the logic of Hilber and Vermeulen (2016) and exploit the fact that a local labour-demand shock should translate into larger housing price increases in cities with more inelastic housing supply. In the first stage, I interact a shift-share labour-demand shock with two predetermined physical constraints on housing supply: terrain slope and the share of developable land already developed at baseline. In the second stage, I use these interactions to instrument for housing price growth and estimate its effect on upward educational mobility. The 1st stage specification is

$$\Delta \ln HP_{cT} = \beta \widehat{\Delta Emp_{cT}} \cdot \mathbf{Constraints}'_c + \lambda \ln HP_{c0} + \boldsymbol{\theta} \cdot \mathbf{Constraints}'_c + inishare_{c0} + \xi_p + \epsilon_c, \\ \mathbf{Constraints}_c = [\%developed_{c0} \quad slope_c]. \quad (1)$$

Here, $\Delta \ln HP_{cT}$ is the change in log housing prices in city c over the period $T \in [2003, 2012]$. The shift-share labour-demand shock is defined as $\widehat{\Delta Emp_{cT}} = \sum_j s_{cj,2003} \ln(1 + g_{jT,-c})$, where $s_{cj,2003}$ is city c ’s initial employment share in industry j and $g_{jT,-c}$ is leave-one-city-out national employment growth in industry j over the same period, computed from national industry employment excluding

⁷<https://www.resdc.cn/doi/doi.aspx?DOIId=54>

⁸Liu et al. (2014) show that the overall classification accuracy of CNLUCC is above 91.2% using random sampling verification of field survey sites.

⁹Cultivated land is excluded because its conversion to urban construction is subject to stringent approval, occupation–reclamation balance requirements, and basic-farmland protection in China, making it less available for marginal urban expansion than otherwise unused land.

city c . The vector **Constraints** $_c$ contains the share of developable land already developed at baseline, $\%developed_{c0}$, and terrain slope, $slope_c$. Both constraint measures are standardised. If housing supply constraints are binding, labour-demand shocks should generate larger housing price increases in more constrained cities, implying positive elements of β . The specification controls for initial housing prices, the two constraint measures, initial industry shares, and province fixed effects. Because the shift-share shock $\widehat{\Delta Emp}_{cT}$ is a linear combination of the initial industry shares, it is absorbed by $inishare_{c0}$ and is therefore not included separately.

The 2nd stage specification is

$$UP_{ct} = \alpha \widehat{\Delta \ln HP}_{cT} + \gamma \ln HP_{c0} + \delta \cdot \mathbf{Constraints}'_c + inishare_{c0} + \xi_p + u_c. \quad (2)$$

The dependent variable UP_{ct} measures upward educational mobility in city c . In the baseline native-sample specification using the 2015 census data, I construct UP_{ct} as the city fixed effect from an individual-level regression for natives born in 1986–1993 whose best-educated parent completed junior high school or below (≤ 9 years of schooling). The outcome equals one if the child attained junior college or above (≥ 15 years of schooling). The first-step regression controls for gender, age, parental industry, and city fixed effects; the estimated city fixed effects are then used as the city-level mobility measure. This construction measures city-level upward mobility net of individual characteristics and parental industry, which proxies for parental skills and socioeconomic background. The coefficient α therefore captures how supply-driven housing price appreciation affects upward educational mobility. For example, a 10% increase in housing prices changes city-level upward mobility by approximately $10 \times \alpha$ percentage points.

I also use the 2016 China Migrants Dynamic Survey (CMDS) to construct analogous upward-mobility measures for co-residing and separated migrant children. Both samples are restricted to children born in 1986–1993 whose best-educated parent completed junior high school or below and who attained junior college or above if upwardly mobile. Migrant families must have arrived in the destination city by 2007, ensuring substantial exposure to housing price changes over 2003–2012, and migrant children are restricted to those who were still in school when migration occurred. The corresponding city fixed effects are constructed from individual-level regressions controlling for children’s gender and age, parents’ employment industries, and additional family-background characteristics. These specifications identify effects within each family-arrangement group rather than by directly comparing co-residing and separated migrant children.

To examine the segregation mechanism, I additionally estimate analogous IV specifications for two outcomes. Using the 2015 Census, I construct city-level net migration ratios for working-age parents, separately by parental education. Using the 2016 CMDS, I estimate city-level parent–child separation among migrant families, again distinguishing low- and high-educated parents.

Two aspects of the specification are worth noting. First, I relate upward mobility to housing price growth over 2003–2012 rather than estimating an annual two-way fixed-effects model, because education is accumulated slowly and housing prices may affect children’s outcomes through multi-year changes in private investment, public education resources, and family arrangements. Second, the dependent variable is a level of upward mobility rather than a temporal change in mobility. This is consistent with the quantitative model, which predicts the probability that children from low-educated families reached junior college education or above as the result of local housing prices and household choices.

4.2 ENDOGENEITY CONCERNS

To obtain a consistent estimate of the coefficient of interest α , the validity of the interacted instrumental variables requires not only first-stage relevance, but also the exclusion restriction that, *conditional on controls*, the two interacted IVs affect UP_{ct} only through $\Delta \ln HP_{cT}$. As shown in [Borusyak et al. \(2025\)](#), when a shift-share instrumental variable (SSIV) is interacted with a unit-level variable, the interaction can be rewritten as a SSIV with the same shocks and adapted exposure weights, where the original shares are multiplied by the unit-level variable. For example, the interacted IV $\widehat{\Delta Emp_{cT}} \times slope_c$ can be rearranged as $\sum_j (slope_c \cdot s_{cj,t=2003}) \ln(1 + g_{jT})$. Following [Borusyak et al. \(2022\)](#) and [Goldsmith-Pinkham et al. \(2020\)](#), the exogeneity of this reconstructed SSIV can be justified either by shock exogeneity or by exposure-share exogeneity. In this setting, I rely primarily on the latter. This is because the data contain only 19 industries from CSY, making it difficult to rely on a large number of independent national industry shocks, and because national industry employment growth may itself reflect equilibrium labor-supply responses that are potentially related to local education opportunities.

The relevant exclusion restriction is therefore that the adapted exposure shares, such as $slope_c \cdot s_{cj,t=2003}$, affect upward mobility only through housing price growth, conditional on initial housing prices, housing constraints, initial industry shares, and province fixed effects. Equivalently, this requires that within cities with similar initial employment shares, unobserved cultural, institutional, or educational factors that affect intergenerational mobility do not vary systematically with local housing supply constraints. This assumption could be violated, for example, if cities with particular initial industry structures and more binding physical constraints also adopted unobserved education policies that independently improved upward mobility.

I provide three appendix exercises to assess this identifying assumption. Appendix Table [B4](#) finds no significant effect on a pre-period mobility measure from the 2005 Census. Appendix Table [B5](#) shows that most direct coefficients on the adapted exposure shares are statistically insignificant conditional on housing-price growth and baseline controls. Appendix Table [B6](#) adds contemporaneous changes in local income, public education spending, education employment, and school capacity; the native-homeowner estimate remains close to the baseline result. These exercises do not isolate a pure household wealth elasticity, but they reduce concerns that the estimate is driven by a specific observable city-level channel.

The first-stage coefficients $\beta = [\beta_1 \ \beta_2]$ should be interpreted as evidence that housing supply constraints amplify the capitalization of labour-demand shocks into housing prices, rather than as standalone causal estimates of the effect of each constraint on housing price growth. This distinction matters because both initial industry composition and housing supply constraints may be correlated with other determinants of housing price changes. For example, cities with higher initial employment in real-estate-related sectors may also have different housing supply conditions, and physical constraints such as $\%developed_{c0}$ and $slope_c$ may partly capture local amenities or other spatial characteristics valued by households. These factors could affect housing price growth even after controlling for initial housing prices. Therefore, while the positive and significant first-stage coefficients provide evidence of first-stage relevance for the interacted IV strategy, I interpret their magnitudes with caution. The main empirical object is the second-stage estimate of α , which uses the predicted component of housing price growth generated by the interaction between labour-demand shocks and predetermined housing supply constraints.

4.3 BASELINE RESULTS: EFFECTS ON UPWARD EDUCATIONAL MOBILITY

Table 1 first establishes the relevance of the two interacted instruments in the full city sample. The coefficients on both interactions are positive and statistically significant. In the preferred specification in column (3), the coefficients are 0.900 for the share of developable land already developed and 0.729 for terrain slope. Thus, a given labour-demand shock is capitalized more strongly into housing prices where supply is more constrained.

Table 2 reports the baseline city-level IV estimates of the effect of supply-driven housing-price appreciation on upward educational mobility. The dependent variable is the city fixed effect from the individual-level upward-mobility regression described in Eq. (2), which nets out observed individual and family-background characteristics. Column (1) focuses on children from low-educated native homeowner families. The estimated coefficient on housing-price growth is 0.193 and statistically significant, implying that a 10% increase in housing prices raises upward mobility by approximately 1.9 percentage points. This positive effect is consistent with a housing-resource channel: housing appreciation increases the resources of incumbent homeowners and may relax constraints on private education investment. Because housing-price appreciation is an equilibrium city-level outcome, however, the estimate does not isolate the homeowner housing-resource channel from concurrent changes in local income, public education resources, and other city-level conditions associated with local growth.

Columns (2)–(3) report the corresponding estimates for migrant children. The coefficients are 0.130 for co-residing migrant children and -0.012 for separated migrant children, and neither is statistically distinguishable from zero. Thus, the migrant-child estimates provide no evidence of mobility gains comparable to those observed among native homeowners, consistent with an asymmetric incidence of housing-price appreciation across incumbent homeowners and migrant families.¹⁰

To complement the analysis, Appendix Table B7 confirms the positive native-homeowner result at the individual level: the coefficient is 0.186 and the KP Wald F statistic is 13.09. The migrant-child coefficients remain imprecise. Appendix Table B6 further shows that the native-homeowner estimate is similar after conditioning on contemporaneous changes in local income and public-education inputs. While these controls may themselves respond to local price growth, the exercise suggests that the baseline estimate is not simply accounted for by these observable city-level changes. Finally, using six biennial waves of China Family Panel Studies from 2010 to 2020, Appendix Tables 1a–1c provide complementary evidence that higher self-reported housing value is associated with greater private education spending among homeowners, particularly on extracurricular tutoring. Taken together, these exercises provide complementary support for interpreting the positive native-homeowner mobility effect as consistent with a household housing-resource channel.

4.4 ADDITIONAL EVIDENCE: MIGRATION AND FAMILY-ARRANGEMENT RESPONSES

Table 3 provides additional evidence on the spatial-exclusion mechanism by applying the same IV strategy to two alternative outcomes. I replace the upward-mobility outcome in Eq. 2 with a city-level net migration ratio in Columns (1)–(3) and a city-level parent–child separation measure in Columns (4)–(6), while retaining the same housing-price treatment, instruments, and controls.

¹⁰The city-level Kleibergen–Paap Wald F statistics are 7.11, 7.43, and 21.11 across the three specifications; the corresponding individual-level statistics are 13.09, 11.83, and 37.85. I therefore report weak-identification-robust Anderson–Rubin tests. These reject the null of no effect for native homeowners ($p = 0.011$), but not for co-residing or separated migrant children ($p = 0.117$ and 0.745 , respectively).

Columns (1)–(3) use the 2015 Census to construct net migration ratios for working-age parents living with at least one child aged 0–18. For each city, net migration is measured as parental inflows minus outflows during 2010–2015, using information on residence one and five years before the survey. The net flow is divided by the corresponding number of parent workers residing in the city in 2010. A 10% increase in housing prices reduces the net migration ratio by 2.6 percentage points in the full sample and by 2.3 percentage points among low-educated parents, while the high-educated estimate is close to zero. Higher housing costs therefore disproportionately discourage low-educated parents from locating in expensive cities.

Columns (4)–(6) use the 2016 CMDS to examine family-arrangement decisions among migrant parents who arrived between 2012 and 2016. The outcome is a city fixed effect from parent-level regressions of whether parents migrated without their children, conditional on household and migration characteristics. A 10% increase in housing prices raises the separation measure by 4.7 percentage points in the full sample, 3.5 percentage points among low-educated parents, and 7.1 percentage points among high-educated parents. Together with the migration estimates, this pattern shows that housing costs shape family sorting along both the decision to migrate and the decision to bring children.¹¹

Taken together, higher prices deter low-educated parental migration and increase separation conditional on migration. Because the observed migrant groups are selected equilibrium outcomes, these estimates cannot be aggregated directly into a national mobility effect. The quantitative model addresses this limitation by jointly accounting for migration, family arrangements, children’s schooling locations, and local public education.

¹¹The CMDS estimates should nevertheless be interpreted cautiously. Because the survey is cross-sectional, it does not track children’s education histories or past schooling locations, limiting the ability to link observed parent–child separation directly to educational exposure. Moreover, the restricted city–education cells are relatively small. These limitations may reduce the precision of the point estimates.

Table 1: Effects of housing constraints on housing price change.

	Dependent variable: $\Delta \ln HP_{c,T}$		
	(1)	(2)	(3)
%developed $\times \widehat{\Delta Emp_{cT}}$	0.8524*** (0.2479)	0.7766*** (0.2502)	0.9004*** (0.2236)
Slope $\times \widehat{\Delta Emp_{cT}}$	0.8575*** (0.2844)	0.8769*** (0.2186)	0.7286*** (0.2105)
Initial lnHP	-0.3037*** (0.0917)	-0.4753*** (0.0958)	-0.6366*** (0.0573)
%developed	-0.2703*** (0.0797)	-0.2413*** (0.0781)	-0.2239*** (0.0666)
Slope	-0.2578*** (0.0862)	-0.2558*** (0.0676)	-0.2124*** (0.0609)
Initial employment shares	No	Yes	Yes
Province FE	No	No	Yes
Excluded IV F-stat.	6.71	8.64	8.77
N	268	268	268

Notes: The dependent variable is logarithm housing price change between 2003 and 2012. The two variables, later used as instruments in 2SLS regression, are the interactions between shift-share labour demand shocks and two physical housing supply constraints: the share of developable land already developed in 2000 and mean terrain slope within 10 km of the city centre, restricted to the city's 2010 administrative boundary. Both physical constraints are standardized within the estimation sample. All columns control for initial housing price; column (2) adds initial employment shares, and column (3) further adds province fixed effects.

Standard errors are clustered at the province level.

*** $p < .01$, ** $p < .05$, * $p < .1$

Table 2: [2SLS: City IM] IV estimates of the effect of housing price change on upward mobility.

	Native	Migrants	
	Homeowners	Co-reside	Separate
	(1)	(2)	(3)
<i>Panel A. 2nd stage: Dependent variable is UP_{ct}</i>			
$\Delta \ln \widehat{HP}_{cT}$	0.1929** (0.0863)	0.1303 (0.4931)	-0.0124 (0.3505)
Initial $\ln HP$	0.1952*** (0.0628)	0.3621 (0.3661)	-0.3041 (0.2458)
%developed	0.0088 (0.0099)	-0.0704 (0.0759)	-0.0018 (0.0590)
Slope	-0.0017 (0.0055)	-0.0830 (0.0524)	0.0573* (0.0314)
<i>Panel B. 1st stage: Dependent variable is $\Delta \ln HP_{c,T}$</i>			
%developed $\times \Delta \widehat{Emp}_{cT}$	0.8472*** (0.2266)	0.7645*** (0.2447)	0.6048* (0.3095)
Slope $\times \Delta \widehat{Emp}_{cT}$	0.6389** (0.2395)	0.8482*** (0.2665)	1.1963*** (0.1976)
Initial employment shares	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Initial UP	Yes	No	No
KP rk LM statistic	9.173	6.199	7.336
KP rk LM p-value	[0.010]	[0.045]	[0.026]
KP rk Wald F-statistic	7.109	7.432	21.112
Anderson–Rubin chi-sq	9.013	4.293	0.590
Anderson–Rubin Wald p-value	[0.011]	[0.117]	[0.745]
N	261	142	140

Notes: The table reports IV estimates of the effects of housing price changes between 2003–2012 on intergenerational education upward mobility in Chinese prefecture cities under Eq. 1. Panel A reports second-stage estimates, and Panel B reports first-stage estimates. Upward mobility (UP) is estimated as a city fixed effect from person-level regressions of educational mobility, controlling for individual and family-background characteristics, including gender, age, and parents' employment industries. A child is coded as moving up if he or she was born into a low-educated family, where the most educated parent attained junior high school or below (≤ 9 years of schooling), and subsequently attained college education or above (≥ 15 years of schooling). All three columns restrict children to the 1986–1993 birth cohorts. Column (1) uses the 2015 census to estimate UP for native children living in local homeowner families. Columns (2)–(3) use CMDS 2016 to estimate UP for co-residing and separated migrant children. Migrant families must have arrived in the destination city by 2007, ensuring at least five years of exposure to housing price changes over 2003–2012; migrant children are restricted to those still in school when migration occurred. Housing price changes are instrumented by interactions between shift-share labour-demand shocks and two physical housing-supply constraints: the share of developable land developed in 2000 and mean terrain slope within 10 km of the city centre, restricted to the city's 2010 administrative boundary. Additional controls, where available, include initial log housing prices in 2003 and the corresponding UP measure from the 2005 census.

Standard errors are clustered at the province level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$.

Table 3: [2SLS: Segregation effects]

IV estimates of the effect of housing price change on migration and parent-child separation.

	Parent net migration ratio			Parent-child separation		
	All (1)	Low edu (2)	High edu (3)	All (4)	Low edu (5)	High edu (6)
<i>Panel A. 2nd stage</i>						
$\Delta \ln \widehat{HP}_{cT}$	-0.2613*** (0.0868)	-0.2348*** (0.0768)	-0.0265 (0.0166)	0.4710*** (0.1721)	0.3462** (0.1611)	0.7076*** (0.2487)
Initial lnHP	-0.2479*** (0.0604)	-0.2265*** (0.0514)	-0.0214* (0.0125)	0.3790*** (0.1248)	0.3100** (0.1234)	0.4368** (0.1764)
%developed	0.0023 (0.0069)	0.0035 (0.0060)	-0.0013 (0.0017)	-0.0332 (0.0213)	-0.0294 (0.0223)	-0.0396 (0.0255)
Slope	0.0068 (0.0061)	0.0071 (0.0056)	-0.0002 (0.0009)	-0.0188 (0.0207)	-0.0088 (0.0184)	-0.0006 (0.0274)
<i>Panel B. 1st stage: Dependent variable is $\Delta \ln \widehat{HP}_{cT}$</i>						
%developed $\times \widehat{\Delta Emp}_{cT}$	0.8975*** (0.2181)	0.8975*** (0.2181)	0.8975*** (0.2181)	0.9789*** (0.2301)	0.9919*** (0.2323)	1.0923*** (0.3326)
Slope $\times \widehat{\Delta Emp}_{cT}$	0.7212*** (0.2124)	0.7212*** (0.2124)	0.7212*** (0.2124)	0.7489*** (0.2181)	0.7198*** (0.2265)	0.8011*** (0.2056)
Initial employment shares	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
KP rk LM statistic	8.927	8.927	8.927	9.165	9.281	9.428
KP rk LM p-value	[0.012]	[0.012]	[0.012]	[0.010]	[0.010]	[0.009]
KP rk Wald F-statistic	9.081	9.081	9.081	9.816	9.724	9.120
Anderson–Rubin chi-sq	10.716	11.898	10.235	12.093	6.748	12.755
Anderson–Rubin Wald p-value	[0.005]	[0.003]	[0.006]	[0.002]	[0.034]	[0.002]
N	268	268	268	259	252	244

Notes: The table reports IV estimates of the effects of housing price changes between 2003–2012 on city-level net migration and parent-child separation in Chinese prefecture cities under Eq. 1. Panel A reports second-stage estimates, and Panel B reports first-stage estimates. Columns (1)–(3) use the 2015 population census to construct city-level net migration ratios for working-age parents currently living with at least one child aged 0–18. Parents are identified as household heads or spouses; net migration is inflows minus outflows over 2010–2015 divided by the corresponding 2010 resident parent base, with one-year moves identified from the one-year residence question and earlier moves within the five-year window identified from the five-year residence question. Columns (4)–(6) use CMDS 2016 to estimate city fixed effects for whether migrant parents left children behind. Housing price changes between 2003 and 2012 are instrumented by interactions between shift-share labour-demand shocks and two physical housing-supply constraints: the share of developable land developed in 2000 and mean terrain slope within 10 km of the city centre, restricted to the city’s 2010 administrative boundary. All regressions control for initial housing price, the two constraint measures, initial industry shares, and province fixed effects.

Standard errors are clustered at the province level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$.

5 A MODEL OF INTERGENERATIONAL MOBILITY

In this section, I develop a quantitative spatial model with an intergenerational life-cycle structure to quantify how housing supply constraints shape the intergenerational incidence of urban growth. Building on work on spatial family migration (Imbert et al., 2026) and spatial intergenerational mobility (Eckert and Kleineberg, 2024), the framework introduces a housing-price incidence margin: housing supply conditions determine how urban growth is capitalized into incumbent housing wealth and the cost of spatial and educational access for migrant families. Three features are central. First, households differ in their initial housing endowments, so house-price appreciation raises the resources of incumbent asset holders while higher destination housing costs tighten the budgets of households seeking to move. Second, parents choose both where to live and whether to co-reside with their children, so a child’s schooling location need not coincide with the parent’s workplace and is endogenously determined by migration and family-arrangement choices. Third, children’s educational attainment depends on private education investment and effective local public education resources, with school-access barriers limiting the resources available to migrant children in destination cities.

Model setup

The economy consists of a finite set of cities $\mathcal{G}$. Households are indexed by their origin city $o \in \mathcal{G}$, parental education type $e \in \{H, L\}$, and initial housing-wealth status $\delta \in \{\mathcal{B}, \mathcal{R}\}$. The education type e captures parents’ early-stage human capital. The housing-wealth state δ indicates whether a household holds an initial urban housing endowment in its origin. Specifically, $\delta = \mathcal{B}$ denotes initial asset holders with housing endowment $E_o^{e\mathcal{B}} > 0$, whereas $\delta = \mathcal{R}$ denotes households without such an endowment, for which $E_o^{e\mathcal{R}} = 0$.¹² Childless adults make only a destination-location choice. Parents choose a family arrangement $f \in \{0, 1\}$ and a destination city $j \in \mathcal{G}$. The family-arrangement choice determines whether the child co-resides with the parent: $f = 1$ denotes co-residence, while $f = 0$ denotes non-co-residence. If parents migrate and choose $f = 0$, the child remains in the origin.

Time is discrete and indexed by t . Individuals pass through two stages of the life cycle, childhood and adulthood. During childhood in period $t - 1$, individual i receives early-stage human capital e , initial housing status δ , and origin city o . The model focuses on the period- t spatial equilibrium, taking these childhood states as predetermined.

Let $H_{ot}^{e\delta\emptyset}$ denote the mass of childless adults of type (e, δ) from origin o , and let $H_{ot}^{e\delta}$ denote the mass of parents of type (e, δ) from origin o . These masses are predetermined in period t . Each city also contains an exogenous, immobile residual population O_j , consisting of individuals aged 65 or above who lie outside the working-age household choice problem and contribute fixed local housing demand.

Upon reaching adulthood in period t , individuals take (e, δ, o) and parental status as given. Their subsequent choices are represented by a nested Fréchet structure over family arrangement and destination city. At each stage, individuals observe the shock associated with that stage, while lower-level choices are summarized by inclusive values. Parents first choose whether to co-reside with their child, $f \in \{0, 1\}$; adults then choose a destination city $j \in \mathcal{G}$. Childless adults only make the destination-location choice. The corresponding shocks are a_{if}^e and s_{ij}^e , whose dispersion parameters may differ by education group $e \in \{H, L\}$. This nested structure, illustrated in Fig. 4, allows the model to be

¹²The initial housing endowment combines several forms of urban owner-occupied housing, of which the windfall generated by public-housing privatisation is one component.

solved by backward induction (Tsivanidis, 2019; Miyauchi et al., 2021; Zhang, 2022; Imbert et al., 2026; Pietrabissa, 2023).

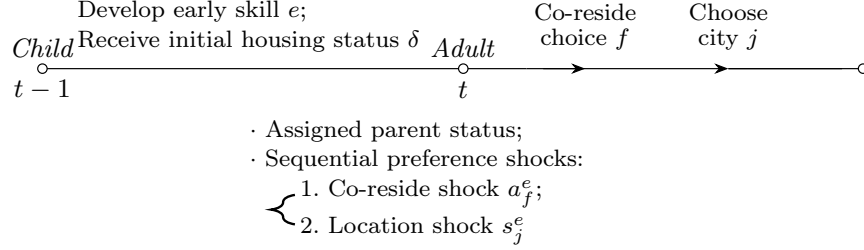


Figure 4: Model timing and household decisions.

Since the analysis focuses on a single period- t spatial equilibrium conditional on these predetermined states, I suppress the time index from all variables below. I first describe the household problem, then specify housing markets, local public education finance, and production in each city. I then define equilibrium using the implied market-clearing conditions.

5.1 HOUSEHOLDS

Utility

For an adult i of type (e, δ) , where e denotes education type and δ denotes initial housing-wealth status, who originates from city o and works, consumes, and resides in city j , adult utility under family arrangement f is

$$U_{ijo}^{e\delta f} = \frac{A_j^{ef} \left(\tilde{\Gamma}_{ijo}^{e\delta f} \right)^{\iota^e \mathbb{1}\{f \neq \emptyset\}} a_{if}^{e \mathbb{1}\{f \neq \emptyset\}} s_{ij}^e}{d_{jo}^{ef}} \left(\frac{c_{ij}}{1 - \alpha^{ef}} \right)^{1 - \alpha^{ef}} \left(\frac{h_{ij} - \bar{h}^f}{\alpha^{ef}} \right)^{\alpha^{ef}}, \quad (3)$$

where $h_{ij} > \bar{h}^f$. The parameter $\bar{h}^f$ is the minimum housing requirement associated with family arrangement f . The Stone–Geary term captures that households must first secure basic housing services before allocating residual resources across numeraire consumption, additional housing services, and, for parents, private education investment. A_j^{ef} denotes local amenities for workers of education type e under family arrangement f . The variable c_{ij} denotes numeraire consumption, and h_{ij} denotes total housing-service consumption. The dummy $\mathbb{1}\{f \neq \emptyset\}$ equals one for adults with a dependent child. The term $\tilde{\Gamma}_{ijo}^{e\delta f}$ is the child’s education-production index, which determines the child’s probability of attaining high education. The parameter $\iota^e > 0$ captures parental altruism by governing the weight that parents place on the child’s human-capital prospects in their utility.

Parents ($f \neq \emptyset$) draw F chet-distributed co-residence shocks a_{if}^e for each family arrangement $f \in \{0, 1\}$, with a_{if}^e independently distributed according to $G_f^e(a) = \exp(-T_f^e a^{-\rho^e})$, where $T_f^e > 0$. All adults draw F chet-distributed location shocks s_{ij}^e for each destination $j \in \mathcal{G}$, with s_{ij}^e independently distributed according to $G_j(s) = \exp(-s^{-\epsilon^e})$. The two sets of shocks are mutually independent. The shape parameters ρ^e and ϵ^e govern the dispersion of preferences over family arrangements and locations, respectively, while T_f^e governs the scale of the family-arrangement preference distribution. I normalise $T_0^e = 1$, so that T_1^e captures the relative scale of the preference distribution for co-residence.

The term d_{jo}^{ef} is an iceberg migration cost that varies by education type and family arrangement:

$$d_{jo}^{ef} = \exp \left[\mu^{ef} \text{dist}_{jo} + \gamma_P^{ef} \mathbb{1}\{p(j) \neq p(o)\} + \gamma_R^{ef} \mathbb{1}\{g(j) \neq g(o)\} + \kappa^{ef} \mathbb{1}\{j \neq o\} \right]. \quad (4)$$

where $d_{oo}^{ef} = 1$. The parameter μ^{ef} captures distance-based migration costs, γ_P^{ef} and γ_R^{ef} capture additional frictions associated with crossing provincial and broader regional boundaries, respectively, and κ^{ef} captures the fixed cost of leaving the origin.

The discretionary housing share α^{ef} varies by education and family arrangement, while the minimum housing requirement $\bar{h}^f$ captures basic housing needs before discretionary spending. For example, $\bar{h}^1 > \bar{h}^0$ would imply the larger housing requirement when parents co-reside with their child.

A type (e, δ) adult choosing family arrangement f and destination city j maximises utility subject to

$$c_{ij} + r_j h_{ij} + \mathbb{1}\{f \neq \emptyset\} K_{ij} = (1 - \tau_j) \omega_j^e + r_o E_o^{e\delta} \equiv Y_{jo}^{e\delta}. \quad (5)$$

On the left-hand side, the term $r_j h_{ij}$ is the annual cost of consuming housing services in destination city j . I abstract from an endogenous destination buy-rent tenure choice, so all households residing in city j face the same housing-service price r_j . The right-hand side of Eq. (5) is the household's effective income, combining after-tax labour income and the annualised housing-service value associated with its initial housing endowment. Initial housing endowments are predetermined by origin and initial housing-wealth status:

$$E_o^{e\mathcal{B}} > 0, \quad E_o^{e\mathcal{R}} = 0.$$

Here, $E_o^{e\delta}$ is an effective housing endowment and r_o is the annualised housing price in the origin. Incumbent households with $\delta = \mathcal{B}$ hold urban housing assets, whereas households with $\delta = \mathcal{R}$ do not.¹³ Given Stone–Geary preferences, the household must first secure the minimum housing requirement $\bar{h}^f$. Residual resources available for numeraire consumption, discretionary housing services, and, for parents, private education investment are written as

$$\tilde{Y}_{jo}^{e\delta f} \equiv Y_{jo}^{e\delta} - r_j \bar{h}^f. \quad (6)$$

An alternative location-parental choice (j, f) is feasible only if $\tilde{Y}_{jo}^{e\delta f} > 0$.

Housing prices therefore enter the household problem asymmetrically across initial housing-wealth groups. A higher destination housing-service price reduces $\tilde{Y}_{jo}^{e\delta f}$ and tightens affordability, especially when co-residence requires a larger minimum housing requirement. For initial asset holders, house-price appreciation can additionally relax education-investment constraints through the housing-wealth channel introduced in the counterfactual analysis.

Production of children's education

Children's educational outcomes depend on parental background, private education investment, and

¹³The term $r_o E_o^{e\delta}$ captures the annualised housing-service value associated with the household's initial housing endowment, rather than a separately liquid capital gain. For an incumbent owner remaining in the origin, changes in this service value are largely offset by corresponding changes in housing-service expenditure. In the productivity counterfactual, I therefore introduce an additional appreciation-based term, $(r_o - r_o^0) E_o$ where r_o^0 denotes the pre-shock annualised housing price in the origin. This term captures, in static model, the housing-wealth channel through which rising housing values relax household resource constraints, including through greater liquidity, collateral value, or lower precautionary saving.

effective public education resources. For the child of a type $e\delta f$ adult from origin o residing in city j , define

$$\tilde{\Gamma}_{ijo}^{e\delta f} = \zeta_e \left(\frac{K_{ij}}{\bar{K}} \right)^\nu \left(\frac{\Lambda_{ijo}^f}{\bar{\Lambda}} \right)^q, \quad (7)$$

where $\zeta_e > 0$ captures baseline intergenerational transmission associated with parental education type e , K_{ij} is private education investment, and Λ_{ijo}^f is the effective public education resource received by the child. I normalise private and public education inputs by their parent-mass-weighted geometric means $\bar{K}$ and $\bar{\Lambda}$ in the calibrated 2005 allocation. These reference levels provide a representative scale for the national parent population and make both input measures unit-free. They are held fixed in all subsequent exercises.

The term $\tilde{\Gamma}_{ijo}^{e\delta f}$ is a latent education-production index that enters parental utility with weight ι^e . The parameters ν and q are the elasticities of the latent education index with respect to private and effective public education inputs, while ζ_e governs its baseline level. For convenience, define

$$\tilde{\nu}^e \equiv \iota^e \nu, \quad \tilde{q}^e \equiv \iota^e q.$$

These adjusted parameters govern the effective importance of private and public education inputs in parental utility.

Importantly, family arrangement determines the child's schooling location and therefore which local public education resources the child can access:

$$\Lambda_{ijo}^f = \left(\Delta_j^{\mathbb{1}\{j \neq o\}} \Lambda_j \right)^{\mathbb{1}\{f=1\}} (\Lambda_o)^{\mathbb{1}\{f=0\}}. \quad (8)$$

Co-residing children attend school in the parent's residence city j . If the parent remains in the origin and co-resides with the child ($f = 1, j = o$), the child receives full local public education resources Λ_o . Importantly, if the parent migrates with the child ($f = 1, j \neq o$), the child receives only a fraction Δ_j of destination public education resources under the institutional setting in China, where $0 < \Delta_j \leq 1$. Non-co-residing children attend school in origin city o ; when $j \neq o$, this corresponds to the left-behind-child arrangement. The access factor Δ_j captures hukou- and enrollment-related barriers faced by migrant children, such as documentation requirements, residence permits, social insurance conditions, and local school-capacity constraints described in Section 2.3.

For adults with a dependent child, private education investment is chosen from residual resources after the minimum housing requirement has been paid. The utility-maximization problem implies

$$K_{ij}^* = \frac{\tilde{\nu}^e}{1 + \tilde{\nu}^e} \tilde{Y}_{jo}^{e\delta f}.$$

Thus, private education investment inherits the same two housing-price channels: higher destination housing-service costs reduce investment by lowering residual resources, while higher origin housing values raise investment for households with initial housing endowments. A larger parental altruism parameter ι^e increases the share of residual resources devoted to private education investment ($\tilde{\nu}^e \equiv \iota^e \nu$).

I map the education-production index into the probability that the child becomes high-educated as in Eq. 9. This probability mapping is used to connect the model to observed educational outcomes.

Since the mapping is monotone in $\tilde{\Gamma}_{ijo}^{e\delta f}$ and bounded between 0 and 1, it preserves the ranking of children's educational prospects while providing a convenient mapping from the latent education index to observed educational attainment probabilities.

$$P_{ijo}^{e\delta f} = \frac{\tilde{\Gamma}_{ijo}^{e\delta f}}{1 + \tilde{\Gamma}_{ijo}^{e\delta f}}. \quad (9)$$

Indirect utility

I solve the household problem by backward induction over destination city j and family arrangement f . This subsection derives the alternative-specific indirect utility conditional on origin o , destination j , and family arrangement f . Given (o, j, f) , households choose numeraire consumption, excess housing services, and, for parents, private education investment. All expressions are defined for feasible alternatives with $\tilde{Y}_{jo}^{e\delta f} > 0$.

For childless adults, $f = \emptyset$, private education investment is absent. Maximising Eq. 3 subject to the discretionary budget gives

$$v_{ijo}^{e\delta\emptyset} = \frac{A_j^{e\emptyset} \tilde{Y}_{jo}^{e\delta\emptyset}}{d_{jo}^{e\emptyset} (r_j)^{\alpha^{e\emptyset}} s_{ij}^e}. \quad (10)$$

Thus, childless adults value destinations with higher amenities ($A_j^{e\emptyset}$) and residual resources ($\tilde{Y}_{jo}^{e\delta\emptyset}$), and prefer lower migration costs ($d_{jo}^{e\emptyset}$) and lower housing-service prices (r_j).

For adults with a dependent child, $f \in \{0, 1\}$, optimal private education investment yields

$$v_{ijo}^{e\delta f} = \chi^e \frac{A_j^{ef} (\Lambda_{ijo}^f)^{\tilde{q}^e} (\tilde{Y}_{jo}^{e\delta f})^{1+\tilde{\nu}^e}}{d_{jo}^{ef} (r_j)^{\alpha^{ef}}} a_{if} s_{ij}^e, \quad (11)$$

where

$$\chi^e \equiv (\zeta_e)^{\iota^e} (\tilde{\nu}^e)^{\tilde{\nu}^e} (1 + \tilde{\nu}^e)^{-(1+\tilde{\nu}^e)} \bar{K}^{-\tilde{\nu}^e} \bar{\Lambda}^{-\tilde{q}^e}.$$

The term χ^e collects factors that are common across all location and family-arrangement alternatives for parental education group e . Conditional on (o, j, f) , parental indirect utility rises with amenities (A_j^{ef}), effective public education (Λ_{ijo}^f), and residual resources ($\tilde{Y}_{jo}^{e\delta f}$), and falls with migration costs (d_{jo}^{ef}) and the destination housing-service price (r_j).

For co-residing parents, $f = 1$, the child lives and attends school in the parent's residence city j , so $\Lambda_{ijo}^1 = \Delta_j^{\mathbb{1}_{\{j \neq o\}}} \Lambda_j$. A child who migrates with the parent therefore receives only a fraction Δ_j of destination public education resources, reflecting migrant school-access barriers. For non-co-residing parents, $f = 0$, the child remains and attends school in the origin city o , so $\Lambda_{ijo}^0 = \Lambda_o$; when $j \neq o$, this is the left-behind-child arrangement. Family arrangements thus determine both the child's schooling location and effective access to public education.

Backward induction I: Location choice

I solve the household problem by backward induction over destination city j and parental arrangement f . This subsection derives the destination-choice probabilities conditional on origin o , education type e , initial housing-wealth status δ , and family arrangement f . Given (o, f) , individuals choose the destination city that gives the highest alternative-specific indirect utility. All expressions are defined

for feasible alternatives with $\tilde{Y}_{jo}^{e\delta f} > 0$.

Let the constant $\gamma_1^e = \Gamma(1 - 1/\epsilon^e)$, where $\Gamma(\cdot)$ is the Gamma function. The parameter ϵ^e governs the dispersion of idiosyncratic location preferences: a larger ϵ^e implies weaker idiosyncratic attachment and more elastic sorting toward destinations with higher deterministic attractiveness.

For childless adults, $f = \emptyset$, define

$$\tilde{w}_{jo}^{e\delta\emptyset} = \frac{A_j^{e\emptyset} \tilde{Y}_{jo}^{e\delta\emptyset}}{d_{jo}^{e\emptyset} (r_j)^{\alpha^{e\emptyset}}}. \quad (12)$$

This term is the deterministic attractiveness of destination j : it increases with amenities ($A_j^{e\emptyset}$) and residual resources ($\tilde{Y}_{jo}^{e\delta\emptyset}$), and decreases with migration costs ($d_{jo}^{e\emptyset}$) and housing-service prices (r_j). Since $v_{ijo}^{e\delta\emptyset} = \tilde{w}_{jo}^{e\delta\emptyset} s_{ij}^e$ and given the properties of the Fréchet-distributed location preference shock s_{ij}^e , the probability that a type $e\delta$ childless adult from origin o chooses destination j is

$$\pi_{j|o}^{e\delta\emptyset} = \frac{(\tilde{w}_{jo}^{e\delta\emptyset})^{\epsilon^e}}{\sum_{n \in \mathcal{G}} (\tilde{w}_{no}^{e\delta\emptyset})^{\epsilon^e}}. \quad (13)$$

Expected utility after integrating over destination choices can be written as

$$\bar{v}_o^{e\delta\emptyset} = \gamma_1^e \left[\sum_{n \in \mathcal{G}} (\tilde{w}_{no}^{e\delta\emptyset})^{\epsilon^e} \right]^{1/\epsilon^e}.$$

For co-residing parents, $f = 1$, define

$$\dot{w}_{jo}^{e\delta 1} = \chi^e \frac{A_j^{e1} \left(\Delta_j^{\mathbb{1}\{j \neq o\}} \Lambda_j \right)^{\tilde{q}^e} \left(\tilde{Y}_{jo}^{e\delta 1} \right)^{1+\tilde{\nu}^e}}{d_{jo}^{e1} (r_j)^{\alpha^{e1}}}.$$

This term captures the attractiveness of destination j when the child co-resides with the parent. It increases with destination amenities (A_j^{e1}), residual resources ($\tilde{Y}_{jo}^{e\delta 1}$), and effective destination public education ($\Delta_j^{\mathbb{1}\{j \neq o\}} \Lambda_j$), and decreases with migration costs (d_{jo}^{e1}) and the housing-service price (r_j). Since the parental-arrangement preference shock a_{i1} is common across destinations conditional on $f = 1$, it does not affect the location choice. The probability that a type $e\delta$ parent from origin o chooses destination j , conditional on co-residing with the child, is

$$\pi_{j|o}^{e\delta 1} = \frac{(\dot{w}_{jo}^{e\delta 1})^{\epsilon^e}}{\sum_{n \in \mathcal{G}} (\dot{w}_{no}^{e\delta 1})^{\epsilon^e}}. \quad (14)$$

Thus the expected utility of a type $e\delta$ co-residing parent after integrating over destination choices is

$$\bar{v}_o^{e\delta 1} = \gamma_1^e \left[\sum_{n \in \mathcal{G}} (\dot{w}_{no}^{e\delta 1})^{\epsilon^e} \right]^{1/\epsilon^e} a_{i1} \equiv \mathcal{U}_o^{e\delta 1} a_{i1}. \quad (15)$$

Similarly, for non-co-residing parents, $f = 0$, define

$$\hat{w}_{jo}^{e\delta 0} = \chi^e \frac{A_j^{e0} (\Lambda_o)^{\tilde{q}^e} \left(\tilde{Y}_{jo}^{e\delta 0} \right)^{1+\tilde{\nu}^e}}{d_{jo}^{e0} (r_j)^{\alpha^{e0}}}.$$

This term captures the attractiveness of destination j when the child remains educated in the origin. It increases with type-specific amenities (A_j^{e0}) and residual resources ($\tilde{Y}_{jo}^{e\delta 0}$), and decreases with migration costs (d_{jo}^{e0}) and the housing-service price (r_j), while the child's public education resources remain tied to the origin (Λ_o); when $j \neq o$, this is the left-behind-child arrangement. Since the parental-arrangement preference shock a_{i0} is common across destinations conditional on $f = 0$, their choice probability of migrating to city j conditional on originating from o is

$$\pi_{j|o}^{e\delta 0} = \frac{(\hat{w}_{jo}^{e\delta 0})^{\epsilon^e}}{\sum_{n \in \mathcal{G}} (\hat{w}_{no}^{e\delta 0})^{\epsilon^e}}. \quad (16)$$

Therefore, the expected utility of a type $e\delta$ non-co-residing parent after integrating over destination choices is

$$\bar{v}_o^{e\delta 0} = \gamma_1^e \left[\sum_{n \in \mathcal{G}} (\hat{w}_{no}^{e\delta 0})^{\epsilon^e} \right]^{1/\epsilon^e} a_{i0} \equiv \mathcal{U}_o^{e\delta 0} a_{i0}. \quad (17)$$

Backward induction II: Parental choice

Finally, I return to the first decision made by adults with a dependent child: whether to co-reside with the child. Parental status is predetermined, so fertility is not a choice margin. At this stage, type $e\delta$ and origin o are given, while destination choices have been integrated out in Eqs. 15 and 17. Parents compare the value of co-residence, $\mathcal{U}_o^{e\delta 1} a_{i1}$, with the value of not co-residing, $\mathcal{U}_o^{e\delta 0} a_{i0}$, after observing the family arrangement preference shock. Since the shocks a_{if}^e are drawn from Fréchet distributions with family-arrangement-specific scale parameters T_f^e and common shape parameter ρ^e , with T_0^e normalized to one, the share of type $e\delta$ parents from origin o choosing arrangement $f \in \{0, 1\}$ is

$$\pi_{f|o}^{e\delta} = \frac{T_f^e (\mathcal{U}_o^{e\delta f})^{\rho^e}}{T_1^e (\mathcal{U}_o^{e\delta 1})^{\rho^e} + T_0^e (\mathcal{U}_o^{e\delta 0})^{\rho^e}}, \quad f \in \{0, 1\}. \quad (18)$$

Co-residence gives the child access to public education resources in the parent's residence city. If the parent migrates with the child, the child faces limited access to destination public education resources relative to incumbent children, captured by the access factor Δ_j . Co-residence is also more housing-intensive, with possibly $\bar{h}^1 > \bar{h}^0$. Higher destination housing prices therefore reduce the deterministic value of co-residence, $\hat{w}_{jo}^{e\delta 1}$, more strongly than the value of not co-residing, $\hat{w}_{jo}^{e\delta 0}$. Through the inclusive values $\mathcal{U}_o^{e\delta 1}$ and $\mathcal{U}_o^{e\delta 0}$, stronger price capitalization in housing-constrained destinations can therefore lower the relative attractiveness of co-residence and migration with children.

Spatial distribution of labour and children

Given the location-choice probabilities in Eqs. 13–16 and the parental-arrangement probabilities in

Eq. 18, the mass of type- e adults working in city j is

$$L_j^e = \sum_{o \in \mathcal{G}} \sum_{\delta} \left[\pi_{j|o}^{e\delta\varnothing} H_o^{e\delta\varnothing} + \sum_{f \in \{0,1\}} \pi_{j|o}^{e\delta f} \pi_{f|o}^{e\delta} H_o^{e\delta} \right]. \quad (19)$$

The first term in the bracket captures the number of childless adults who choose destination city j , while the second captures the number of parents who first choose family arrangement f and then choose destination city j conditional on f .

Children may either migrate with their parents or remain in the origin. I do not explicitly model couples. Instead, each parental decision unit is associated with φ dependent-child equivalents, where φ is calibrated to match the child-to-parent ratio in the data.

The mass of children attending school in city j is

$$m_j = \varphi \sum_e \sum_{\delta} \left[\sum_{o \in \mathcal{G}} \pi_{j|o}^{e\delta 1} \pi_{1|o}^{e\delta} H_o^{e\delta} + \pi_{0|j}^{e\delta} H_j^{e\delta} \right]. \quad (20)$$

The first term in the bracket captures co-residing children who live and attend school in city j with their parents, including both local families and migrants from origins $o \neq j$. The second term captures non-co-residing children whose origin is city j and who remain there while their parents choose not to co-reside and thus may work elsewhere. Note that among children attending school in city j , the mass of migrant children who originates from another city $o \neq j$ can be expressed as

$$m_j^M = \varphi \sum_e \sum_{\delta} \sum_{o \in \mathcal{G} \setminus \{j\}} \pi_{j|o}^{e\delta 1} \pi_{1|o}^{e\delta} H_o^{e\delta}. \quad (21)$$

The mass of local children is then $m_j^L = m_j - m_j^M$, where the superscript L on m_j^L denotes local rather than low-educated children.

5.2 HOUSING MARKETS

Given the Stone–Geary housing preference, housing demand consists of a minimum housing requirement and discretionary housing services. For childless adults,

$$h_{jo}^{e\delta\varnothing} = \bar{h}^{\varnothing} + \frac{\alpha^{e\varnothing} \tilde{Y}_{jo}^{e\delta\varnothing}}{r_j}. \quad (22)$$

For parents, optimal private education investment absorbs the fraction $\tilde{\nu}^e/(1 + \tilde{\nu}^e)$ of residual resources. Hence, for $f \in \{0,1\}$,

$$h_{jo}^{e\delta f} = \bar{h}^f + \frac{\alpha^{ef}}{1 + \tilde{\nu}^e} \frac{\tilde{Y}_{jo}^{e\delta f}}{r_j}. \quad (23)$$

In both cases, the first term is the basic housing requirement and the second term is discretionary housing demand.

Aggregate housing demand in each destination city j is

$$F_j^D = \sum_{o,e,\delta} \pi_{j|o}^{e\delta\varnothing} H_o^{e\delta\varnothing} h_{jo}^{e\delta\varnothing} + \sum_{o,e,\delta} \sum_{f \in \{0,1\}} \pi_{j|o}^{e\delta f} \pi_{f|o}^{e\delta} H_o^{e\delta} h_{jo}^{e\delta f} + O_j h^O. \quad (24)$$

The first term aggregates housing demand from childless adults originating across all cities $o \in \mathcal{G}$, the second from parents originating across all cities $o \in \mathcal{G}$ whose housing demand depends on family arrangement choice f , and the final term from the exogenous immobile residual population O_j , with per-capita housing demand h^O .

Residential developers combine capital with inelastically supplied local land using a Cobb–Douglas technology.¹⁴ Perfect competition implies the inverse housing supply function

$$r_j = l_j (F_j^S)^{1/\eta_j}, \quad (25)$$

where l_j captures city-specific construction costs, and $\eta_j > 0$ is the housing-supply elasticity. A lower η_j means that housing supply is more constrained and that housing costs rise more strongly with local housing demand. Thus, η_j governs how strongly increases in local housing demand are capitalized into prices rather than absorbed through housing quantity; this is the housing-supply margin varied in the main counterfactual. The housing market clears in each city when aggregate demand equals aggregate supply:

$$F_j^D = F_j^S.$$

5.3 GOVERNMENTS

The economy has a decentralised public education system. Local governments levy a tax at rate τ_j on adult labour income and use the revenue to finance local public education. I take τ_j as exogenously given, allowing local fiscal capacity to differ across cities without modelling strategic tax competition among local governments.¹⁵

Local public education resources are financed by the city tax base and allocated to children attending school in the city. Let m_j^L and m_j^M denote the masses of local and migrant children attending school in city j , respectively. Local children receive the full per-pupil public education resource Λ_j , while migrant children receive a fraction Δ_j of this amount. The local government budget constraint is

$$\Lambda_j m_j^L + \Delta_j \Lambda_j m_j^M = \tau_j \sum_e \omega_j^e L_j^e, \quad (26)$$

where $0 < \Delta_j \leq 1$. The right-hand side is local income tax revenue.

Thus, public education resources depend on local fiscal capacity, the local tax base, and the composition of school-age children. The school-access factor Δ_j captures hukou- and enrollment-related barriers faced by migrant children: compared with local children, migrant children receive

¹⁴See Appendix Section D.1 for details.

¹⁵Land sales constitute an important source of local fiscal revenue in China (Xiong, 2023; Gyourko et al., 2022). I abstract from land-sale revenues because the empirical evidence suggests that land revenues are often directed towards growth-oriented infrastructure rather than education or public health spending (Ding et al., 2014).

fewer effective public education resources per pupil in the destination city. Because employment (L_j^e) and the masses of local and migrant pupils (m_j^L and m_j^M) are endogenous, migration, co-residence, and skill sorting feed back into local public education resources in equilibrium.

5.4 PRODUCTION

Each city hosts perfectly competitive firms that produce a single final good. Workers with different education types are imperfect substitutes. Output in city j is

$$X_j = \left[(B_j^H)^{\frac{1}{\vartheta}} (L_j^H)^{\frac{\vartheta-1}{\vartheta}} + (B_j^L)^{\frac{1}{\vartheta}} (L_j^L)^{\frac{\vartheta-1}{\vartheta}} \right]^{\frac{\vartheta}{\vartheta-1}}, \quad (27)$$

where $\vartheta > 1$ is the elasticity of substitution between high- and low-educated labour, L_j^e is type- e employment in city j , and B_j^e is a city- and education-specific productivity fundamental. Changes in B_j^e are the model's local growth shifters: higher productivity raises labour demand and wages and, through household location choices, local housing demand; housing-supply elasticity then determines how strongly this growth is capitalized into housing prices.

With the final-good price normalised to one, perfect competition implies that pre-tax wages equal marginal products. The high-skill wage premium is

$$\psi_j \equiv \frac{\omega_j^H}{\omega_j^L} = \left(\frac{B_j^H}{B_j^L} \right)^{\frac{1}{\vartheta}} \left(\frac{L_j^H}{L_j^L} \right)^{-\frac{1}{\vartheta}}. \quad (28)$$

This downward-sloping relative demand curve limits complete high-skilled sorting: as high-educated workers concentrate in a city, their relative wage falls, partially offsetting the attractiveness of that destination.

5.5 EQUILIBRIUM AND DRIVERS OF UPWARD MOBILITY

Equilibrium

Given structural parameters $\{\alpha^{ef}, \bar{h}^f, \zeta_e, \nu, q, \iota^e, T_f^e, \rho^e, \epsilon^e, \mu^{ef}, \gamma_P^{ef}, \gamma_R^{ef}, \kappa^{ef}, \vartheta\}$, initial adult populations $\{H_o^{e\delta\emptyset}, H_o^{e\delta}\}$, and local fundamentals $\{A_j^{ef}, B_j^e, O_j, E_o^{e\delta}, \eta_j, l_j, \tau_j, \text{dist}_{jo}, \Delta_j\}$, a competitive equilibrium consists of endogenous prices and public education resources $\{r_j, \omega_j^e, \Lambda_j\}$, allocations $\{L_j^e, m_j, m_j^M, F_j^D, F_j^S\}$, and choice probabilities $\{\pi_{f|o}^{e\delta}, \pi_{j|o}^{e\delta f}\}$ that simultaneously solve the household problem, firms' problem, local governments' budget constraint, and housing-market-clearing conditions:

- i) *Household optimisation.* Taking prices, taxes, public education resources, and local fundamentals as given, households maximise utility subject to the budget constraint and the feasibility condition $\tilde{Y}_{jo}^{e\delta f} > 0$. This pins down family-arrangement and location-choice probabilities in Eqs. (13)–(16) and (18), as well as optimal consumption, housing services, and private education investment.
- ii) *Firm optimisation.* Taking production fundamentals and local labour supplies as given, perfectly competitive firms choose labour demand. Wages equal marginal products under Eq. (27),

implying the relative wage condition in Eq. (28).

- iii) *Local government budget balance.* Taking local tax rates τ_j as given, local governments finance public education from local labour income taxes. Public education resources satisfy Eq. (26), where migrant children receive a fraction Δ_j of the public education resources received by local children.
- iv) *Housing market clearing.* Given household housing demand and the inverse housing supply function, housing markets clear in every city: $F_j^D = F_j^S$, with aggregate demand given by Eq. (24) and inverse supply given by Eq. (25).

City-level intergenerational mobility

For parental education type e , the model aggregates household-level education probabilities across children attending school in city j . The resulting city-level probability of high educational attainment is

$$\bar{P}_j^e = \frac{\sum_{\delta, n \in g} P_{nj}^{e\delta 0} \pi_{n|j}^{e\delta 0} \pi_{0|j}^{e\delta} H_j^{e\delta} + \sum_{\delta, o \in g} P_{jo}^{e\delta 1} \pi_{j|o}^{e\delta 1} \pi_{1|o}^{e\delta} H_o^{e\delta}}{\sum_{\delta, n \in g} \pi_{n|j}^{e\delta 0} \pi_{0|j}^{e\delta} H_j^{e\delta} + \sum_{\delta, o \in g} \pi_{j|o}^{e\delta 1} \pi_{1|o}^{e\delta} H_o^{e\delta}}. \quad (29)$$

The numerator sums the model-implied probabilities of high education across children attending school in city j , weighted by their equilibrium masses, while the denominator is the corresponding child mass. The first term in each part captures non-co-residing children whose origin is j and whose parents may work in any city n ; the second captures co-residing children whose parents originate from city o and live with them in j . City-level mobility is therefore indexed by the child's schooling location rather than the parent's workplace or origin. This distinction is central because migration can separate the location where parents work from the location where their children accumulate human capital. For children of low-educated parents, $\bar{P}_j^L$ measures upward mobility; for children of high-educated parents, $\bar{P}_j^H$ measures educational persistence.

Aggregate intergenerational mobility

The model also implies national education outcomes by aggregating city-level probabilities across schooling locations. Let m_j^e denote the mass of children attending school in city j with parental education type e . The national high-education probability for children with parental education type e is

$$\bar{P}^e = \frac{\sum_j m_j^e \bar{P}_j^e}{\sum_j m_j^e}. \quad (30)$$

For children of low-educated parents, $\bar{P}^L$ measures aggregate upward mobility; for children of high-educated parents, $\bar{P}^H$ measures aggregate educational persistence. The counterfactual analysis focuses primarily on $\bar{P}^L$. Unlike a city-level comparison, this national object allows both children's outcomes and their endogenous allocation across schooling locations to adjust, which is necessary for evaluating the aggregate implications of housing constraints.

Aggregating over parental education types gives the high-education share of the next generation:

$$\bar{P} = \frac{\sum_{e,j} m_j^e \bar{P}_j^e}{\sum_{e,j} m_j^e}. \quad (31)$$

This aggregate captures both differences in education outcomes across household types and the en-

dogenous allocation of children across local education markets.

6 MODEL QUANTIFICATION

This section describes how I quantify the baseline economy.¹⁶ The procedure has four steps. First, I quantify the location-invariant parameters governing housing demand, child human-capital production, nested choices, migration costs, and production: $\{\alpha^{ef}, \bar{h}^f, \zeta_e, \nu, q, \ell^e, \epsilon^e, \rho^e, T_1^e, \mu^{ef}, \gamma_P^{ef}, \gamma_R^{ef}, \kappa^{ef}, \vartheta\}$. Second, I construct location-specific primitives, including initial populations by education and housing-endowment status, housing endowments $\{E_o^{e\delta}\}$, housing-supply elasticities $\{\eta_j\}$, housing-cost shifters $\{l_j\}$, fiscal capacity $\{\tau_j\}$, school-access wedges $\{\Delta_j\}$, and residual housing demand. Third, given these objects, I recover production fundamentals $\{B_j^e\}$ and education- and family-arrangement-specific amenities $\{A_j^{ef}\}$ that rationalize the baseline equilibrium. Finally, I assess the model fit using non-targeted moments.¹⁷

6.1 PARAMETERS

Stone–Geary housing demand $\{\alpha^{ef}, \bar{h}^f\}$

I estimate the Stone–Geary housing-demand parameters using the 2013–2014 Urban Household Survey (UHS). As Fang and Huang (2022) notes, UHS housing-expenditure measures became more reliable after 2013, when imputed rents for owner-occupied housing began to be included as housing costs. I construct empirical counterparts to the model’s three family arrangements, $f \in \{\emptyset, 0, 1\}$: two-member households with neither a co-resident dependent child nor a child studying away, two-member households with no co-resident dependent child and at least one child studying away, and three-member households with a co-resident child aged 18 or below, respectively. I use the same education classification as in the quantitative model: L denotes junior high school or below and H denotes high school or above. Housing services are measured by floor space. For household i in city j and year t , h_{ijt}^{ef} and Y_{ijt}^{ef} denote floor space and reported disposable income per working-age adult. I construct the common city–year rental-equivalent price r_{jt} as the median imputed rent per unit of floor space.

For each family arrangement f , I jointly estimate the Engel curves for high- and low-education households:

$$s_{ijt}^{ef} = \beta^{ef} + (1 - \beta^{ef}) \bar{h}^f \frac{r_{jt}}{Y_{ijt}^{ef}} + u_{ijt}^{ef}, \quad s_{ijt}^{ef} \equiv \frac{r_{jt} h_{ijt}^{ef}}{Y_{ijt}^{ef}}.$$

This estimating equation follows from Eqs. (22) and (23).¹⁸ For each f , nonlinear least squares jointly estimates the two education-specific parameters, β^{Hf} and β^{Lf} , and the common effective subsistence housing requirement, $\bar{h}^f$, by minimising the sum of squared residuals in this equation. Thus, the two intercepts identify the education-specific discretionary housing parameters, while the restricted slopes, $(1 - \beta^{ef})\bar{h}^f$, jointly identify the common $\bar{h}^f$.

¹⁶The model is static, and the baseline economy does not correspond to a single calendar year. It represents the schooling environment faced by the 1986–1993 birth cohorts, with different structural objects disciplined using the data that best measure the relevant economic environment or predetermined initial conditions during the 2000s.

¹⁷Appendix Table F1 summarizes the location-invariant parameters, and Appendix Table F2 summarizes the construction of the local primitives.

¹⁸For either household type, write housing demand as $h = \bar{h} + \beta \tilde{Y}/r$, where $\tilde{Y} = Y - r\bar{h}$. Substitution gives $h = \beta Y/r + (1 - \beta)\bar{h}$; multiplying by r/Y yields the estimating equation.

For childless households, the Engel-curve intercept satisfies $\beta^{e\varnothing} = \alpha^{e\varnothing}$. For parents, private education investment absorbs the fraction $\tilde{\nu}^e/(1+\tilde{\nu}^e)$ of residual resources. Hence, $\beta^{ef} = \alpha^{ef}/(1+\tilde{\nu}^e)$ for $f \in \{0, 1\}$. I therefore recover

$$\hat{\alpha}^{e\varnothing} = \hat{\beta}^{e\varnothing}, \quad \hat{\alpha}^{ef} = (1 + \tilde{\nu}^e) \hat{\beta}^{ef}, \quad f \in \{0, 1\}.$$

The estimated effective subsistence requirements are similar for childless households and parents whose child lives away, with $\bar{h}^\varnothing = 25.77$ and $\bar{h}^0 = 25.27$. The corresponding requirement is substantially higher under co-residence, at $\bar{h}^1 = 30.05$. This pattern is consistent with a co-residing child increasing basic space needs and thereby strengthening the housing-cost constraint on family migration. The estimated α^{ef} is higher for high-education households, especially under co-residence, but this does not necessarily imply a higher average housing expenditure share, which also depends on income and housing prices through the Stone–Geary term $\bar{h}^f r_{jt}/Y_{ijt}^{ef}$.

Table 4: Stone–Geary Housing-Demand Parameters

	$f = \varnothing$	$f = 0$	$f = 1$
α^{Hf}	0.07	0.05	0.05
α^{Lf}	0.06	0.04	0.02
$\bar{h}^f$	25.77	25.27	30.05

Notes: Parameters are estimated using the 2013–2014 UHS for Shanghai, Guangdong, Sichuan, and Liaoning. Floor space and disposable income are expressed per working-age adult. The effective subsistence requirement $\bar{h}^f$, measured in square metres per working-age adult, is common across education types within each family arrangement. For $f = \varnothing$, $\alpha^{e\varnothing} = \beta^{e\varnothing}$; for $f \in \{0, 1\}$, $\alpha^{ef} = (1 + \tilde{\nu}^e)\beta^{ef}$, with $\tilde{\nu}^e = 0.08$ in the baseline calibration.

Education-production parameters $\{\zeta_e, \nu, q\}$

As in Eq. 7, children’s educational outcomes depend on parental background, private education investment, and effective public education resources:

$$\tilde{\Gamma}_{ijo}^{e\delta f} = \zeta_e \left(\frac{K_{ij}}{\bar{K}} \right)^\nu \left(\frac{\Lambda_{ijo}^f}{\bar{\Lambda}} \right)^q.$$

The parameter ζ_e captures baseline intergenerational transmission associated with parental education type e , ν governs the importance of private education investment, and q governs the importance of effective public education resources.

I discipline the private-investment elasticity ν and the altruism parameter ι in two steps. I first calibrate the composite parameter $\tilde{\nu}^e \equiv \iota^e \nu$ using observed private education spending shares; I then use external evidence on ν to recover ι . Since [Chi and Qian \(2016\)](#) reports aggregate education-expenditure shares rather than education-group-specific mean shares that map directly into the model, the baseline calibration uses a common composite parameter across parental education groups, $\tilde{\nu}^H = \tilde{\nu}^L \equiv \tilde{\nu}$.¹⁹ In the model, the optimality condition for parents implies that the fraction of residual

¹⁹This baseline parsimonious restriction ensures that differences in private education investment across education groups arise from household resources and housing constraints rather than imposed differences in preferences or investment productivity.

resources $\tilde{Y}_{jo}^{edf}$ allocated to private education investment is $\tilde{\nu}/(1 + \tilde{\nu})$. I calibrate this share using the total education-expenditure-to-income ratios reported by [Chi and Qian \(2016\)](#) for 2007 and 2011. Let $\hat{s}_K$ denote this empirical education-expenditure share. Averaging the reported ratios of 6.07 percent in 2007 and 8.86 percent in 2011 gives $\hat{s}_K = 0.075$. The moment condition $\tilde{\nu}/(1 + \tilde{\nu}) = \hat{s}_K$ implies $\tilde{\nu} = 0.08$.

Then, to separately discipline the primitive private-investment elasticity ν , I use external evidence from [Attanasio et al. \(2020\)](#). Estimating this parameter is empirically challenging because parental investment may respond to unobserved shocks to child development and other household characteristics that also affect children’s educational outcomes. I therefore rely on an externally identified estimate from the existing literature. [Attanasio et al. \(2020\)](#) explicitly treats parental material investment as endogenous and addresses this concern using a control-function approach, with local prices of child-investment goods as excluded instruments. Their estimated cognition production function is close to Cobb–Douglas, with an investment elasticity of 0.19 for children aged 12. I therefore use $\nu = 0.19$ as an external benchmark.²⁰ Combined with $\tilde{\nu} = 0.08$, this implies an altruism parameter $\iota = \tilde{\nu}/\nu \approx 0.42$. Since the baseline uses common values of $\tilde{\nu}$ and ν across parental education groups, it also implies $\iota^H = \iota^L = 0.42$.

I next calibrate q using external causal evidence on the effect of public school spending on long-run educational attainment. Directly estimating q in my setting is difficult because credible exogenous shocks to city-level public education spending in China are hard to find, and observed spending is confounded with local fiscal capacity, household sorting, school quality, and migrant children’s access barriers. I therefore rely on quasi-experimental evidence from the school-finance literature to discipline its magnitude. Since

$$P_{ijo}^{edf} = \frac{\tilde{\Gamma}_{ijo}^{edf}}{1 + \tilde{\Gamma}_{ijo}^{edf}},$$

we have $\text{logit}(P_{ijo}^{edf}) = \log \tilde{\Gamma}_{ijo}^{edf}$. Hence q can be interpreted as the elasticity of the log odds of high educational attainment with respect to effective public education resources. I use the quasi-experimental estimate from [Hyman \(2017\)](#), who studies Michigan’s school finance reform and finds that a \$1,000, or approximately 10 percent, increase in school spending raises college enrollment by 3 percentage points. Using the baseline college enrollment rate of 44.8 percent, this estimate implies

$$\hat{q} = \frac{\text{logit}(0.448 + 0.030) - \text{logit}(0.448)}{\log(1 + 0.1)} \approx 1.27.$$

Because this evidence comes from a U.S. school-finance setting and may not map one-for-one into public education resources in Chinese cities, I use $q = 1.2$ as a conservative baseline calibration. This magnitude is also consistent with the meta-analysis in [Jackson and Mackevicius \(2024\)](#), which finds that a \$1,000 per-pupil spending increase sustained over four years raises college-going by 2.8 percentage points. In the absence of directly comparable causal estimates from developing-country settings, I view $q = 1.2$ as a reasonable externally disciplined input.

Finally, given the calibrated values of ν and q , I estimate $\{\zeta_e\}_{e \in \{H,L\}}$ using educational-attainment moments from the 2016 CMDS. For a household of type (e, δ, o) choosing destination j and family

²⁰The smaller magnitude of ν relative to the public-education elasticity q is consistent with private education spending playing a supplementary role to formal schooling. Appendix Section B shows that the housing-wealth-induced increase in such spending is concentrated particularly in extracurricular tutoring.

arrangement f , the model implies

$$P_{jo}^{e\delta f} = \text{logit}^{-1} \left[\log \zeta_e + \nu \log \left(\frac{K_{jo}^{e\delta f}}{\bar{K}} \right) + q \log \left(\frac{\Lambda_{jo}^f}{\bar{\Lambda}} \right) \right], \quad (32)$$

where $K_{jo}^{e\delta f}$ is optimal private education investment and Λ_{jo}^f is the effective public education resource defined in Eq. (8). For each parental education group, I choose ζ_e to match the survey-weighted high-education rate among children to its model-implied counterpart. The resulting estimates are $\hat{\zeta}_H = 0.818$ and $\hat{\zeta}_L = 0.490$. See Appendix Section C.4 for details.

Family-arrangement choice preference dispersion $\{\rho^e\}$

The parameter ρ^e governs the dispersion of idiosyncratic preferences over family arrangements. Among migrant parents, the relevant choice is whether to bring the child to the destination and co-reside, $f = 1$, or to migrate without the child and leave the child in the origin, $f = 0$. From Eq. 18,

$$\log \left(\frac{\pi_{1|o}^{e\delta}}{\pi_{0|o}^{e\delta}} \right) = \log T_1^e + \rho^e \log \left(\frac{\mathcal{U}_o^{e\delta 1}}{\mathcal{U}_o^{e\delta 0}} \right),$$

so that ρ^e governs the elasticity of family-arrangement odds with respect to the relative inclusive value of co-residence.

Estimating this elasticity directly is challenging because the relative inclusive value is a generated equilibrium object and may therefore be subject to measurement error and endogeneity. I instead externally calibrate ρ^e using Imbert et al. (2026), who estimate the corresponding middle-nest elasticity in a closely related nested migration model for China. Their IV strategy exploits gravity-weighted exposure to destination living costs predicted by geographic constraints to urban expansion. I use their estimate of 3.253 and set $\rho^H = \rho^L = 3.253$ in the baseline. Though I impose a common ρ across education groups, this restriction does not imply identical family-arrangement responses, since the relative values $\mathcal{U}_o^{e\delta 1}/\mathcal{U}_o^{e\delta 0}$ remain education-specific through wages, housing demand, migration costs, location preferences, and the returns to children's education.

Other Parameters

I borrow other parameters from existing literature, as summarized in Appendix Table F1.

6.2 CALIBRATED LOCAL CHARACTERISTICS

Initial housing endowments $\{E_o^{e\delta}\}$

I use the 2000 Census to calibrate city-specific initial housing endowments. For each origin city o and education group e , I construct $E_o^{e\delta}$ as the total floor space of owner-occupied housing among local urban owner-type households, divided by the number of working-age adults in those households.²¹ Thus, $E_o^{e\delta}$ is a housing-service-equivalent endowment measured in square metres per owner adult. Households without an initial urban owner endowment, including urban renters and rural households, are assigned $E_o^{e\delta} = 0$. In the household budget, this endowment generates the annualised resource

²¹ Owner-occupied housing comprises self-built housing, purchased commercial housing, purchased economic housing, and purchased former public housing.

value $r_o E_o^{e\delta}$, where r_o is the annualised housing price in the origin. Appendix Section C.1 provides further details and describes the public-housing privatisation component used in the counterfactual analysis.

Housing supply elasticities $\{\eta_j\}$ and construction-cost shifters $\{l_j\}$

I write the inverse housing supply curve as

$$r_j = l_j (F_j^S)^{1/\eta_j}, \quad (33)$$

where η_j is the city-specific housing-supply elasticity and l_j is a city-specific construction-cost shifter. A lower η_j implies a more constrained housing market and a stronger price response to a given increase in local housing demand.

Following Liang et al. (2024) and the approach in Diamond (2017) of imposing economically admissible housing-supply elasticities, I allow inverse supply elasticities to vary with predetermined physical constraints. I parametrise the log inverse housing-supply elasticity as

$$\log\left(\frac{1}{\eta_j}\right) = \theta_0 + \theta_D \text{Developed}_j + \theta_S \text{Slope}_j, \quad (34)$$

where Developed_j is the baseline share of developable land already developed and Slope_j is terrain slope. Both variables are standardized. This parametrisation ensures positive housing-supply elasticities for all cities while allowing supply responsiveness to vary systematically with local physical constraints.

Equivalently, $1/\eta_j = \exp(\theta_0 + \theta_D \text{Developed}_j + \theta_S \text{Slope}_j)$. Taking first differences of Eq. (33) implies $\Delta \ln r_{jT} = (1/\eta_j) \Delta \ln F_{jT}^S$. Since consistent city-level residential floor-space stock is not observed, I use the log change in permanent population between 2005 and 2015, $\Delta \ln Q_{jT}$, as a proxy for population-induced housing-demand growth. The estimating equation is therefore

$$\Delta \ln r_{jT} = \exp(\theta_0 + \theta_D \text{Developed}_j + \theta_S \text{Slope}_j) \Delta \ln Q_{jT} + X_j' \Gamma + \gamma_p + u_j. \quad (35)$$

Because population growth is endogenous, I estimate Eq. (35) using nonlinear IV/GMM, with the shift-share local labour-demand shock Z_{jT} and its interactions with Developed_j and Slope_j providing excluded instruments. The vector X_j includes log initial housing price and the two physical supply constraints, and province fixed effects γ_p . The identifying assumption is that, conditional on these controls, the predicted labour-demand shock shifts local housing demand but does not directly affect changes in local construction costs or unobserved per-capita housing-service growth. The estimated parameters are $\hat{\theta}_0 = -0.807$, $\hat{\theta}_D = 0.800$, and $\hat{\theta}_S = 0.757$. See Appendix Section C.2 for details.

Given the estimated coefficients, I construct the city-specific housing-supply elasticity as $\hat{\eta}_j = \exp[-(\hat{\theta}_0 + \hat{\theta}_D \text{Developed}_j + \hat{\theta}_S \text{Slope}_j)]$.²² I then recover the city-specific construction-cost shifter from the 2005 housing-market baseline:

$$\hat{l}_j = \frac{r_{j0}}{(F_{j0}^D)^{1/\hat{\eta}_j}}, \quad (36)$$

²²To limit the influence of extreme tail predictions implied by the exponential parametrization, the baseline calibration caps $1/\hat{\eta}_j$ at its 99th percentile.

where r_{j0} is the annualized housing price in 2005 and F_{j0}^D is the model-implied aggregate housing demand evaluated at the 2005 housing-market baseline.²³

Effective fiscal parameter τ_j

I calibrate the effective fiscal parameter τ_j using the ratio of observed local government education expenditure to the model-implied formal wage bill in 2005. Education expenditure is obtained from prefecture-level data in the China City Statistical Yearbook. The wage bill is constructed from the model’s 2005 labour allocation and city–education-specific mean wages estimated from the Census, with monthly wages converted to annual values. Since the model abstracts from the different sources of education finance, including local revenues and intergovernmental transfers, I interpret τ_j as an exogenous reduced-form fiscal ratio rather than a statutory tax rate. It is calibrated once from the 2005 baseline and held fixed in the equilibrium and counterfactual analyses.

Education-access wedge Δ_j

I treat Δ_j as an exogenous, city-specific parameter measuring the share of local public-education resources effectively accessible to an average migrant child relative to a local child. It summarizes *hukou*-related barriers, including documentation, residence, and enrollment requirements. Because the allocation of local education spending across migrant and local children is not directly observed, I follow Sieg et al. (2023) and proxy Δ_j by the share of migrant children aged 6–18 enrolled in local public schools in the 2011 China Migrants Dynamic Survey (CMDs), restricting the sample to children residing in the destination city and attending public, private, or migrant schools. This enrollment share is informative about access because, given the quality advantage and lower cost of local public schools, eligible migrant families are less likely to opt instead for migrant or private schools. Across the model cities, the resulting measure has a mean of 0.889 and ranges from 0.486 to 0.997.²⁴

Others

The initial worker distribution $\{H_o^{e\delta\varnothing}, H_o^{e\delta}\}$ and the residual population $\{O_j\}$ are calibrated from the 2005 Census. The model covers 270 prefecture-level cities with complete model inputs, which together account for about 88.0% of the national working-age adult population. These cities contain 721,675,277 working-age adults and a residual population of 101,209,685. The effective per-capita housing demand of the residual population, h^O , is calibrated to 26.09. See Appendix C.3 for details.

6.3 INVERSION OF LOCAL FUNDAMENTALS

Production fundamentals $\{B_j^e\}$

I recover city–education-specific production fundamentals from the production side. Following the inversion logic in Fang and Huang (2022), I use observed city–education wage bills and employment together with firms’ first-order conditions under the CES production technology. This disciplines production fundamentals using wage and employment moments.

²³I convert housing asset prices into annual housing-service prices using a common rent-to-price factor of 4 percent. This corresponds to a price-to-annual-rent ratio of 25, close to the centre of the quality-adjusted ratios documented by Wu et al. (2012) for major Chinese cities in 2000s.

²⁴To reduce small-sample noise, I shrink each city’s raw public-school enrollment share toward the corresponding province mean using a prior weight of 10; cities without direct observations are assigned the province mean.

With the final-good price normalised to one, perfect competition implies

$$\omega_j^e = X_j^{1/\vartheta} (B_j^e)^{1/\vartheta} (L_j^e)^{-1/\vartheta}, \quad e \in \{H, L\}.$$

Zero profit implies

$$X_j = \omega_j^H L_j^H + \omega_j^L L_j^L.$$

Define the wage-bill share of education group e in city j as

$$\Xi_j^e = \frac{\omega_j^e L_j^e}{\omega_j^H L_j^H + \omega_j^L L_j^L}.$$

Combining the marginal-product condition with zero profit gives

$$B_j^e = \Xi_j^e (\omega_j^e)^{\vartheta-1}, \quad e \in \{H, L\}.$$

In the empirical implementation, I use individual wage records in the 2005 census to construct city-education employment and wage bills, since wage information is only available in the 2005 census. For each city and education group, I compute ω_j^e as the total wage bill of workers of type e divided by employment L_j^e . I then apply the inversion above and normalise the recovered productivities so that the employment-weighted mean of B_j^e equals one within each education group.

Amenity inversion and migration costs

I recover destination amenities and estimate migration costs using a nested inversion procedure in the spirit of [Fan \(2019\)](#). The estimation is conducted separately by parental-education group. I use the family-arrangement index $f \in \{\emptyset, 0, 1\}$, where $f = \emptyset$ denotes a childless adult, $f = 0$ denotes a parent who does not co-reside with the child, and $f = 1$ denotes a parent who co-resides with the child. In my model, A_j^{ef} absorbs education- and family-arrangement-specific residual destination attractiveness, while migration costs are disciplined by how households from different origins sort across destinations.

I parametrise migration costs from o to j as

$$d_{jo}^{ef} = \exp \left[\mu^{ef} \text{dist}_{jo} + \gamma_P^{ef} \mathbf{1}\{p(j) \neq p(o)\} + \gamma_R^{ef} \mathbf{1}\{g(j) \neq g(o)\} + \kappa^{ef} \mathbf{1}\{j \neq o\} \right], \quad (37)$$

where dist_{jo} denotes bilateral transport time, while $p(j)$ and $g(j)$ denote the province and broader geographic region containing city j . I normalise the migration cost of remaining in the origin to one, $d_{oo}^{ef} = 1$. The parameters μ^{ef} , γ_P^{ef} , and γ_R^{ef} capture, respectively, the transport-time gradient and the additional frictions associated with crossing provincial and broader regional boundaries for family arrangement f . The parameter κ^{ef} captures the fixed cost of leaving the origin for that family arrangement.

Let

$$\varsigma^e \equiv \left(\left\{ \mu^{ef}, \gamma_P^{ef}, \gamma_R^{ef}, \kappa^{ef} \right\}_{f \in \{\emptyset, 0, 1\}} \right)$$

denote the vector of migration-cost parameters. I normalise the family-arrangement scale parameter for non-co-residence to $T_0^e = 1$. Conditional on a candidate migration-cost vector ς^e , the inner problem solves for T_1^e and the family-arrangement-specific amenities jointly. For each trial value of T_1^e , the

inversion recovers $\{A_j^{ef}\}_{f \in \{\emptyset, 0, 1\}}$ to match the relative destination distribution within each family arrangement. The algorithm then updates T_1^e until the model-implied aggregate co-residence share among parents matches its empirical counterpart.

More specifically, for childless adults, model-implied destination employment is

$$L_j^{e\emptyset, \text{model}} = \sum_{o \in \mathcal{G}} \sum_{\delta} H_o^{e\delta\emptyset} \pi_{j|o}^{e\delta\emptyset}. \quad (38)$$

For parents choosing family arrangement $f \in \{0, 1\}$, model-implied destination employment is

$$L_j^{ef, \text{model}} = \sum_{o \in \mathcal{G}} \sum_{\delta} H_o^{e\delta} \pi_{f|o}^{e\delta} \pi_{j|o}^{e\delta f}. \quad (39)$$

The choice probabilities in Eqs. (38) and (39) depend on amenities, migration costs, and, for parents, the family-arrangement scale parameter. The inner inversion chooses the three amenity vectors so that the model matches the relative destination distribution within each family arrangement:

$$L_j^{ef, \text{model}} = L_j^{ef, \text{data}}, \quad f \in \{\emptyset, 0, 1\}.$$

Amenities are identified only up to a family-arrangement-specific multiplicative scale, and are therefore normalised to have geometric mean one.

I denote the resulting inverted amenities and co-residence scale by

$$\hat{A}^e(\varsigma^e), \quad \hat{T}_1^e(\varsigma^e).$$

The outer problem selects the migration-cost parameters using origin-conditioned bilateral choices:

$$\varsigma^e = \arg \max_{\varsigma^e} \ell_{\text{bil}}^e \left(\hat{A}^e(\varsigma^e), \hat{T}_1^e(\varsigma^e), \varsigma^e \right), \quad \hat{A}^e = \hat{A}^e(\varsigma^e). \quad (40)$$

The likelihood uses origin-destination choices for childless adults and joint origin-destination and family-arrangement choices for parents, including same-origin stayers. Appendix E.1 provides details on the nested estimation procedure, and Appendix C.5 describes the resulting fit.

The different margins of the household choice problem discipline different objects. The destination distribution within each family arrangement disciplines the relative inverted amenities. Conditional on these amenities, the transport and boundary parameters are disciplined by how bilateral flows vary with distance and administrative boundaries. The fixed-cost parameters κ^{ef} are disciplined by the mover-stayer margin within each family arrangement. The parameter T_1^e is disciplined by the aggregate incidence of co-residence among parents. The destination distribution of co-residing children is determined by the interaction of co-residence choices, migration costs, housing costs, and effective access to destination public education.

The results in Table 5 indicate higher migration frictions for low-skilled households, especially along the distance and fixed-cost margins. For both education groups, the fixed cost of leaving the origin is highest for co-residing parents, consistent with additional barriers to migrating with children.

Table 5: Migration-Cost and Family-Arrangement Parameters

Education	Mode	T_1^e	μ^{ef}	γ_P^{ef}	γ_R^{ef}	κ^{ef}
High-skilled	Childless	1.337	0.020	0.455	0.164	1.297
High-skilled	Parent, separate	1.337	0.018	0.332	0.209	1.495
High-skilled	Parent, co-reside	1.337	0.019	0.718	0.137	1.645
Low-skilled	Childless	1.158	0.031	0.418	0.228	1.644
Low-skilled	Parent, separate	1.158	0.028	0.334	0.242	1.726
Low-skilled	Parent, co-reside	1.158	0.031	0.672	0.239	1.773

Note: Parameters are estimated using the 2005 Census. The estimation is conducted separately by education group. T_1^e is the relative scale of the co-residence preference, with T_0^e normalised to one; it is chosen inside the inversion to match the aggregate co-residence share. The remaining parameters govern family-arrangement-specific migration costs: μ^{ef} is the transport-time gradient, γ_P^{ef} and γ_R^{ef} capture provincial- and regional-boundary frictions, and κ^{ef} is the fixed cost of leaving the origin.

6.4 BASELINE MODEL FIT

The calibrated primitives pin down a unique baseline equilibrium that closely reproduces the city-level employment, housing prices, wages, and public education resources used in calibration. These comparisons assess the internal consistency of the calibrated model. I then assess untargeted child outcomes to test whether the calibrated model predicts educational differences across cities. For children of low-educated incumbent homeowners, the child-count-weighted correlation between observed and model-implied city-level upward mobility is 0.67. For the children generation as a whole, the corresponding correlation for the city-level high-education share is 0.78. The first comparison tests mobility conditional on housing ownership; the second tests the spatial distribution of next-generation education. Neither is a calibration target. I interpret these correlations as evidence about spatial variation, not exact fit in levels, given differences in cohort timing and city measurement. See Appendix Section E.3 for more details.

7 COUNTERFACTUALS

In this section, I use the calibrated model to quantify three dimensions of the link between housing markets and intergenerational upward mobility. First, I examine how housing-supply elasticity changes the incidence of a common productivity shock on upward mobility. Second, I study how unequal access to local public education affects migration, parent-child co-residence, and mobility among migrant families. Third, I remove the housing endowment created by public-housing privatisation to isolate its long-run wealth effect. Together, these exercises distinguish the aggregate effects of each counterfactual from its distributional consequences across initial asset holders, non-asset households, and migrant families.

7.1 RELAXING HOUSING CONSTRAINTS

I first examine how housing-supply elasticity affects the incidence of a common urban-growth shock. I raise η_j by 25% in all cities or separately in initially less- and more-constrained cities, defined by whether calibrated elasticity is above or below the median. For each regime, I re-invert l_j so that the complete pre-shock equilibrium coincides with the calibrated baseline. The exercise therefore changes

the transmission of growth without changing the initial allocation. I then apply the same nationwide 10% wage-equivalent productivity shock:

$$B_j^{e,cf} = (1.1)^{\vartheta-1} B_j^e.$$

Because the shock is common across cities, the exercise holds fixed the spatial distribution of productivity growth.

To capture the housing-wealth channel identified in the reduced-form analysis, initial asset holders receive the annualised capital gain

$$(r_o - r_o^0)E_o^{e\beta},$$

where r_o^0 is the corresponding pre-shock housing price. This term is zero before the shock and becomes positive only when the value of the initial housing asset rises. Housing-supply relaxation therefore has two opposing effects: it reduces the capital gain received by initial asset holders, but lowers destination housing costs for non-asset and migrant households.

Table 6 reports the response to the productivity shock under calibrated supply and the change in that response under each supply reform. Under calibrated supply, the shock raises low-educated wages by 9.487%, house prices by 0.943%, housing floor space by 1.103%, and public-education resources per effective pupil by 9.478%. Raising η_j by 25% in initially more-constrained cities reduces the house-price response by 0.128 percentage points and increases the housing-quantity response by 0.012 percentage points. The corresponding changes when supply is relaxed in less-constrained cities are only -0.019 and $+0.002$ percentage points. The price effect is therefore concentrated in initially more-constrained cities, while wage and public-resource responses change by less than 0.01 percentage points.

The weaker price response redistributes private-investment gains across initial housing-wealth groups. In more-constrained cities, the private-investment response of initial asset holders falls by 0.100 percentage points because their housing capital gain is smaller. The response of initial non-asset households rises by 0.073 percentage points as lower housing costs leave more resources available for education. The corresponding mobility response falls by 0.002 percentage points among children of initial asset holders and rises by 0.004 percentage points among children of initial non-asset households.

The main adjustment occurs through migration and family arrangements. Relaxing supply in more-constrained cities raises the migration response of low-educated parents by 0.010 percentage points and reduces the migrant-child separation response by 0.095 percentage points. The mobility response among migrants who co-reside with their children rises by 0.102 percentage points, whereas the response among migrants whose children remain at origin rises by only 0.001 percentage points. Because migration and family arrangements are endogenous, these group-specific responses also reflect selection into migration and co-residence. The larger response of separation than migration nevertheless points to the relevant margin: housing costs matter primarily for the family composition of migration—whether parents who move are able to bring their children with them.

Figure 5 varies housing-supply elasticity in initially more-constrained cities from 0 to 30% above its calibrated value. At a 30% increase, the house-price response falls by 0.149 percentage points, parental migration rises by 0.012 percentage points, and migrant-child separation falls by 0.112 percentage points relative to calibrated supply. The mobility response among co-residing migrant children rises by 0.120 percentage points, while aggregate upward mobility increases by only 0.003 percentage points.

More elastic supply therefore has a limited effect on average mobility but a meaningful effect on its incidence: it shifts part of the gains from urban growth away from initial asset holders and toward non-asset and migrant families.

Sensitivity Analysis

The incidence of housing-supply relaxation depends on how readily households adjust their location and family arrangements. I therefore jointly scale the location-choice parameters, ϵ^e , and the family-arrangement parameters, ρ^e , by $\pm 10\%$ around their calibrated values. At each value, I re-invert the housing-supply shifter so that the complete pre-shock equilibrium is unchanged, raise housing-supply elasticity by 25% in initially more-constrained cities, and apply the same nationwide productivity shock. The aggregate mobility effect remains small, changing by only 0.002–0.004 percentage points across the parameterizations. The reform nevertheless continues to shift incidence away from initial asset holders and toward migrant children who co-reside with their parents. Appendix Section C.6 reports the full sensitivity results.

Table 6: Housing-supply elasticity and the incidence of common productivity growth

	(1)	Change relative to (1)		
	Baseline response	25% supply relaxation in:		All cities
		Less constrained	More constrained	
<i>Panel A. Equilibrium transmission of productivity growth (%)</i>				
Low-education wage response	+9.4873	+0.0001	-0.0067	-0.0066
House-price response	+0.9432	-0.0191	-0.1278	-0.1457
Housing floor-space response	+1.1025	+0.0023	+0.0118	+0.0140
Public education resources per effective pupil	+9.4776	+0.0000	-0.0094	-0.0094
<i>Panel B. Private education investment (%)</i>				
Initial asset holders	+10.1618	-0.0068	-0.1002	-0.1065
Initial non-asset households	+15.3017	+0.0108	+0.0732	+0.0833
<i>Panel C. Spatial and family adjustment (pp)</i>				
Low-education parent migration	-0.0273	-0.0006	+0.0105	+0.0099
Migrant-child separation	-1.5052	+0.0001	-0.0951	-0.0949
<i>Panel D. Upward-mobility effects (pp)</i>				
Aggregate upward mobility	+2.9176	+0.0003	+0.0029	+0.0033
<i>By initial housing-asset status</i>				
Children of initial asset holders	+2.8545	-0.0002	-0.0015	-0.0017
Children of initial non-asset households	+2.9277	+0.0004	+0.0037	+0.0041
<i>By realised location and family arrangement</i>				
Stayers, initial asset holders	+2.7927	-0.0004	-0.0027	-0.0031
Stayers, initial non-asset households	+2.8886	+0.0005	-0.0004	+0.0000
Migrants co-residing with children	+4.2599	+0.0023	+0.1023	+0.1041
Migrants whose children remain at origin	+2.8726	+0.0000	+0.0008	+0.0008

Notes: All scenarios apply the same nationwide 10% wage-equivalent productivity shock. The first column reports the shock response under calibrated housing supply; the remaining columns subtract this response from the response when supply elasticity is raised by 25% in the indicated cities. The housing-supply shifter is re-anchored in each regime to preserve the full pre-shock equilibrium. In each shock equilibrium, initial asset holders receive the annualized capital gain $(r_o - r_o^0)E_o$ on predetermined origin housing in addition to $r_o E_o$; the gain is zero in the corresponding pre-shock equilibrium. Less- and more-constrained cities have calibrated housing-supply elasticities at or above and below the median, respectively. Price, housing quantity, wage, and public-resource log changes use fixed calibrated-baseline housing demand, housing supply, low-education employment, and effective-pupil masses as weights. Outcomes by initial housing-asset status follow fixed pre-shock cohorts while allowing their destination and family arrangements to adjust. Outcomes by realised location and family arrangement are equilibrium conditional means and therefore reflect both within-group outcome changes and endogenous selection into each realised group; this composition response is part of the mechanism quantified by the model. Investment changes are percentages; mobility, migration, and separation changes are percentage points. The separation rate is conditional on parental migration.

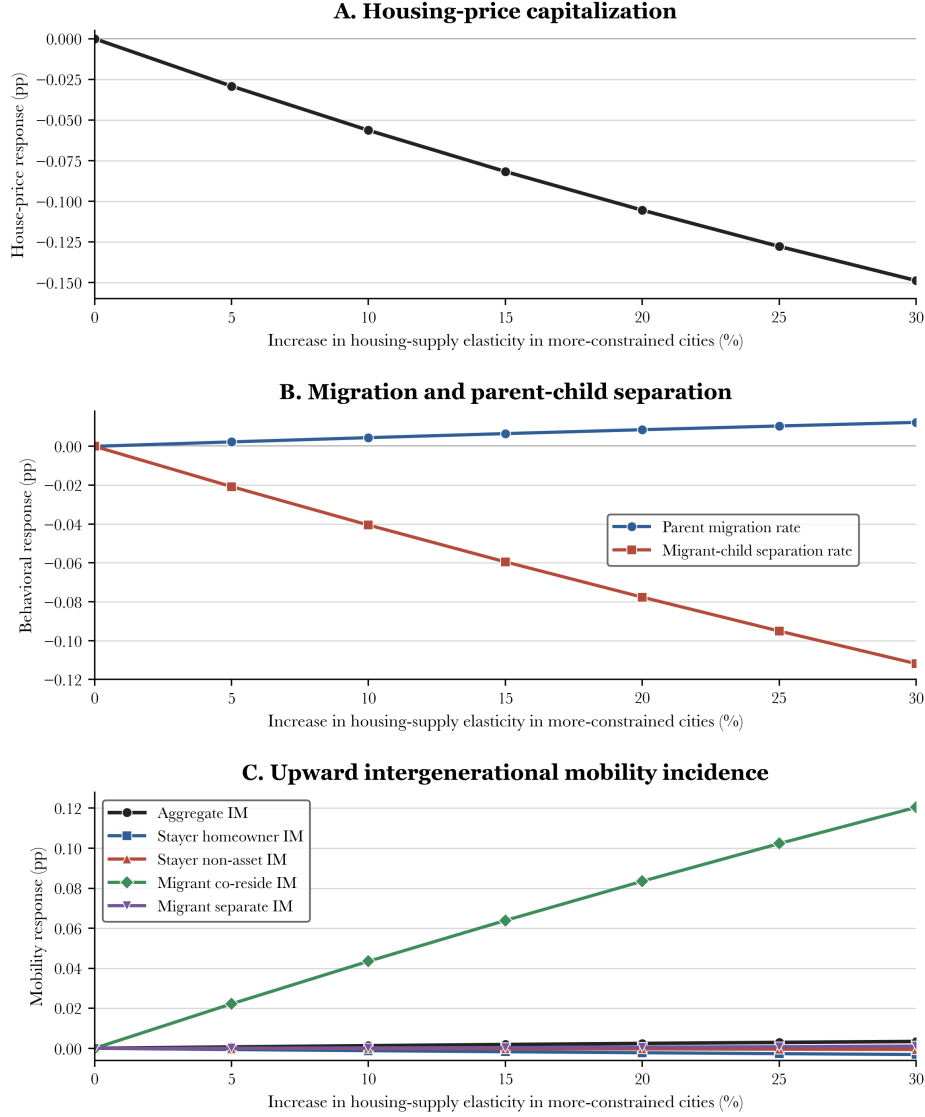


Figure 5: Housing-supply relaxation and the incidence of productivity growth

Notes: The figure raises housing-supply elasticity, η_j , by 0–30% relative to its calibrated value in initially more-constrained cities, while leaving it unchanged elsewhere. Initially more-constrained cities are those with below-median calibrated housing-supply elasticities. At each reform intensity, the housing-supply shifter l_j is re-inverted to preserve the complete calibrated pre-shock equilibrium. The same nationwide 10% wage-equivalent productivity shock is then applied, and the full spatial equilibrium is re-solved. All panels report the change in the shock response relative to that under calibrated housing supply, in percentage points. Panel A reports the house-price response. Panel B reports low-educated parental migration and migrant-child separation, with the separation rate conditional on parental migration. Panel C reports upward mobility among children of low-educated parents, both in the aggregate and by realised location and family arrangement. These realised groups are endogenous equilibrium outcomes; their responses therefore incorporate both within-group changes and compositional adjustment.

7.2 EQUALISING PUBLIC EDUCATION ACCESS

I next quantify the effects of eliminating migrant children’s unequal access to local public education. I set the city-specific access factor to $\Delta_j = 1$ so that migrant children who co-reside with their parents receive the same effective public-education resources as local children. I first equalise access nationwide and then repeat the exercise separately in less- and more-constrained cities. All other structural parameters and local fundamentals are held fixed, and the full spatial equilibrium is resolved.

This counterfactual operates on a different margin from the housing-supply exercise. Housing-supply relaxation lowers the pecuniary cost of bringing children to a destination, whereas equal access raises the educational return to doing so. Both margins affect migrant families’ location and family-arrangement choices, although housing supply operates through affordability while school access operates through the educational return to bringing children.

Table 7 reports the results. Nationwide equalisation raises upward mobility among children of low-educated parents by 0.182 percentage points. It increases the migration rate of low-educated parents by 0.473 percentage points and reduces the separation rate among migrant families by 6.733 percentage points. Thus, unequal school access affects not only the resources available to children already living at the destination, but also the incentive to migrate with children.

To separate these channels, I decompose the aggregate mobility effect into a redistribution effect and a residual general-equilibrium and sorting effect. Holding fixed the baseline allocation of families, prices, private investment, and local fiscal revenue, reallocating the fixed education budget under equal access raises mobility by 0.032 percentage points. The remaining 0.150 percentage points arise through endogenous migration, destination choice, and co-residence responses. Approximately 83% of the aggregate gain therefore reflects equilibrium adjustment rather than the mechanical improvement in access for the baseline allocation of families.

The effects are concentrated in more-constrained cities. Equalising access only in these cities raises aggregate mobility by 0.148 percentage points, compared with 0.035 percentage points in less-constrained cities. It also reduces separation by 5.249 percentage points and raises parental migration by 0.346 percentage points. These larger effects reflect the concentration of migrant children and access gaps in more-constrained destinations. Equal-access reform therefore has its greatest effect where the educational return to bringing children to the destination is most limited.

Equal access lowers public-education resources per effective pupil because the effective pupil mass expands faster than local education revenue. Under nationwide equalisation, resources per effective pupil fall by 0.852% in less-constrained cities and 2.556% in more-constrained cities. A reform confined to more-constrained cities lowers resources there by 2.595% while raising them by 0.310% in less-constrained cities through equilibrium reallocation. Nevertheless, the improved access of migrant children dominates this dilution, so aggregate upward mobility rises.

Sensitivity Analysis

The equal-access gain depends on the public-education productivity parameter, q , which is calibrated using evidence outside China. I therefore re-estimate the model to match the same baseline moments at alternative values of q . Varying q by $\pm 10\%$ around its calibrated value changes the nationwide mobility gain from 0.154 to 0.212 percentage points, compared with 0.182 percentage points at the baseline calibration. The gain remains positive and of a similar order of magnitude throughout this range. Appendix Section C.6 extends this exercise by jointly varying q and the responsiveness of

location and family-arrangement choices over the same range.

Table 7: Equilibrium effects of equalising public education access

	Cities with access equalised		
	All cities	Less constrained	More constrained
<i>Panel A. Migration and family responses (pp change)</i>			
Low-educated parent migration rate	+0.473	+0.128	+0.346
Migrant-child separation rate	-6.733	-1.724	-5.249
<i>Panel B. Public education resources per effective pupil (% change)</i>			
Less-constrained cities	-0.852	-1.164	+0.310
More-constrained cities	-2.556	+0.036	-2.595
<i>Panel C. Upward-mobility effects and decomposition (pp change)</i>			
Aggregate IM	+0.182	+0.035	+0.148
Redistribution effect	+0.032	+0.006	+0.026
General-equilibrium and sorting effect	+0.150	+0.029	+0.122

Notes: The table reports changes from the calibrated baseline when migrant children’s access to local public education is equalised in the indicated cities. Equal access is implemented by setting the migrant school-access factor to $\Delta_j = 1$; Δ_j remains at its calibrated value elsewhere. Less- and more-constrained cities have calibrated housing-supply elasticities at or above and below the median, respectively. All other structural parameters and local fundamentals are held fixed, and the full spatial equilibrium is re-solved. Panel A reports changes in the national migration and migrant-child separation rates among low-educated parents. The migration rate is the share of low-educated parents living outside their origin, and the separation rate is the share of low-educated migrant parents whose children remain at origin. Panel B reports changes in Λ_j averaged within each city group using fixed baseline pupil-mass weights. The redistribution effect in Panel C holds baseline migration, family arrangements, destination choices, wages, house prices, private education investment, and local fiscal revenue fixed, and reallocates the fixed education budget using the policy access factor and the baseline pupil distribution. The general-equilibrium and sorting effect is the residual between the full-equilibrium effect and this frozen-allocation effect. Panels A and C report percentage-point changes; Panel B reports percent changes.

7.3 REMOVING THE PUBLIC-HOUSING PRIVATISATION WINDFALL

I finally examine the long-run consequences of the housing-wealth windfall created by public-housing privatisation. The baseline housing endowment of initial asset holders, E_o^{eB} , includes housing acquired through both market purchases and the privatisation of former public housing. To isolate the estimated privatisation-induced component, I set

$$E_o^{eB,cf} = \max \{E_o^{eB} - E_o^{e,priv}, 0\}.$$

Households retain their initial housing-status type, all other structural parameters and local fundamentals are held fixed, and the full spatial equilibrium is re-solved. Among low-educated owner parents, the counterfactual removes 15.4% of their aggregate baseline housing endowment.

Table 8 shows that the effect operates primarily through the resources of initial asset holders. Removing the windfall reduces private education investment among initial asset holders by 7.1% but leaves investment among non-asset households essentially unchanged. Upward mobility falls by 0.252 percentage points among asset holders’ children, compared with 0.012 percentage points among non-asset children, and by 0.045 percentage points in the aggregate. The privatisation-induced endowment therefore generated persistent mobility gains by relaxing household resource constraints, with effects on both private education investment and family mobility.

The same resource channel also affects spatial and family adjustment. Removing the windfall lowers the migration rate of asset-holding parents by 0.369 percentage points and raises their separation rate

conditional on migration by 2.874 percentage points. With fewer resources, initial asset holders are less able to finance migration and, especially, the higher housing requirement associated with bringing a child to the destination. Thus, the privatisation-induced endowment facilitated both parental mobility and family co-residence.

General-equilibrium spillovers are small. The national house price falls by 0.138%, while non-asset households' migration, separation, private investment, and mobility change only slightly. Because these households receive no direct endowment shock, their responses arise solely through equilibrium prices and sorting. The intergenerational incidence is therefore concentrated among initial asset-holding households. Overall, the results show that the housing-asset transfer generated by public-housing privatisation can have persistent intergenerational consequences, implying that evaluations focused on contemporaneous housing outcomes may understate its longer-run distributional effects.

Table 8: Effects of removing the public-housing privatisation windfall

	Baseline	No privatisation windfall	Change
<i>Panel A. Housing wealth and private education investment</i>			
Initial housing endowment, asset holders	32.156	27.215	-15.368%
Private education investment, asset holders	488.709	453.936	-7.115%
Private education investment, non-asset households	246.085	246.093	+0.004%
<i>Panel B. Spatial and family adjustment</i>			
Parent migration rate, asset holders	4.872%	4.503%	-0.369 pp
Parent migration rate, non-asset households	5.429%	5.403%	-0.026 pp
Migrant-child separation rate, asset holders	40.777%	43.651%	+2.874 pp
Migrant-child separation rate, non-asset households	63.604%	63.519%	-0.085 pp
House price	76.550	76.444	-0.138%
<i>Panel C. Upward-mobility effects</i>			
Aggregate IM	32.108%	32.063%	-0.045 pp
Initial asset-holder IM	37.728%	37.476%	-0.252 pp
Initial non-asset IM	31.208%	31.196%	-0.012 pp

Notes: The table compares the calibrated baseline with a counterfactual that removes the privatisation component of initial housing endowments, setting $E_o^{eB,cf} = \max\{E_o^{eB} - E_o^{e,priv}, 0\}$ for initial asset holders. Housing endowments are measured in square metres per owner adult, private education investment in annual RMB per parental decision unit, and house prices in annual RMB per square metre. All household outcomes refer to low-educated parents or their children and are pooled nationally. The migration rate is the share of parents living outside their origin; the migrant-child separation rate is the share of migrant parents whose children remain at origin. The national house price uses fixed baseline housing-demand weights. Changes in rates are reported in percentage points; all other changes are reported in percent.

8 CONCLUSION

This paper studies how housing constraints shape intergenerational mobility by changing both household resources and access to high-opportunity cities. The reduced-form evidence reveals sharply different effects across families. Supply-driven house-price appreciation increases upward educational mobility among children of low-educated incumbent homeowners, consistent with a housing-resource channel, while reducing the migration of low-educated parents and increasing parent-child separation among low-educated migrant families. By amplifying the capitalization of local growth into housing prices, housing constraints therefore affect not only where households locate, but also which families are able to share in the opportunities generated by productive cities.

The quantitative analysis shows that substantial distributional consequences of housing supply constraints can arise even when changes in the national average are small. Making housing supply

more elastic primarily changes the incidence of urban growth: weaker price capitalisation facilitates parental migration and co-residence and improves outcomes particularly for migrant children who co-reside with their parents, while generating only a negligible increase in aggregate upward mobility. Equalising migrant children’s access to local public education, particularly in more supply-constrained cities, raises aggregate mobility, with most of the gain arising through general-equilibrium and sorting responses—including endogenous migration, co-residence, and destination choices—rather than the mechanical redistribution of a fixed education budget. The public-housing privatisation counterfactual further shows that housing wealth can have persistent intergenerational effects concentrated among the children of initial asset holders.

Taken together, the results show that housing constraints shape intergenerational mobility through their interaction with family sorting and access to local public goods. Higher housing costs deter migrant families from accessing productive cities and make parent–child co-residence more costly, while barriers to local public education further reduce the returns to bringing children to those destinations. For incumbent homeowners, by contrast, the capitalization of local growth into housing values can increase household resources and investment in children. While the empirical analysis exploits predetermined physical constraints for identification, the same mechanism may also apply to regulatory constraints insofar as they reduce housing-supply elasticity, such as construction-land quotas, floor-area restrictions, and height limits.

This paper focuses on educational mobility and therefore does not quantify the full effects of housing constraints on next-generation income, wealth, or welfare. Future work could extend the framework to incorporate wealth transmission and housing inheritance, thereby providing a more complete account of how housing constraints shape intergenerational outcomes.

REFERENCES

- Ahlfeldt, G. M., Redding, S. J., Sturm, D. M., and Wolf, N. (2015). The economics of density: Evidence from the berlin wall. *Econometrica*, 83(6):2127–2189.
- Alesina, A., Hohmann, S., Michalopoulos, S., and Papaioannou, E. (2021). Intergenerational mobility in africa. *Econometrica*, 89(1):1–35.
- Angst, S., Rosen, J., De Gregorio, S., and Painter, G. (2023). How do renters survive unaffordability? household-level impacts of rent burden in los angeles. *Journal of Urban Affairs*, pages 1–24.
- Attanasio, O., Meghir, C., and Nix, E. (2020). Human capital development and parental investment in india. *The review of economic studies*, 87(6):2511–2541.
- Binder, A. J., Risch, M., and Voorheis, J. L. (2026). Housing capital and intergenerational mobility in the united states. Technical report, National Bureau of Economic Research.
- Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *The Review of Economic Studies*, 89(1):181–213.
- Borusyak, K., Hull, P., and Jaravel, X. (2025). A practical guide to shift-share instruments. *Journal of Economic Perspectives*, 39(1):181–204.
- Bryan, G. and Morten, M. (2019). The aggregate productivity effects of internal migration: Evidence from indonesia. *Journal of Political Economy*, 127(5):2229–2268.
- Burchfield, M., Overman, H. G., Puga, D., and Turner, M. A. (2006). Causes of sprawl: A portrait from space. *The Quarterly Journal of Economics*, 121(2):587–633.
- Card, D., Domnisoru, C., and Taylor, L. (2022). The intergenerational transmission of human capital: Evidence from the golden age of upward mobility. *Journal of Labor Economics*, 40(S1):S39–S95.
- Charles, K. K., Hurst, E., and Notowidigdo, M. J. (2015). Housing booms and busts, labor market opportunities, and college attendance. Technical report, National Bureau of Economic Research.
- Chen, J., Hu, M., and Lin, Z. (2019). Does housing unaffordability crowd out elites in chinese superstar cities? *Journal of Housing Economics*, 45:101571.
- Chen, K. and Wen, Y. (2017). The great housing boom of china. *American Economic Journal: Macroeconomics*, 9(2):73–114.
- Chetty, R., Hendren, N., Kline, P., and Saez, E. (2014). Where is the land of opportunity? the geography of intergenerational mobility in the united states. *The Quarterly Journal of Economics*, 129(4):1553–1623.
- Chi, W. and Qian, X. (2016). Human capital investment in children: An empirical study of household child education expenditure in china, 2007 and 2011. *China Economic Review*, 37:52–65.
- Corak, M. (2013). Income inequality, equality of opportunity, and intergenerational mobility. *Journal of Economic Perspectives*, 27(3):79–102.
- Couture, V., Gaubert, C., Handbury, J., and Hurst, E. (2019). Income growth and the distributional effects of urban spatial sorting. Technical report, National Bureau of Economic Research.
- Diamond, R. (2016). The determinants and welfare implications of us workers’ diverging location choices by skill: 1980–2000. *American Economic Review*, 106(3):479–524.
- Diamond, R. (2017). Housing supply elasticity and rent extraction by state and local governments. *American Economic Journal: Economic Policy*, 9(1):74–111.
- Diamond, R. and Gaubert, C. (2022). Spatial sorting and inequality. *Annual Review of Economics*, 14:795–819.
- Ding, C., Niu, Y., and Lichtenberg, E. (2014). Spending preferences of local officials with off-budget land revenues of chinese cities. *China Economic Review*, 31:265–276.
- Dong, Y. (2016). A note on geographical constraints and housing markets in china. *Journal of Housing Economics*, 33:15–21.
- Duranton, G. and Puga, D. (2023). Urban growth and its aggregate implications. *Econometrica*,

- 91(6):2219–2259.
- Eckert, F. and Kleineberg, T. (2024). The geography of opportunity: Education, work, and intergenerational mobility across us counties.
- Fan, J. (2019). Internal geography, labor mobility, and the distributional impacts of trade. *American Economic Journal: Macroeconomics*, 11(3):252–288.
- Fan, Y. and Yavas, A. (2020). How does mortgage debt affect household consumption? micro evidence from china. *Real Estate Economics*, 48(1):43–88.
- Fan, Y., Yi, J., and Zhang, J. (2015). The great gatsby curve in china: Cross-sectional inequality and intergenerational mobility. *Working paper, CUHK, Hongkong*.
- Fan, Y., Yi, J., and Zhang, J. (2021). Rising intergenerational income persistence in china. *American Economic Journal: Economic Policy*, 13(1):202–230.
- Fang, H., Gu, Q., Xiong, W., and Zhou, L.-A. (2016). Demystifying the chinese housing boom. *NBER macroeconomics annual*, 30(1):105–166.
- Fang, M., Han, L., Huang, Z., Lu, M., and Zhang, L. (2021). Regional convergence or just an illusion? place-based land policy and spatial misallocation. *Place-based Land Policy and Spatial Misallocation (May 25, 2021)*.
- Fang, M. and Huang, Z. (2022). Migration, housing constraints, and inequality: A quantitative analysis of china. *Labour Economics*, 78:102200.
- Favilukis, J., Mabile, P., and Van Nieuwerburgh, S. (2023). Affordable housing and city welfare. *The Review of Economic Studies*, 90(1):293–330.
- Feng, Q. and He, Q. (2022). Does parental migration increase upward intergenerational mobility? evidence from rural china. *Economic Modelling*, 115:105955.
- Fletcher, J. and Han, J. (2019). Intergenerational mobility in education: Variation in geography and time. *Journal of Human Capital*, 13(4):585–634.
- Fogli, A. and Guerrieri, V. (2019). The end of the american dream? inequality and segregation in us cities. Technical report, National Bureau of Economic Research.
- Fu, Q. (2016). The persistence of power despite the changing meaning of homeownership: An age-period-cohort analysis of urban housing tenure in china, 1989–2011. *Urban Studies*, 53(6):1225–1243.
- Gan, J. (2010). Housing wealth and consumption growth: Evidence from a large panel of households. *The review of financial studies*, 23(6):2229–2267.
- Ganong, P. and Shoag, D. (2017). Why has regional income convergence in the us declined? *Journal of Urban Economics*, 102:76–90.
- Glaeser, E. and Gyourko, J. (2005). Urban decline and durable housing. *Journal of political economy*, 113(2):345–375.
- Glaeser, E. and Gyourko, J. (2018). The economic implications of housing supply. *Journal of economic perspectives*, 32(1):3–30.
- Glaeser, E. L., Gyourko, J., and Saks, R. E. (2005). Why have housing prices gone up? *American Economic Review*, 95(2):329–333.
- Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.
- Grujters, R. J., Chan, T. W., and Ermisch, J. (2019). Trends in educational mobility: How does china compare to europe and the united states? *Chinese Journal of Sociology*, 5(2):214–240.
- Gyourko, J., Mayer, C., and Sinai, T. (2013). Superstar cities. *American Economic Journal: Economic Policy*, 5(4):167–199.
- Gyourko, J., Shen, Y., Wu, J., and Zhang, R. (2022). Land finance in china: Analysis and review. *China Economic Review*, 76:101868.
- Hilber, C. A. and Mayer, C. (2009). Why do households without children support local public schools? linking house price capitalization to school spending. *Journal of Urban Economics*, 65(1):74–90.

- Hilber, C. A. and Vermeulen, W. (2016). The impact of supply constraints on house prices in england. *The Economic Journal*, 126(591):358–405.
- Hsieh, C.-T. and Moretti, E. (2019). Housing constraints and spatial misallocation. *American Economic Journal: Macroeconomics*, 11(2):1–39.
- Huang, B., Xu, C., Wang, D., and Hu, X. (2021). Compulsory education finance in china. *China Review*, 21(4):259–290.
- Huang, Y. and Yi, C. (2011). Second home ownership in transitional urban china. *Housing Studies*, 26(03):423–447.
- Hyman, J. (2017). Does money matter in the long run? effects of school spending on educational attainment. *American Economic Journal: Economic Policy*, 9(4):256–280.
- Imbert, C., Monras, J., Seror, M., and Zylberberg, Y. (2026). Floating population: migration with(out) family and the spatial distribution of economic activity. In *progress*. Federal Reserve Bank of San Francisco.
- Ioannides, Y. M. and Ngai, L. R. (2025). Housing and inequality. *Journal of Economic Literature*, 63(3):916–963.
- Jackson, C. K. and Mackevicius, C. L. (2024). What impacts can we expect from school spending policy? evidence from evaluations in the united states. *American Economic Journal: Applied Economics*, 16(1):412–446.
- Jia, J., Gu, J., and Ma, G. (2021). Real estate boom and firm productivity: Evidence from china. *Oxford Bulletin of Economics and Statistics*, 83(5):1218–1242.
- Jia, J., Guo, Q., and Zhang, J. (2014). Fiscal decentralization and local expenditure policy in china. *China Economic Review*, 28:107–122.
- Krueger, A. (2012). The rise and consequences of inequality. *Presentation made to the Center for American Progress, January 12th*.
- Krusell, P., Ohanian, L. E., Ríos-Rull, J.-V., and Violante, G. L. (2000). Capital-skill complementarity and inequality: A macroeconomic analysis. *Econometrica*, 68(5):1029–1053.
- Lai, R. N. and Van Order, R. A. (2020). A tale of two countries: comparing the us and chinese housing markets. *The Journal of Real Estate Finance and Economics*, 61:505–547.
- Lee, M.-H. (2011). Migration and children’s welfare in china: The schooling and health of children left behind. *The Journal of Developing Areas*, pages 165–182.
- Li, B. and Shin, H. B. (2013). Intergenerational housing support between retired old parents and their children in urban china. *Urban Studies*, 50(16):3225–3242.
- Li, F. and Xiao, J. J. (2020). Losing the future: Household wealth from urban housing demolition and children’s human capital in china. *China Economic Review*, 63:101533.
- Li, Q., Liu, G., and Zang, W. (2015). The health of left-behind children in rural china. *China Economic Review*, 36:367–376.
- Liang, W., Lu, M., and Zhang, H. (2016). Housing prices raise wages: Estimating the unexpected effects of land supply regulation in china. *Journal of Housing Economics*, 33:70–81.
- Liang, W., Song, R., and Timmins, C. (2024). Frictional sorting: The impacts of dual constraints on mobility and housing supply in china. *International Economic Review*, 65(4):1747–1776.
- Liu, J., Kuang, W., Zhang, Z., Xu, X., Qin, Y., Ning, J., Zhou, W., Zhang, S., Li, R., Yan, C., et al. (2014). Spatiotemporal characteristics, patterns, and causes of land-use changes in china since the late 1980s. *Journal of Geographical sciences*, 24:195–210.
- Liu, Y., Martinez-Vazquez, J., and Wu, A. M. (2017). Fiscal decentralization, equalization, and intra-provincial inequality in china. *International Tax and Public Finance*, 24:248–281.
- Lovenheim, M. F. (2011). The effect of liquid housing wealth on college enrollment. *Journal of Labor Economics*, 29(4):741–771.
- Lü, X. (2015). Intergovernmental transfers and local education provision—evaluating china’s 8-7

- national plan for poverty reduction. *China Economic Review*, 33:200–211.
- Miyauchi, Y., Nakajima, K., and Redding, S. J. (2021). Consumption access and agglomeration: evidence from smartphone data.
- Monte, F., Redding, S. J., and Rossi-Hansberg, E. (2018). Commuting, migration, and local employment elasticities. *American Economic Review*, 108(12):3855–3890.
- Newman, S. J. and Holupka, C. S. (2014). Housing affordability and investments in children. *Journal of Housing Economics*, 24:89–100.
- Painter, G., Yang, X., and Zhong, N. (2022). Housing wealth as precautionary saving: Evidence from urban china. *Journal of Financial and Quantitative Analysis*, 57(2):761–789.
- Peng, C., Qiu, W., Song, Q., and Huang, B. (2019). Housing wealth appreciation and consumption: evidence from china. *Journal of the Asia Pacific Economy*, 24(4):556–577.
- Peng, Y., Tian, C., and Wen, H. (2021). How does school district adjustment affect housing prices: An empirical investigation from hangzhou, china. *China Economic Review*, 69:101683.
- Pietrabissa, G. (2023). School access and city structure. In *progress*.
- Rozelle, S. and Hell, N. (2020). *Invisible China: How the urban-rural divide threatens China’s rise*. University of Chicago Press.
- Saiz, A. (2010). The geographic determinants of housing supply. *The Quarterly Journal of Economics*, 125(3):1253–1296.
- Saks, R. E. (2008). Job creation and housing construction: Constraints on metropolitan area employment growth. *Journal of Urban Economics*, 64(1):178–195.
- Sieg, H., Yoon, C., and Zhang, J. (2020). The impact of migration controls on urban fiscal policies and the intergenerational transmission of human capital in china. Technical report, National Bureau of Economic Research.
- Sieg, H., Yoon, C., and Zhang, J. (2023). The impact of local fiscal and migration policies on human capital accumulation and inequality in china. *International Economic Review*, 64(1):57–93.
- State Council (1994). *Guo wu yuan guan yu shen hua cheng zhen zhu fang zhi du gai ge de tong zhi*, volume Document of State Council No. 43. Gai ge chu ban she [Reform Press], Beijing. Notice of State Council on deepening up housing reform in cities and towns. Office of Leading Group of Housing System Reform of the State Council and Institution of China Housing System Reform.
- Takeda, K. (2022). *The geography of structural transformation: Effects on inequality and mobility*. Centre for Economic Performance, London School of Economics and Political . . .
- Tan, Y., Wang, Z., and Zhang, Q. (2020). Land-use regulation and the intensive margin of housing supply. *Journal of Urban Economics*, 115:103199.
- Tian, L., Yan, Y., Wu, Y., and Shao, L. (2020). Breaking the land monopoly: Can collective land reform alleviate the housing shortage in china’s mega-cities? *Cities*, 106:102878.
- Tombe, T. and Zhu, X. (2019). Trade, migration, and productivity: A quantitative analysis of china. *American Economic Review*, 109(5):1843–1872.
- Tsivanidis, N. (2019). Evaluating the impact of urban transit infrastructure: Evidence from bogota’s transmilenio. Technical report, UC Berkeley (mimeo), 2020.[Google Scholar].
- Wang, S., Chan, S. H., and Xu, B. (2012). The estimation and determinants of the price elasticity of housing supply: Evidence from china. *Journal of Real Estate Research*, 34(3):311–344.
- Wang, S.-Y. (2011). State misallocation and housing prices: theory and evidence from china. *American Economic Review*, 101(5):2081–2107.
- Wang, W. (2014). Decomposing inequality in compulsory education finance in china: 1998–2008. *Public Finance and Management*, 14(4):437–458.
- Wolf, C. K. (2023). The missing intercept: A demand equivalence approach. *American Economic Review*, 113(8):2232–2269.
- Wrenn, D. H., Yi, J., and Zhang, B. (2019). House prices and marriage entry in china. *Regional*

- Science and Urban Economics*, 74:118–130.
- Wu, J., Gyourko, J., and Deng, Y. (2012). Evaluating conditions in major chinese housing markets. *Regional Science and Urban Economics*, 42(3):531–543.
- Xie, Y., Dong, H., Zhou, X., and Song, X. (2022). Trends in social mobility in postrevolution china. *Proceedings of the National Academy of Sciences*, 119(7):e2117471119.
- Xiong, W. (2023). Derisking real estate in china’s hybrid economy. Technical report, National Bureau of Economic Research.
- Xu, X., Liu, J., Zhang, S., Li, R., Yan, C., and Wu, S. (2018). China’s multi-period land use land cover remote sensing monitoring data set (cnlucc). *Resource and Environment Data Cloud Platform: Beijing, China*.
- Zhang, H. (2022). Consumer cities: the role of housing variety.
- Zhang, J., Li, H., Lin, J., Zheng, W., Li, H., and Chen, Z. (2020). Meta-analysis of the relationship between high quality basic education resources and housing prices. *Land Use Policy*, 99:104843.
- Zheng, A. and Graham, J. (2022). Public education inequality and intergenerational mobility. *American Economic Journal: Macroeconomics*, 14(3):250–282.

A APPENDIX FIGURES

Declining intergenerational education mobility in China net of industrialization [\[back\]](#)

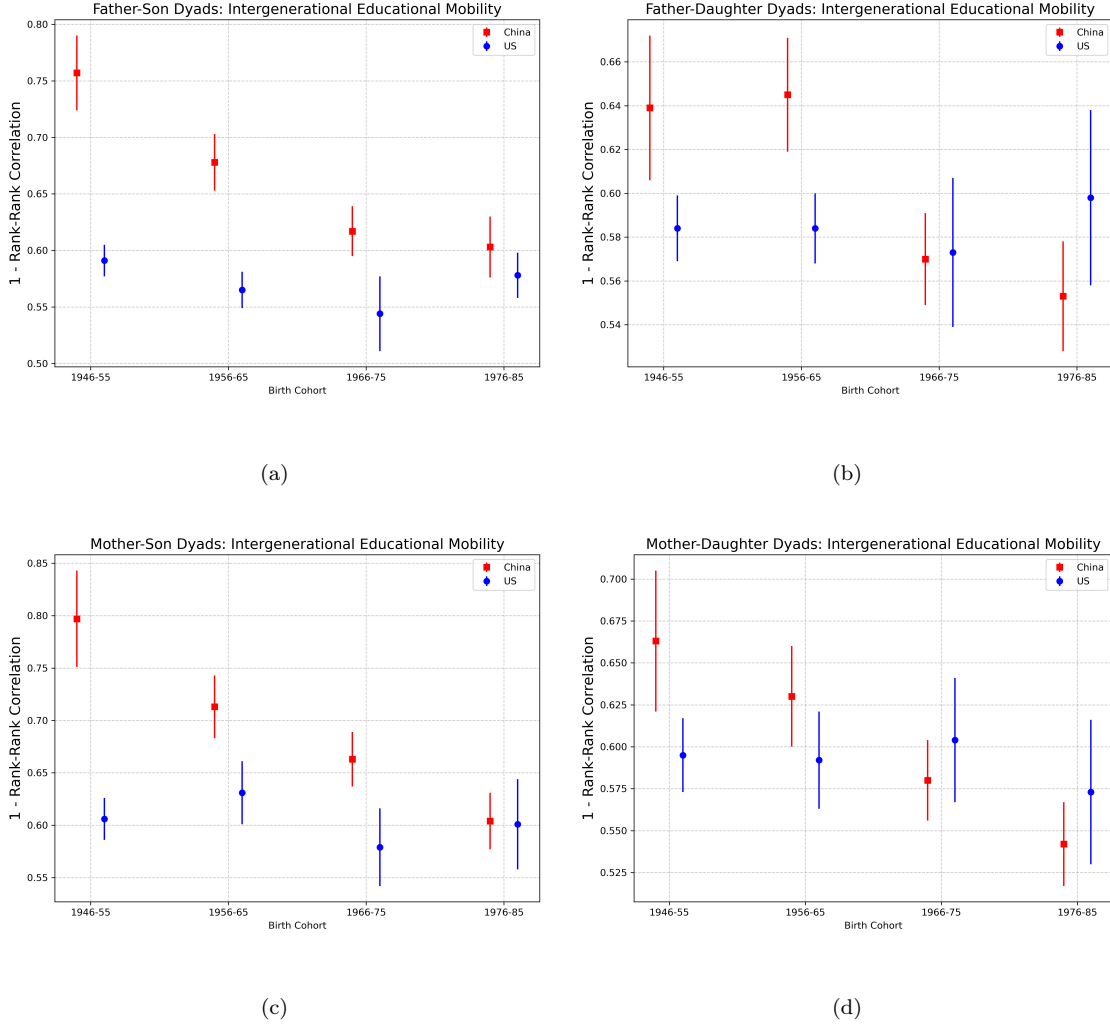


Figure A1: The declining trend of intergenerational education mobility among nonfarm origin population in China as compared to U.S. over the recent four decades compiled with data from [Xie et al. \(2022\)](#). The x-axis denotes each of the four age cohorts: 1946–1955, 1956–1965, 1966–1975 and 1976–1985. The y-axis is calculated as “1 - rank-rank correlation”, a higher value of which denotes higher educational mobility. The rank-rank correlation captures the correlation of education achievement between parents and children based on their education rank among peers. Original data sources used in [Xie et al. \(2022\)](#) are: GSS 1972–2018; NLS–YM 1966–1981; NLS–YW 1968–1993; NLSY79 1979–2012; NSFH 1987; OCG II (only males); PSID (SRC sample) 1968–2017; SIPP 1986, 1987, 1988; LHSCCC 1996; CGSS 2005, 2006, 2008, 2010, 2011, 2012, 2013, 2015, 2017, 2018. See [Xie et al. \(2022\)](#) for further details on the estimation strategies.

B APPENDIX TABLES

Wealth effects

The three tables below (1a–1c) highlights the “wealth effects” mechanism of housing price change by analyzing the effects of housing value/rent on households’ education spending per school-age child, using six biennial waves of China Family Panel Studies (CFPS) data from 2010 to 2020. For columns (2)–(8), the regressions include household fixed effects and province-by-year fixed effects to account for households’ time-invariant characteristics and province-specific temporal variations in education costs. As an interpretation, the coefficient 0.5724 in column (2) of Table 1a indicates that a 1,000 RMB increase in self-reported housing market value corresponds to a 0.5724 RMB increase in homeowners’ total education expenses per child within the same year. Column (3)–(6) decomposes education expenses into 4 main categories, showing that the housing appreciation mainly impacts on homeowners’ spending on kids’ extracurricular tutoring. Similar “wealth effects” of housing appreciation are observed for other expenses, such as food and recreational activities, as shown in columns (7) and (8) in Table 1a. In Table 1c, a placebo regression is performed exclusively for homeowners aged 50+ without school-age children in the sample. As expected, while housing appreciation shows positive effects on expense categories such as food and recreational activities, no significant effect is found on their education expenses.

Table B1: The effects of **housing value** on households’ private education spending per child over 2010-2020.

	Total education expenses per child		Education expenses by categories					
	(1)	(2)	Tuition	Textbooks	Tutoring	Necessities	Food expenses	Recreational expenses
			(3)	(4)	(5)	(6)	(7)	(8)
Housing value	0.8896*** (0.3187)	0.5724** (0.2699)	0.2367 (0.2222)	-0.0262 (0.0375)	0.6036*** (0.1837)	-0.0008 (0.0538)	1.8164*** (0.6310)	0.7267*** (0.1401)
Wage	7.0930* (3.6842)	6.0386** (2.7666)	3.3398* (1.7829)	0.3061 (0.3783)	2.0451 (3.4034)	0.0847 (0.6994)	12.5919** (6.3464)	3.2520** (1.5890)
Province-year FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs	14720	14689	14689	7671	14689	7671	55159	55174

Notes: This table displays the effect of homeowners’ self-reported housing market value (in 000 RMB) on households’ annual education spending per child (in RMB) using China Family Panel Studies (CFPS) panel data between 2010–2020 (6 waves of biennially surveys) controlling for family wage (in 000 RMB). Column (1)–(2) capture the effects of housing value on total education spending per child, in which column (2) controls for province-by-year fixed effects. Column (3)–(6) evaluate the effects of housing value on four main categories of education expenses: tuition and school selection fees; textbooks and stationery; extracurricular tutoring; accomodation, food and transports (Necessities). Column (7)–(8) document the effects of housing price on households’ annual spending on food and recreational activities respectively, as a comparison for education spending. The 2016, 2018 and 2020 rounds of survey give no information about the expenses on textbooks and stationery, and necessities.

Standard errors clustered at household and district level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$

Table B2: The effects of annual **rent** on renter households' private education spending per child over 2010-2020.

	Education expenses by categories							
	Total education expenses per child		Tuition	Textbooks	Tutoring	Necessities	Food expenses	Recreational expenses
	(1)	(2)						
Rent	-2.0487 (1.5562)	-3.4388 (2.6233)	-1.2559 (1.7128)	3.2464 (5.2700)	0.3739 (1.5627)	-8.5818 (5.9656)	15.7256 (17.3694)	4.9098 (6.1732)
Wage	23.0863*** (7.0917)	23.2322** (8.9193)	19.2352** (7.5070)	0.9791 (1.1615)	1.9156 (4.5712)	-2.7409 (2.0758)	77.4041*** (21.4717)	4.7648 (3.7489)
Province-year FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs	979	924	924	321	924	321	3747	3700

Notes: This table displays the effect of renter households' self-reported housing market value (in 000 RMB) on households' annual education spending per child (in RMB) using China Family Panel Studies (CFPS) panel data between 2010–2020 (6 waves of biennially surveys) controlling for family wage (in 000 RMB). Column (1)-(2) capture the effects of rent level on total education spending per child, in which column (2) controls for province-by-year fixed effects. Column (3)-(6) evaluate the effects of rent on four main categories of education expenses: tuition and school selection fees; textbooks and stationery; extracurricular tutoring; accomodation, food and transports (Necessities). Column (7)-(8) document the effects of housing price on households' annual spending on food and recreational activities respectively, as a comparison for education spending. The 2016, 2018 and 2020 rounds of survey give no information about the expenses on textbooks and stationery, and necessities.

Standard errors clustered at household and district level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$

Table B3: Placebo effects of **housing value** on education spending for childless households' aged 50+ (2010-2020).

	Total education expenses		Food expenses	Recreational expenses
	(1)	(2)	(3)	(4)
Housing value	0.2788 (0.2483)	0.2873 (0.2874)	1.8877** (0.8861)	0.4452*** (0.0976)
Wage	14.9671** (6.4399)	16.7672** (6.8319)	12.4915 (19.3106)	22.0482** (10.8672)
Province-year FE	No	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Obs	7308	7282	7245	7267

Notes: This table displays the placebo effect of homeowners' self-reported housing market value (in 000 RMB) on childless households' annual education spending per child (in RMB) using China Family Panel Studies (CFPS) panel data between 2010–2020 (6 waves of biennially surveys) controlling for family wage (in 000 RMB). Only households in which household head is ≥ 50 years old without school-age child are included in the regression. Column (1)-(2) capture the effects of housing value on total education spending per child, in which column (2) controls for province-by-year fixed effects. Column (3)-(4) document the effects of housing price on households' annual spending on food and recreational activities respectively, as a comparison for education spending. The 2016, 2018 and 2020 rounds of survey give no information about the expenses on textbooks and stationery, and necessities.

Standard errors clustered at household and district level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$

Reduced-form

Table B4: Placebo test using the 2005 census city IM measure

	(1)	(2)
<i>Panel A. 2nd stage: Dependent variable is placebo $UP_{c,2005}$</i>		
$\Delta \ln \widehat{HP}_{cT}$	0.0506 (0.0358)	0.0430 (0.0355)
Initial $\ln HP$	0.0669*** (0.0219)	0.0612** (0.0252)
%developed	0.0023 (0.0059)	0.0016 (0.0058)
Slope	-0.0036 (0.0041)	-0.0035 (0.0036)
Δ Gov pc edu spending		0.0056 (0.0153)
Δ GDP per capita		-0.0273* (0.0159)
Δ Education employment		-0.0020 (0.0116)
Δ Primary schools per capita		0.0034 (0.0053)
Δ Middle schools per capita		0.0073 (0.0264)
<i>Panel B. 1st stage: Dependent variable is $\Delta \ln HP_{c,T}$</i>		
%developed $\times \Delta \widehat{Emp}_{cT}$	0.8975*** (0.2181)	0.8958*** (0.2234)
Slope $\times \Delta \widehat{Emp}_{cT}$	0.7212*** (0.2124)	0.8043*** (0.2062)
Initial employment shares	Yes	Yes
Province FE	Yes	Yes
Auxiliary controls	No	Yes
KP rk Wald F-statistic	9.081	10.109
N	268	268

Notes: The table reports placebo 2SLS regressions using a city IM measure estimated from the 2005 census. The placebo IM is constructed for native children in homeowner families from pre-treatment birth cohorts born in 1976–1981, so that their education should already be largely completed before the 2003–2012 housing-price shock. Column (1) uses the baseline city controls from the main specification. Column (2) additionally controls for contemporaneous changes in government education spending per capita, GDP per capita, education employment, and the numbers of primary and middle schools per capita.

Standard errors are clustered at the province level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$.

Table B5: Mechanism test on the effects of housing price on intergenerational mobility measures

	Native		Migrants			
	Homeowners		Co-reside		Separate	
	(1)		(2)		(3)	
$\Delta \ln HP_{c,T}$	0.108*	(0.055)	0.762*	(0.428)	-0.239	(0.381)
Initial $\ln HP$	0.114*	(0.063)	0.507	(0.334)	-0.460	(0.502)
%developed	0.513	(1.109)	-8.767	(10.792)	4.489	(6.538)
Slope	-0.575	(0.859)	-10.181	(9.381)	-2.770	(11.493)
%developed $\times$ Ind.1	-0.639	(1.103)	11.744	(10.555)	-3.786	(6.397)
%developed $\times$ Ind.2	0.450	(1.433)	6.965	(15.020)	3.669	(9.010)
%developed $\times$ Ind.3	1.511	(3.010)	2.241	(20.542)	-11.585	(20.839)
%developed $\times$ Ind.4	-0.910	(1.320)	6.402	(10.823)	-6.991	(8.528)
%developed $\times$ Ind.5	-0.483	(1.135)	11.192	(10.130)	-5.623	(7.482)
%developed $\times$ Ind.6	-0.491	(1.117)	8.947	(10.976)	-5.372	(6.896)
%developed $\times$ Ind.7	0.762	(1.991)	5.094	(14.293)	-0.329	(11.763)
%developed $\times$ Ind.8	1.664	(2.149)	29.299	(21.587)	-24.511	(20.897)
%developed $\times$ Ind.9	-0.579	(1.199)	7.287	(11.845)	-4.776	(6.716)
%developed $\times$ Ind.10	-3.839	(2.319)	17.113	(22.447)	22.464	(14.692)
%developed $\times$ Ind.11	0.022	(1.108)	12.645	(8.845)	-13.594	(9.299)
%developed $\times$ Ind.12	-0.677	(0.990)	13.811	(12.443)	-2.484	(7.019)
%developed $\times$ Ind.13	0.654	(1.485)	19.517	(30.266)	22.691	(17.830)
%developed $\times$ Ind.14	-2.273	(1.881)	-1.032	(14.847)	8.937	(19.984)
%developed $\times$ Ind.15	-0.380	(1.348)	8.895	(18.034)	-12.244	(10.577)
%developed $\times$ Ind.16	-0.306	(1.705)	-2.266	(8.826)	-6.126	(14.426)
%developed $\times$ Ind.17	-1.581	(1.494)	3.067	(10.889)	-16.562	(17.129)
%developed $\times$ Ind.18	-0.681	(1.147)	7.744	(10.026)	-3.917	(6.762)
Slope $\times$ Ind.1	0.061	(1.086)	12.069	(8.980)	8.738	(13.367)
Slope $\times$ Ind.2	2.697	(1.721)	-6.139	(16.107)	-2.271	(14.311)
Slope $\times$ Ind.3	2.670	(2.598)	26.028	(27.919)	-80.429	(50.987)
Slope $\times$ Ind.4	0.342	(0.928)	4.774	(10.970)	1.077	(12.546)
Slope $\times$ Ind.5	0.919	(0.836)	12.662	(8.502)	-4.077	(12.259)
Slope $\times$ Ind.6	0.531	(0.880)	11.130	(9.314)	1.638	(11.393)
Slope $\times$ Ind.7	0.953	(1.681)	18.294*	(10.333)	16.399	(25.249)
Slope $\times$ Ind.8	6.585*	(3.514)	3.572	(26.433)	-23.101	(28.758)
Slope $\times$ Ind.9	0.968	(0.777)	8.589	(9.413)	4.917	(11.610)
Slope $\times$ Ind.10	-2.715	(1.815)	39.878	(24.893)	14.254	(21.915)
Slope $\times$ Ind.11	1.216	(0.859)	16.053*	(8.383)	-6.813	(11.778)
Slope $\times$ Ind.12	0.433	(0.834)	13.514	(9.584)	4.393	(9.880)
Slope $\times$ Ind.13	0.739	(0.978)	1.320	(32.707)	29.839*	(15.372)
Slope $\times$ Ind.14	1.003	(1.811)	-3.256	(21.188)	22.187	(26.422)
Slope $\times$ Ind.15	1.019	(1.176)	9.563	(17.073)	14.750	(20.109)
Slope $\times$ Ind.16	-0.048	(1.540)	8.949	(11.434)	-2.412	(19.068)
Slope $\times$ Ind.17	-2.386*	(1.312)	-14.468	(20.824)	15.078	(23.309)
Slope $\times$ Ind.18	0.500	(0.892)	9.825	(9.182)	0.704	(11.380)
Initial employment shares	Yes		Yes		Yes	
Province FE	Yes		Yes		Yes	
Initial UP	Yes		No		No	
N	261		142		140	

Notes: The table reports direct-effect checks for the adapted exposure shares used in the shift-share instruments. The dependent variables are city-level upward mobility measures. Each column regresses upward mobility on housing price growth, baseline controls, initial employment shares, province fixed effects, and the full set of interactions between initial industry shares and the two physical housing-supply constraints: the share of developable land developed in 2000 and mean terrain slope within 10 km of the city centre, restricted to the city's 2010 administrative boundary. Column (1) additionally controls for the corresponding 2005 census UP measure.

Standard errors are clustered at the province level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$.

Table B6: The effects of housing price on intergenerational mobility measures with auxiliary controls

	Native	Migrants	
	Homeowners (1)	Co-reside (2)	Separate (3)
<i>Panel A. 2nd stage: Dependent variable is UP_{ct}</i>			
$\Delta \ln \widehat{HP}_{cT}$	0.1905** (0.0881)	0.0092 (0.5748)	0.1049 (0.4018)
Initial $\ln HP$	0.1912*** (0.0609)	0.2588 (0.4258)	-0.1849 (0.2676)
%developed	0.0113 (0.0103)	-0.0762 (0.0786)	-0.0146 (0.0668)
Slope	-0.0030 (0.0061)	-0.0544 (0.0615)	0.0209 (0.0291)
Δ Gov pc edu spending	0.0522 (0.0507)	-0.2607 (0.1629)	0.2581 (0.2233)
Δ GDP per capita	-0.0105 (0.0414)	0.1462 (0.2145)	-0.0837 (0.2096)
Δ Education employment	-0.0043 (0.0503)	0.0196 (0.3789)	-0.2110 (0.3815)
Δ Primary schools per capita	-0.0153 (0.0140)	0.1709 (0.1246)	-0.0840 (0.1267)
Δ Middle schools per capita	0.0025 (0.0530)	-0.1402 (0.2956)	-0.3929 (0.3293)
<i>Panel B. 1st stage: Dependent variable is $\Delta \ln HP_{c,T}$</i>			
%developed $\times \Delta \widehat{Emp}_{cT}$	0.8367*** (0.2293)	0.7827*** (0.2700)	0.5726* (0.3230)
Slope $\times \Delta \widehat{Emp}_{cT}$	0.7299*** (0.2256)	0.8886*** (0.2353)	1.1803*** (0.2375)
Initial employment shares	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Initial UP	Yes	No	No
KP rk LM statistic	9.457	6.644	7.106
KP rk LM p-value	[0.009]	[0.036]	[0.029]
KP rk Wald F-statistic	7.935	8.613	13.938
Anderson–Rubin chi-sq	9.544	2.815	1.058
Anderson–Rubin Wald p-value	[0.008]	[0.245]	[0.589]
N	261	142	140

Notes: The table reports IV estimates of the effects of housing price changes between 2003–2012 on upward mobility after controlling for contemporaneous changes in government education spending per capita, GDP per capita, education-sector employment, and the numbers of primary and middle schools per capita. These auxiliary controls check whether the estimated effect is driven by changes in local public education spending, broad city-level income growth, or public education capacity rather than household-side responses to housing wealth. Housing price changes are instrumented by interactions between two physical housing-supply constraints and shift-share labour-demand shocks.

Standard errors are clustered at the province level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$.

Individual-level regression results

Table B7: [2SLS: *Individual IM*] IV estimates of the effect of housing price change on upward mobility.

	Native	Migrants	
	Homeowners (1)	Co-reside (2)	Separate (3)
<i>Panel A. 2nd stage: Dependent variable is individual upward mobility</i>			
$\Delta \ln \widehat{HP}_{c,T}$	0.1862*** (0.0614)	0.1613 (0.3260)	-0.2000 (0.1609)
Initial lnHP	0.1829*** (0.0441)	0.4566* (0.2666)	-0.3798* (0.2096)
%developed	0.0033 (0.0081)	-0.1783** (0.0694)	-0.0394 (0.0470)
Slope	0.0002 (0.0046)	-0.0699** (0.0289)	0.0220 (0.0398)
<i>Panel B. 1st stage: Dependent variable is $\Delta \ln \widehat{HP}_{c,T}$</i>			
%developed $\times \Delta \widehat{Emp}_{cT}$	0.8081*** (0.2156)	1.0118*** (0.2099)	0.9038*** (0.2901)
Slope $\times \Delta \widehat{Emp}_{cT}$	0.8330*** (0.1901)	0.9107*** (0.3054)	1.6450*** (0.1899)
Initial employment shares	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Initial UP	Yes	No	No
KP rk LM statistic	8.112	9.722	5.345
KP rk LM p-value	[0.017]	[0.008]	[0.069]
KP rk Wald F-statistic	13.086	11.831	37.848
Anderson–Rubin chi-sq	7.110	22.189	1.634
Anderson–Rubin Wald p-value	[0.029]	[0.000]	[0.442]
N	18786	745	646

Notes: The table reports individual-level IV estimates. The dependent variable equals one if a child born into a low-educated family attains college education or above. Column (1) uses the 2015 census native-homeowner child sample. Columns (2)–(3) use CMDS 2016 migrant child samples with CMDS sampling weights. All columns use the 1986–1993 birth cohorts and control for individual and family-background characteristics, initial housing price, standardized physical constraints, initial employment shares, and province fixed effects. Housing price changes over 2003–2012 are instrumented by interactions between shift-share labour-demand shocks and two physical housing-supply constraints: the share of developable land developed in 2000 and mean terrain slope within 10 km of the city centre, restricted to the city’s 2010 administrative boundary.

Standard errors are clustered at the province level in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$.

C QUANTIFICATION DETAILS

C.1 INITIAL HOUSING ENDOWMENTS

This subsection complements Section 6.2 in the main paper by describing how I construct the initial housing endowments $\{E_o^{e\delta}\}$ that enter the household budget constraint in the quantitative model.

Initial housing status δ is predetermined by origin. Households with $\delta = \mathcal{B}$ are local urban owner-type households, while $\delta = \mathcal{R}$ comprises urban renters and rural households. In the household budget constraint, this endowment enters through the annualised resource value $r_o E_o^{e\delta}$, with $E_o^{e\mathcal{B}} > 0$ and $E_o^{e\mathcal{R}} = 0$. The calibrated object $E_o^{e\mathcal{B}}$ is a quantity-based housing-service-equivalent endowment: the average initial floor space per working-age adult in type- e owner households from origin o . Population masses by initial housing status are constructed separately when building the baseline population distribution.

The construction uses the 2000 Census. Among local urban households, I classify self-built housing, purchased commercial housing, purchased economic housing, and purchased former public housing as owner-type housing. Self-built units enter the baseline housing-service endowment; the model treats all owner-type housing as a common housing-service-equivalent asset and abstracts from differences in marketability across housing types.

For each household i of education type e in origin city o , let q_{io} denote total housing floor space and let n_i^A denote the number of working-age adults in the household. Since the model budget constraint is written at the adult decision-unit level, I construct the city-education-specific initial housing endowment by allocating total urban owner-occupied housing space across urban owner adults:

$$E_o^{e\mathcal{B}} = \frac{\sum_{i \in (o, e, \text{urban owner})} q_{io}}{\sum_{i \in (o, e, \text{urban owner})} n_i^A}, \quad E_o^{e\mathcal{R}} = 0.$$

The set $i \in (o, e, \text{urban owner})$ includes households of education type e in origin city o living in self-owned housing. This construction treats $\delta = \mathcal{B}$ as the initial urban owner state and calibrates the corresponding housing-service endowment using the average housing space available per owner adult.

This baseline measure is intended to capture the general housing-wealth channel. It does not require all owner-occupied housing to have been received as a costless transfer. Rather, the model takes initial housing status as a predetermined state and uses $E_o^{e\mathcal{B}}$ to summarise the collateralizable housing wealth available to owner-type households at the beginning of the decision problem. This is distinct from the privatisation-induced component of housing wealth, which I construct separately for counterfactual exercises.

To isolate the component of initial housing wealth generated by public-housing privatisation, I estimate the discount received by households that purchased former public rental housing. For each city, I estimate a hedonic model using market-based owner-occupied units. Specifically, I regress the log purchase price per square metre on housing characteristics, including floor space and its higher-order terms, number of rooms, building-storey categories, wall materials, kitchen type, cooking-energy source, bath facilities, toilet type, tap-water availability, building age and its higher-order terms, city fixed effects, and an indicator for economic housing. I then use the estimated coefficients to predict the market price per square metre, $\hat{P}_{io}$, for purchased former public housing units.

For a household i of education type e in origin city o living in purchased former public housing, I define the privatisation discount per square metre as

$$d_{io} = \max \left\{ \frac{\hat{P}_{io} - P_{io}}{\hat{P}_{io}}, 0 \right\},$$

where P_{io} is the actual purchase price per square metre paid by the household. The privatisation-

induced housing surplus of household i , expressed in housing-service units, is $d_{io}q_{io}$. I aggregate this surplus to the city-education level by allocating it across all urban owner adults in the same city-education cell:

$$E_o^{e,\text{priv}} = \frac{\sum_{i \in (o,e,\text{public purchase})} d_{io}q_{io}}{\sum_{i \in (o,e,\text{urban owner})} n_i^A}.$$

This object measures the average privatisation-induced component of initial housing wealth among urban owner adults. The denominator is the number of urban owner adults, rather than only adults in purchased former public housing, because the model does not separately distinguish former-public-housing owners from other urban owner-type households. The estimated public-housing surplus is therefore averaged within each city-education owner cell. I use it in counterfactual exercises that remove the housing-wealth advantage generated by public-housing privatisation:

$$E_o^{e\mathcal{B},\text{cf}} = \max \{ E_o^{e\mathcal{B}} - E_o^{e,\text{priv}}, 0 \}.$$

Thus, the baseline endowment captures the general wealth effect of initial urban homeownership, while the counterfactual component isolates the part of that endowment attributable to the public-housing privatisation legacy.

Several limitations should be noted. First, housing prices in the census are reported in categories rather than exact values; I use category midpoints and code the open-ended top category at its lower bound. Second, the census does not report exact within-city locations, so the hedonic model used to estimate privatisation discounts cannot fully adjust for neighbourhood-level price differences. Third, public-housing allocation depended on work-unit status, age, job rank, Party membership, and social networks. These determinants are not separately modelled; instead, their implications are summarised by the observed initial housing status δ , the baseline housing endowment $E_o^{e\mathcal{B}}$, and the privatisation-induced component $E_o^{e,\text{priv}}$.

Figure C1 reports the distribution of the inferred city-level baseline housing endowment $E_o^{e\mathcal{B}}$ for high- and low-educated urban owner adults. Figure C2 maps the corresponding city-specific values by education group. Figure C3 reports the distribution of the privatisation-induced component $E_o^{e,\text{priv}}$, and Figure C4 maps its spatial distribution.

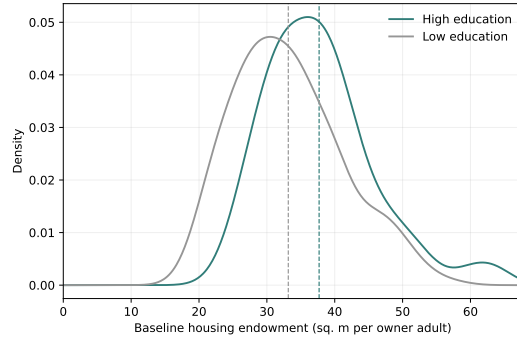


Figure C1: Kernel density distribution of inferred city-level initial housing endowments for high- and low-educated urban owner adults. The measure is constructed as total urban owner-occupied housing space divided by the number of working-age adults in urban owner households within each city–education cell. Mean values for each group are marked by dashed lines. High education corresponds to high-school attainment or above, and low education refers to below high school.

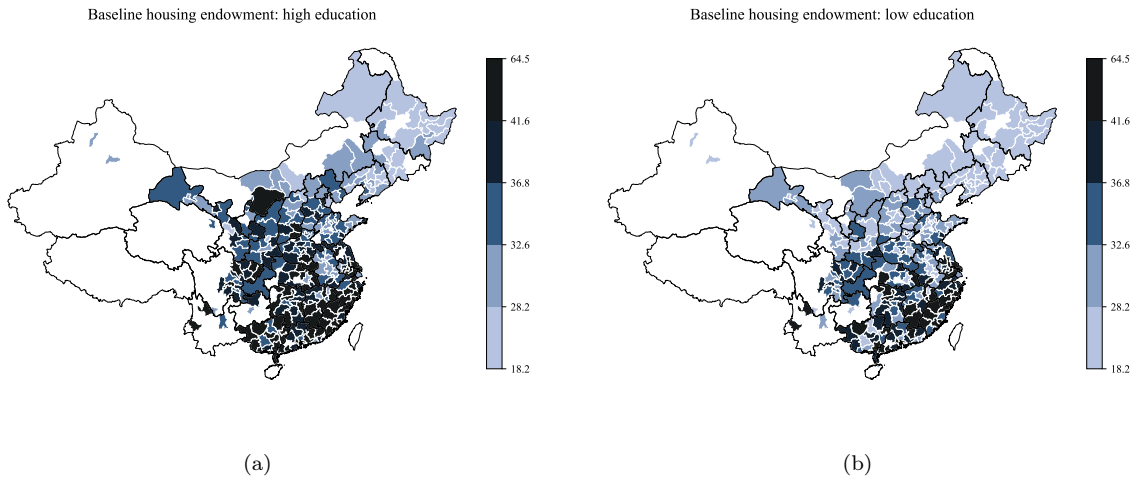


Figure C2: **Spatial distribution of initial housing endowments.** Panel (a) displays the inferred city-specific initial housing endowment for high-educated urban owner adults. Panel (b) displays the corresponding measure for low-educated urban owner adults. The measure is constructed as total urban owner-occupied housing space divided by the number of working-age adults in urban owner households within each city–education cell. Details are provided in Appendix C.1.

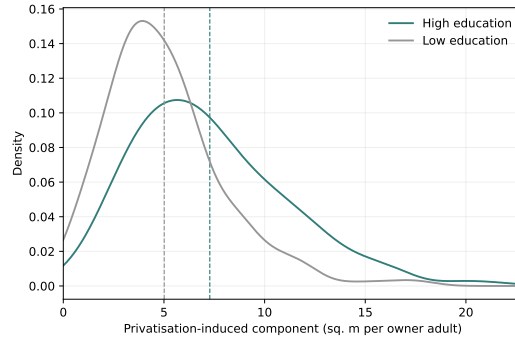


Figure C3: Kernel density distribution of the inferred city-level privatisation-induced component of initial housing endowments for high- and low-educated urban owner adults. The measure is constructed as the estimated housing surplus from purchased former public housing, identified from the privatisation discount, divided by the number of working-age adults in urban owner households within each city–education cell. Mean values for each group are marked by dashed lines. High education corresponds to high-school attainment or above, and low education refers to below high school.

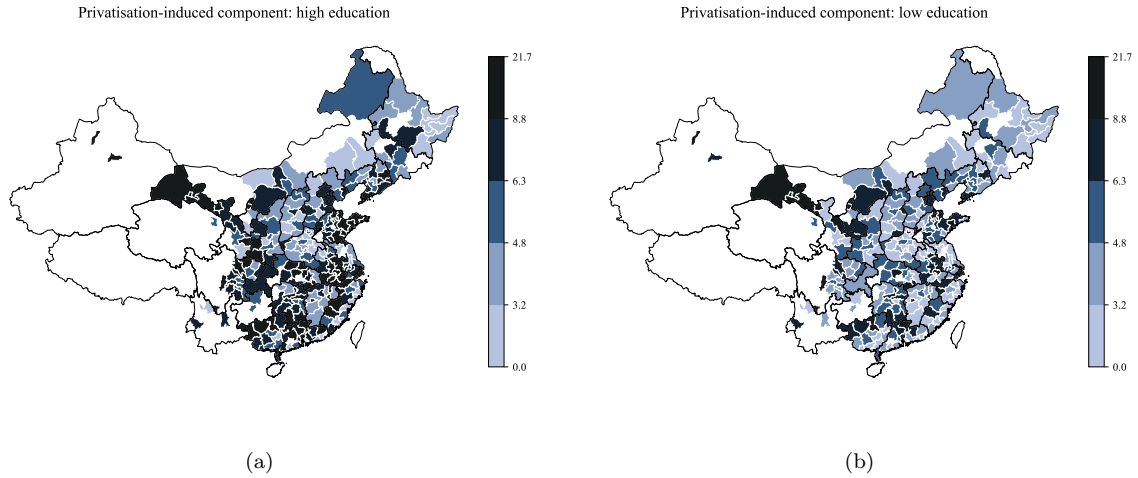


Figure C4: **Spatial distribution of the privatisation-induced component of initial housing endowments.** Panel (a) displays the inferred city-specific privatisation-induced component of initial housing endowments for high-educated urban owner adults. Panel (b) displays the corresponding measure for low-educated urban owner adults. The measure is constructed as the estimated housing surplus from purchased former public housing, identified from the privatisation discount, divided by the number of working-age adults in urban owner households within each city–education cell. Details are provided in Appendix C.1.

C.2 SUPPLY ELASTICITY

This section details the estimation of the city-specific supply elasticities used in Section 6.2.

Table C1: GMM estimates of housing-supply elasticities.

	Robust SE	Clustered SE
<i>Dependent variable is $\Delta \ln r_{jT}$</i>		
θ_0	-0.8073 (0.5617)	-0.8073 (0.8446)
θ_D	0.8004*** (0.2858)	0.8004* (0.4097)
θ_S	0.7568*** (0.1594)	0.7568*** (0.1661)
Initial lnHP	-0.5860*** (0.0682)	-0.5860*** (0.0886)
Developed	0.0107 (0.0177)	0.0107 (0.0259)
Slope	-0.0021 (0.0250)	-0.0021 (0.0277)
Province FE	Yes	Yes
Cluster at province	No	Yes
Joint test: $\theta_D = \theta_S = 0$ p-value	[0.000]	[0.000]
N	277	277

Notes: The table reports nonlinear IV/GMM estimates of Eq. 35, where $Developed_j$ and $Slope_j$ are standardized physical housing-supply constraints. The endogenous quantity term is permanent-population growth, $\Delta \ln Q_{jT}$, measured between 2005 and 2015. Identification uses the shift-share local labour-demand shock ΔEmp_{cT} and its interactions with the two physical constraints as excluded instruments: ΔEmp_{cT} , $Developed_j \times \Delta Emp_{cT}$, and $Slope_j \times \Delta Emp_{cT}$. The vector X_j includes log initial housing price and the two physical supply constraints, and γ_p denotes province fixed effects. Robust and province-clustered standard errors are computed using the GMM sandwich variance estimator. *** $p < .01$, ** $p < .05$, * $p < .1$.

C.3 CALIBRATION OF INITIAL ADULT POPULATIONS FROM THE 2005 CENSUS

I use the 2005 Population Census to calibrate the initial population objects $\bar{H}$, φ and $\{H_o^{e\delta\varnothing}, H_o^{e\delta}, O_j\}$ in Table F1–F2. The census records current residence, hukou registration, education, household structure, fertility information for women, and housing source. These variables allow me to construct, for each prefecture, the distribution of adults by origin, education, parenthood, and initial housing status.

The unit of observation is the individual. I restrict the adult population to ages 19–64 and define dependent children as household members aged 0–18. Cities are measured at the prefecture level, with the municipality districts of Beijing, Tianjin, Shanghai, and Chongqing collapsed to a single prefecture-level market in each case. I assign each individual an origin city using the hukou registration prefecture. This delivers a partition between locals, whose current city coincides with origin, and migrants, whose origin differs from current residence.

I split adults into two education groups to match the model. Low education consists of junior secondary schooling or below, while high education consists of senior secondary schooling or above. I

then classify each working-age adult as either a parent or a childless adult. An adult is coded as a parent if she is a woman aged 19–54 with at least one surviving child, or if the adult is the household head or spouse in a household containing either a directly observed child aged 0–18 or such a maternal parent signal. The age restriction applies only to the surviving-child signal, because the Census reports neither the ages nor the locations of children identified from women’s fertility histories. All remaining working-age adults are classified as childless. This procedure identifies mothers and their co-residing spouses while preserving the model’s adult parental decision unit.

The residual immobile population O_j consists of individuals aged 65 or above, who lie outside the working-age population represented in the household choice problem. This group is treated as exogenous and immobile in the model and contributes only to residual local housing demand. I calibrate h^O as the total floor space of households containing no working-age adult, divided by the total elderly population, thereby avoiding double counting housing jointly occupied with working-age households.

The census does not directly report whether migrants own housing in their place of origin. I therefore infer the initial housing-status split δ from the homeownership distribution among same-origin locals. For each origin city and education-parenthood cell, I compute the owner share among adults whose current city equals origin, and apply that share to the corresponding total mass of adults from that origin. This yields the decomposition of $H_o^{e\delta\emptyset}$ and $H_o^{e\delta}$ into owner-type and non-owner types. When a city-origin cell has no local observations in a given group, I use the corresponding national owner share as a fallback.

Finally, I set the population scale parameter $\bar{H}$ equal to the weighted total number of adults aged 19–64 in the 2005 Census. I calibrate φ , the number of dependent-child equivalents per parental unit, as the weighted ratio of all children aged 0–18 to all parental adults under the preceding rule. This yields $\varphi = 0.658$. The numerator is not restricted to directly observed co-residing children: that restriction would omit dependent children living apart from their parents. This normalization maps the national child-to-parent ratio into the model’s parental decision unit.

C.4 EDUCATION PRODUCTION PARAMETERS

This section complements Section 6.1 in the main paper by describing the estimation of $\{\zeta_e\}_{e \in \{H,L\}}$. Given the externally calibrated values of ν and q , I estimate these parameters using educational-attainment moments from the 2016 CMDS.

For a household of type (e, δ, o) choosing destination j and family arrangement f , the model-implied probability of high educational attainment is

$$P_{jo}^{e\delta f}(\zeta_e) = \text{logit}^{-1} \left[\log \zeta_e + \nu \log \left(\frac{K_{jo}^{e\delta f}}{\bar{K}} \right) + q \log \left(\frac{\Lambda_{jo}^f}{\bar{\Lambda}} \right) \right], \quad (41)$$

where $K_{jo}^{e\delta f}$ is the model-implied optimal private education investment and Λ_{jo}^f is the effective public-education resource defined in Eq. (8).

The fixed reference levels are $\bar{K} = 308.12$ and $\bar{\Lambda} = 809.34$, the parent-mass-weighted geometric means of the two model-implied inputs across feasible parent decision units on the common 2005 model support. They are held fixed throughout estimation, inversion, and the counterfactuals.

I assign destination wages, housing costs, and public education resources at the observed destination-

city level. Because the origin city is unobserved, I replace origin housing-endowment income and, for non-co-residing children, origin public education resources with their baseline-parent-mass-weighted averages across model cities in the reported origin, computed separately by parental education and initial housing status. Let $P_i(\zeta_e)$ denote the model-implied attainment probability evaluated using these inputs for child i .

Let D_i equal one if child i attains high education, and let ϖ_i denote the CMDS survey weight. For each parental education group e , I construct the moment condition

$$g_e(\zeta_e) = \frac{\sum_{i:e_i=e} \varpi_i [D_i - P_i(\zeta_e)]}{\sum_{i:e_i=e} \varpi_i}. \quad (42)$$

I estimate ζ_e by solving $g_e(\zeta_e) = 0$. This exactly identified moment matches the survey-weighted high-education rate within each parental education group to its model-implied counterpart. The resulting estimates are

$$\hat{\zeta}_H = 0.818, \quad \hat{\zeta}_L = 0.490.$$

The corresponding survey-weighted high-education rates are 0.581 for children of high-educated parents and 0.406 for children of low-educated parents, both of which are matched by construction.

C.5 INVERTED AMENITIES AND MODEL FIT

The nested procedure described in Section 6.3 and Appendix E.1 recovers family-arrangement-specific destination amenities $\{A_j^{ef}\}_{f \in \{\emptyset, 0, 1\}}$ for each education group. These amenities capture residual destination attractiveness after accounting for wages, housing costs, public-education resources, and migration frictions. Because the amenities are normalized separately by family arrangement, their levels are not directly comparable across f . I therefore focus on the model-implied spatial distribution of children.

For education group e , let $m_j^{e, \text{all}}$ denote the mass of children attending school in city j , including both co-residing and non-co-residing children. Let $m_j^{e, \text{mig}}$ denote the mass of children of parents who reside outside their origin city. The corresponding city shares are

$$s_j^{e, \text{all}} = \frac{m_j^{e, \text{all}}}{\sum_{k \in \mathcal{G}} m_k^{e, \text{all}}}, \quad s_j^{e, \text{mig}} = \frac{m_j^{e, \text{mig}}}{\sum_{k \in \mathcal{G}} m_k^{e, \text{mig}}}.$$

Figure C5 compares the observed and model-implied schooling-city distributions. The left panel reports the distribution of all children across schooling cities, while the right panel reports the distribution of children of migrant parents. Each panel reports separately the distributions for children of high- and low-educated parents. Marker size is proportional to the observed child mass in the corresponding city.

The model reproduces the all-child distribution closely, with a weighted correlation of 1.00 for both education groups. The corresponding weighted correlations for the distribution of children of migrant parents are 0.97 for children of high-educated parents and 0.90 for children of low-educated parents. These distributions are not added as separate moments to the outer objective. They are implied by the estimated bilateral location and family-arrangement choices and summarize how the fitted migration block translates into the spatial distribution of children.

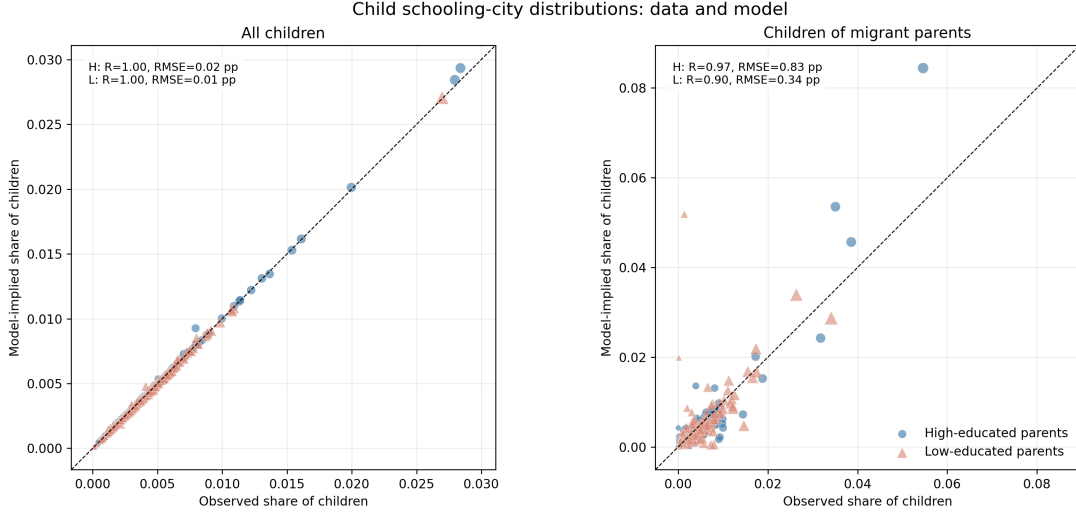


Figure C5: Model fit: children's schooling-city distribution

Notes: The left panel compares the observed and model-implied distribution of all children across schooling cities. The right panel compares the distribution of children of migrant parents. Points are shown separately for children of high- and low-educated parents. Marker size is proportional to the observed number of children in the city. The dashed line is the 45-degree line.

C.6 PARAMETER SENSITIVITY

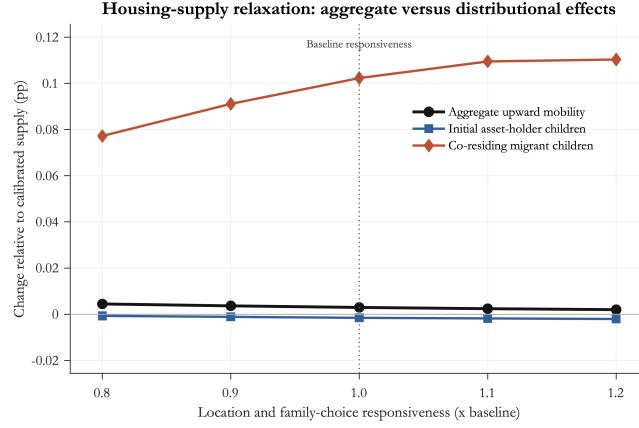
This section complements Sections 7.1 and 7.2 by assessing whether the counterfactual conclusions are robust to uncertainty in household sorting and public-education productivity parameters. The sorting parameters, ϵ^e and ρ^e , govern how strongly households adjust destination and parent-child co-residence choices. The public-education productivity parameter, q , governs how effectively public resources translate into children's educational outcomes.

For the housing-supply counterfactual, I jointly scale ϵ^e and ρ^e by 0.9, 1, and 1.1 relative to their calibrated values. At each value, I raise housing-supply elasticity by 25% in initially more-constrained cities and re-invert l_j so that the complete pre-shock equilibrium coincides with the corresponding baseline. I then apply the same nationwide 10% wage-equivalent productivity shock under calibrated and relaxed supply. Figure C6 reports the relaxed-minus-calibrated change in each shock response.

Figure C6 shows that the incidence of housing-supply relaxation is robust to the sorting parameters. Across the parameterizations considered, the aggregate mobility effect is only 0.002–0.004 percentage points. The mobility response among children of initial asset holders falls by 0.001–0.002 percentage points, whereas the response among migrant children who co-reside with their parents rises by 0.077–0.110 percentage points. More responsive sorting changes the magnitude of these effects but not their incidence: more elastic supply weakens the capitalization gains of initial asset holders and lowers the housing-cost barrier to parent-child co-residence.

For the equal-access counterfactual, I vary q and jointly scale ϵ^e and ρ^e by 0.9, 1, and 1.1 relative to their calibrated values. At each parameter combination, I re-calibrate education productivity and re-invert the migration and family-arrangement objects to preserve the same baseline Chinese moments. I then solve the corresponding baseline equilibrium and nationwide equal-access equilibrium. A higher q strengthens the mobility return to equal access, while more responsive sorting amplifies the migration

Figure C6: Sensitivity of the incidence of housing-supply relaxation

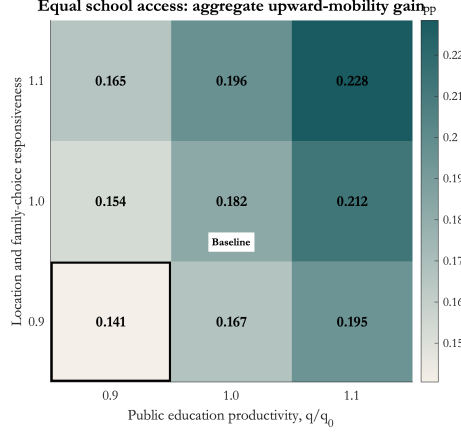


Notes: The horizontal axis applies a common multiplier to the location-choice parameters, ϵ^e , and family-arrangement parameters, ρ^e . At each value, housing-supply elasticity is increased by 25% only in initially more-constrained cities, and l_j is re-inverted to preserve the corresponding complete baseline equilibrium. The same common 10% wage-equivalent productivity shock is then applied under calibrated and relaxed supply. Each line reports the relaxed-minus-calibrated change in the shock response, in percentage points.

and co-residence channel through which access affects children.

Figure C7 shows that the equal-access gain remains positive across the parameter grid. The gain rises with both public-education productivity and the responsiveness of household sorting, but the qualitative conclusion is unchanged: equalising school access generates an aggregate mobility gain, whereas housing-supply relaxation primarily changes the distribution of the gains from urban growth across households.

Figure C7: Sensitivity of the equal-school-access counterfactual



Notes: Each cell reports the percentage-point increase in aggregate upward mobility from equalising migrant children’s access to local public education. The horizontal axis scales q relative to its calibrated value, while the vertical axis applies a common multiplier to ϵ^e and ρ^e . For each parameter pair, education-productivity, migration, and family-arrangement objects are re-calibrated to match the baseline moments, and the full spatial equilibrium is re-solved. The outlined cell denotes the calibrated parameter vector.

D MODEL APPENDIX

D.1 HOUSING MARKET

This section provides a microfoundation for the inverse housing-supply curve in Eq. 25. In each city j , competitive residential developers produce housing services using mobile capital K_j and local land N_j according to a constant-returns Cobb–Douglas technology:

$$F_j^S = \phi_j K_j^{\beta_k} N_j^{\beta_l}, \quad \beta_k + \beta_l = 1,$$

where ϕ_j denotes city-specific housing-sector productivity.

Capital is rented at the common price p^k . The effective price of local land increases with aggregate housing development according to

$$p_j^l = \bar{p}_j^l (F_j^S)^{\gamma_j^l}, \quad \gamma_j^l > 0,$$

where $\bar{p}_j^l$ captures the baseline level of local land costs. A larger γ_j^l implies that marginal land costs rise more rapidly as development expands, capturing local housing-supply constraints.

Conditional on producing F_j^S , cost minimization under the Cobb–Douglas technology implies the unit cost of housing services

$$r_j = \frac{(p^k)^{\beta_k} (p_j^l)^{\beta_l}}{\phi_j \beta_k^{\beta_k} \beta_l^{\beta_l}}.$$

Substituting the land-price schedule yields

$$r_j = (p^k)^{\beta_k} \frac{(\bar{p}_j^l)^{\beta_l}}{\phi_j \beta_k^{\beta_k} \beta_l^{\beta_l}} (F_j^S)^{\gamma_j^l \beta_l}.$$

Define the housing-supply elasticity η_j by

$$\frac{1}{\eta_j} \equiv \gamma_j^l \beta_l.$$

This decomposition provides only a microfoundation for the reduced-form supply elasticity: the quantitative implementation does not separately identify γ_j^l or β_l , but instead directly estimates their composite implication, $1/\eta_j$.

Since the capital-price component $(p^k)^{\beta_k}$ is common across cities, I normalize it to one and define the city-specific housing-supply cost shifter as

$$l_j \equiv \frac{(\bar{p}_j^l)^{\beta_l}}{\phi_j \beta_k^{\beta_k} \beta_l^{\beta_l}}.$$

The inverse housing-supply curve then becomes

$$r_j = l_j (F_j^S)^{1/\eta_j},$$

which corresponds to Eq. 25. Thus, a lower η_j implies a steeper inverse supply curve and a larger housing-cost response to a given increase in housing demand.

In the quantitative implementation, I directly parametrize the composite inverse supply elasticity as

$$\log \left(\frac{1}{\eta_j} \right) = \theta_0 + \theta_D \text{Developed}_j + \theta_S \text{Slope}_j,$$

and estimate θ_0 , θ_D , and θ_S as described in Section 6.2 and Appendix Section C.2. I do not separately calibrate ϕ_j , $\bar{p}_j^l$, γ_j^l , or the factor shares. Given the estimated η_j , I recover the composite shifter l_j from the 2005 baseline housing-market equilibrium and hold it fixed in the counterfactuals.

E ALGORITHMS

E.1 AMENITY INVERSION AND MIGRATION COST PARAMETERS ESTIMATION

This section provides details on the nested procedure described in Section 6.3. The procedure is implemented separately by education group. It combines an inner inversion of family-arrangement-specific destination amenities with an outer search over migration-cost parameters.

Throughout this section, $f \in \{\emptyset, 0, 1\}$, where $f = \emptyset$ denotes childless adults, $f = 0$ denotes parents who do not co-reside with the child, and $f = 1$ denotes parents who co-reside with the child. For a given education group e , migration costs are parameterized as

$$d_{jo}^{ef} = \exp \left[\mu^{ef} \text{dist}_{jo} + \gamma_P^{ef} \mathbf{1}\{p(j) \neq p(o)\} + \gamma_R^{ef} \mathbf{1}\{g(j) \neq g(o)\} + \kappa^{ef} \mathbf{1}\{j \neq o\} \right],$$

with $d_{oo}^{ef} = 1$. The outer parameter vector is

$$\varsigma^e \equiv \left(\left\{ \mu^{ef}, \gamma_P^{ef}, \gamma_R^{ef}, \kappa^{ef} \right\}_{f \in \{\emptyset, 0, 1\}} \right).$$

The co-residence scale T_1^e is not searched over in the outer loop. Instead, given a candidate ς^e and the normalization $T_0^e = 1$, the inner problem solves for T_1^e and the amenity vectors jointly. For each trial value of T_1^e , the inversion recovers $\{A_j^{ef}\}_{f \in \{\emptyset, 0, 1\}}$ to match the destination distribution within each family arrangement. The algorithm then updates T_1^e until the model-implied aggregate co-residence share among parents matches its empirical counterpart.

For any candidate ς^e and T_1^e , model-predicted destination masses are

$$\begin{aligned} L_j^{e\emptyset, \text{model}} &= \sum_{o \in \mathcal{G}} \sum_{\delta \in \{B, R\}} H_o^{e\delta\emptyset} \pi_{j|o}^{e\delta\emptyset}, \\ L_j^{ef, \text{model}} &= \sum_{o \in \mathcal{G}} \sum_{\delta \in \{B, R\}} H_o^{e\delta} \pi_{f|o}^{e\delta} \pi_{j|o}^{e\delta f}, \quad f \in \{0, 1\}. \end{aligned} \quad (43)$$

The corresponding data targets are constructed by aggregating the 2005 Census bilateral cells to destination totals within each family arrangement.

For each $f \in \{\emptyset, 0, 1\}$, the amenity inversion chooses $\{A_j^{ef}\}_{j \in \mathcal{G}}$ so that the model matches the observed destination distribution within that family arrangement:

$$\frac{L_j^{ef, \text{model}}}{\sum_{k \in \mathcal{G}} L_k^{ef, \text{model}}} = \frac{L_j^{ef, \text{data}}}{\sum_{k \in \mathcal{G}} L_k^{ef, \text{data}}}, \quad j \in \mathcal{G}, \quad f \in \{\emptyset, 0, 1\}.$$

Because destination amenities are identified only up to a family-arrangement-specific multiplicative scale, I impose the normalization

$$\frac{1}{J} \sum_{j \in \mathcal{G}} \log A_j^{ef} = 0, \quad f \in \{\emptyset, 0, 1\}.$$

The aggregate co-residence share in the data is

$$s_{\text{core}}^{e, \text{data}} = \frac{\sum_{o \in \mathcal{G}} \sum_{j \in \mathcal{G}} M_{jo}^{e1, \text{data}}}{\sum_{o \in \mathcal{G}} \sum_{j \in \mathcal{G}} \left(M_{jo}^{e0, \text{data}} + M_{jo}^{e1, \text{data}} \right)},$$

where $M_{jo}^{ef, \text{data}}$ denotes the observed origin-destination cell count for parents choosing family arrangement f , including same-origin stayers. For each trial value of T_1^e , after the amenity inversion has recovered $\{A_j^{ef}\}_{f \in \{\emptyset, 0, 1\}}$, the model counterpart is

$$s_{\text{core}}^{e, \text{model}} = \frac{\sum_{j \in \mathcal{G}} L_j^{e1, \text{model}}}{\sum_{j \in \mathcal{G}} \left(L_j^{e0, \text{model}} + L_j^{e1, \text{model}} \right)}.$$

The inner loop updates T_1^e until

$$s_{\text{core}}^{e,\text{model}} = s_{\text{core}}^{e,\text{data}}.$$

I denote the resulting objects by

$$\hat{T}_1^e(\varsigma^e), \quad \hat{A}^e(\varsigma^e) = \left\{ \hat{A}_j^{ef}(\varsigma^e) \right\}_{j \in \mathcal{G}, f \in \{\emptyset, 0, 1\}}.$$

The outer problem then uses origin-conditioned bilateral choices. Initial housing-wealth status δ is a predetermined state in the household problem but is not directly observed for Census bilateral flows. I therefore integrate over its calibrated origin-specific distribution when constructing model choice probabilities.

Define

$$H_o^{e\emptyset} \equiv \sum_{\delta \in \{B, R\}} H_o^{e\delta\emptyset}, \quad H_o^e \equiv \sum_{\delta \in \{B, R\}} H_o^{e\delta}.$$

For childless adults, the model-implied origin-conditioned probability of choosing destination j is

$$P_{j|o}^{e\emptyset} = \frac{\sum_{\delta \in \{B, R\}} H_o^{e\delta\emptyset} \pi_{j|o}^{e\delta\emptyset}}{H_o^{e\emptyset}}. \quad (44)$$

For parents, the model-implied joint probability of choosing family arrangement f and destination j is

$$P_{fj|o}^e = \frac{\sum_{\delta \in \{B, R\}} H_o^{e\delta} \pi_{f|o}^{e\delta} \pi_{j|o}^{e\delta f}}{H_o^e}, \quad f \in \{0, 1\}. \quad (45)$$

Let $M_{jo}^{e\emptyset, \text{data}}$ denote the observed bilateral cell count of childless adults, and let $M_{jo}^{ef, \text{data}}$ denote the corresponding parental cell count under family arrangement f . The grouped bilateral-choice likelihood is

$$\begin{aligned} \ell_{\text{bil}}^e &= \sum_{o \in \mathcal{G}} \sum_{j \in \mathcal{G}} M_{jo}^{e\emptyset, \text{data}} \log P_{j|o}^{e\emptyset} \\ &\quad + \sum_{o \in \mathcal{G}} \sum_{f \in \{0, 1\}} \sum_{j \in \mathcal{G}} M_{jo}^{ef, \text{data}} \log P_{fj|o}^e. \end{aligned} \quad (46)$$

Same-origin stayers are included in the likelihood. Cells with zero observed counts make no contribution to the objective.

The profile estimator is

$$\hat{\varsigma}^e = \arg \max_{\varsigma^e} \ell_{\text{bil}}^e \left(\hat{A}^e(\varsigma^e), \hat{T}_1^e(\varsigma^e), \varsigma^e \right), \quad (47)$$

with

$$\hat{A}^e = \hat{A}^e(\hat{\varsigma}^e), \quad \hat{T}_1^e = \hat{T}_1^e(\hat{\varsigma}^e).$$

Thus, each evaluation of the outer objective requires solving for the co-residence scale and inverting the three family-arrangement-specific amenity vectors.

Algorithm 1: Amenity inversion and migration cost parameter estimation

Inputs : Calibrated structural parameters and baseline equilibrium objects; predetermined origin populations $\{H_o^{e\delta\emptyset}, H_o^{e\delta}\}$; observed destination totals $\{L_j^{ef,\text{data}}\}_{f \in \{\emptyset, 0, 1\}}$; observed bilateral cell counts $\{M_{jo}^{e\emptyset,\text{data}}, M_{jo}^{ef,\text{data}}\}$; and bilateral transport times and province/region indicators.

Output: Estimated migration-cost parameters $\hat{\varsigma}^e$, co-residence scale $\hat{T}_1^e$, and inverted amenities $\{\hat{A}_j^{ef}\}_{j \in \mathcal{G}, f \in \{\emptyset, 0, 1\}}$, for $e \in \{H, L\}$.

for each education group $e \in \{H, L\}$ **do**

- Initialize the outer search over ς^e ;
- while** the bilateral likelihood has not been maximized **do**
 - Given the current ς^e , construct $\{d_{jo}^{ef}\}_{f \in \{\emptyset, 0, 1\}}$;
 - while** $s_{\text{core}}^{e,\text{model}} \neq s_{\text{core}}^{e,\text{data}}$ **do**
 - Choose a trial value of T_1^e , with $T_0^e = 1$;
 - while** the model destination distributions do not match the data within each family arrangement **do**
 - Solve household choices under the current $(\varsigma^e, T_1^e, \{A_j^{ef}\}_{j,f})$;
 - Aggregate choices to obtain $\{L_j^{ef,\text{model}}\}_{j,f}$;
 - Update the amenity vectors and re-impose $(1/J) \sum_j \log A_j^{ef} = 0$ for each $f \in \{\emptyset, 0, 1\}$;
 - Compute $s_{\text{core}}^{e,\text{model}}$ and update T_1^e ;
 - Construct the origin-conditioned choice probabilities $\{P_{j|o}^{e\emptyset}, P_{fj|o}^e\}$ by integrating over δ ;
 - Evaluate ℓ_{bil}^e and update ς^e ;
- Store $\hat{\varsigma}^e$, $\hat{T}_1^e$, and $\{\hat{A}_j^{ef}\}_{j \in \mathcal{G}, f \in \{\emptyset, 0, 1\}}$;

E.2 SOLVING FOR THE EQUILIBRIUM

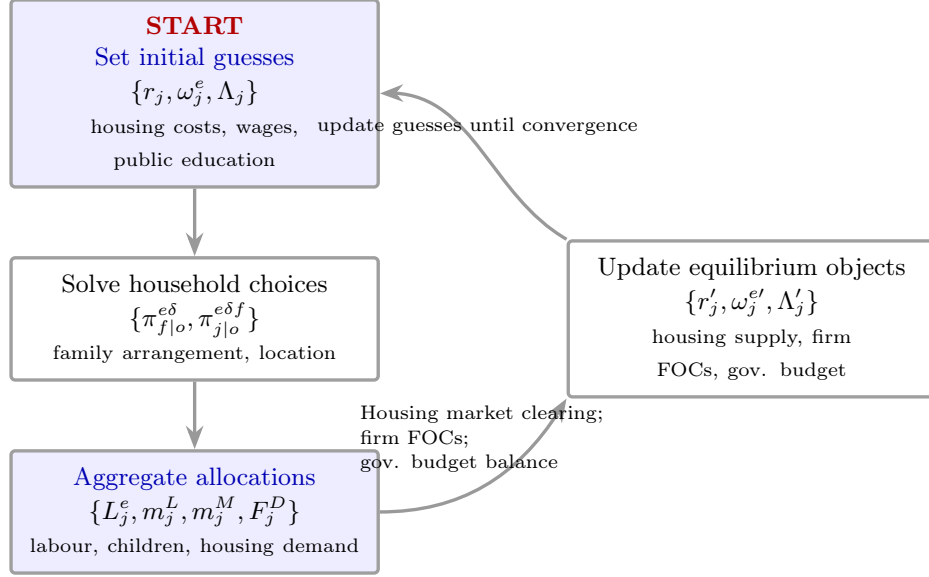


Figure E1: **Equilibrium solution algorithm.** The diagram summarizes the fixed-point iteration for target endogenous variables in the model including housing prices, education-specific wages, and public-education resources.

The fixed point algorithm used to solve the equilibrium is as summarized in the diagram above. Given the calibrated primitives, local fundamentals, and initial adult populations by types, I initialise city-level housing-service prices r_j , education-specific wages ω_j^e , and public-education resources Λ_j at their observed baseline counterparts. Holding these objects fixed within an iteration, I solve households' destination and family-arrangement choices using Eqs. (13)–(16) and (18), along with their implied housing services and private education investment. I then aggregate these choices into local employment L_j^e using Eq. (19), total and migrant child populations using Eqs. (20) and (21) (with local children given by the difference), and housing demand F_j^D using Eq. (24).

The implied labour supplies update wages through firms' marginal-product conditions in Eq. (27). Imposing $F_j^S = F_j^D$ in Eq. (25) gives updated housing prices, while Eq. (26) gives updated public-education resources from employment, the current wage guesses, and effective pupil population $m_j^L + \Delta_j m_j^M$. Each new value is combined with the previous guess using damped updates. I repeat these steps until the maximum log difference between the implied and updated values of r_j , ω_j^e , and Λ_j is below 10^{-8} .

E.3 MODEL FIT

This section provides additional detail on the model fit reported in Section 6.4 of the main paper. Figure E2 compares observed city-level outcomes with their baseline-equilibrium counterparts. The model closely reproduces housing prices, wages, employment, and public education resources. The

baseline equilibrium satisfies the model's market-clearing and government-budget conditions to numerical precision and is robust to dispersed starting values.

Figure E3 examines the model-implied spatial distribution of children. The model nearly exactly reproduces the overall distribution of children and also closely tracks the distribution of children of migrant parents, including those with low-educated parents, who are central to the analysis. The weighted correlations between observed and model-implied distributions for migrant-parent children are 0.97 for high-educated parents and 0.90 for low-educated parents. Importantly, these distributions are not separately targeted in the outer migration-cost objective.

Figure E4 evaluates two untargted educational outcomes. Because the calibration matches aggregate attainment by parental education rather than city-specific child outcomes, these comparisons provide independent diagnostics of spatial fit. The first panel compares upward mobility among children of low-educated incumbent homeowners, while the second compares the next generation's high-education share, which aggregates parental composition, sorting, family arrangements, and education inputs. The observed-child-count-weighted correlations are 0.67 and 0.78, respectively. The model therefore captures a substantial amount of cross-city variation in educational outcomes that is not directly targeted in the calibration.

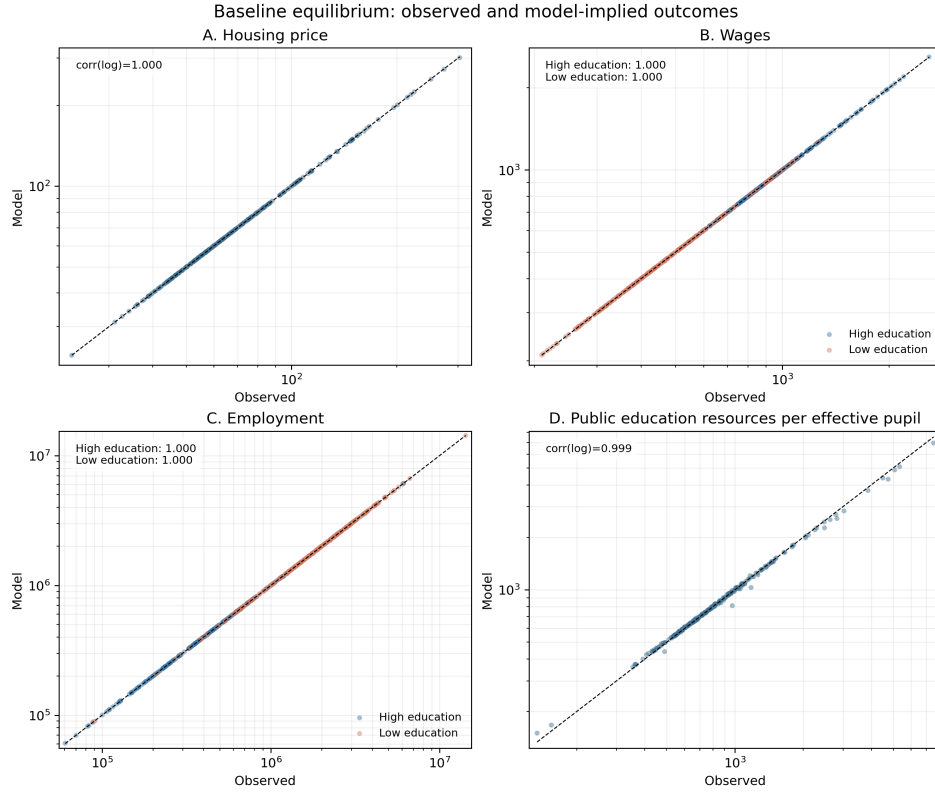


Figure E2: **Baseline reproduction of endogenous city outcomes.** Each panel compares an observed city outcome with its independently solved baseline-equilibrium counterpart. The dashed line is the 45-degree line.

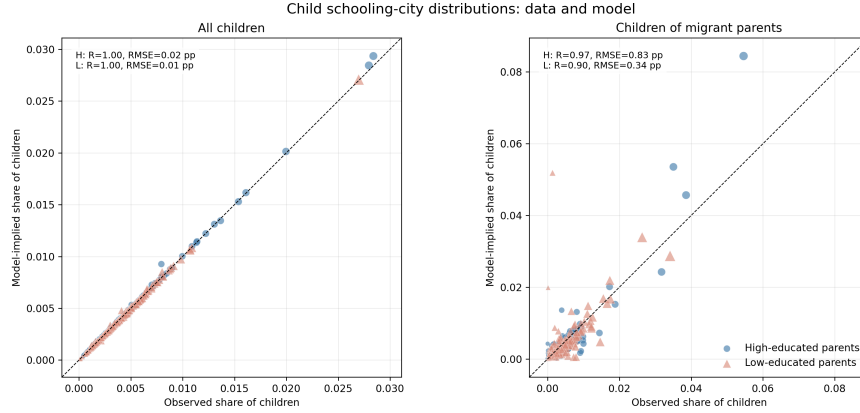


Figure E3: **Children’s schooling-city distributions: data and model.** The left panel compares the distribution of all children across schooling cities. The right panel reports children of migrant parents. Marker size is proportional to the observed child mass, and the dashed line is the 45-degree line.

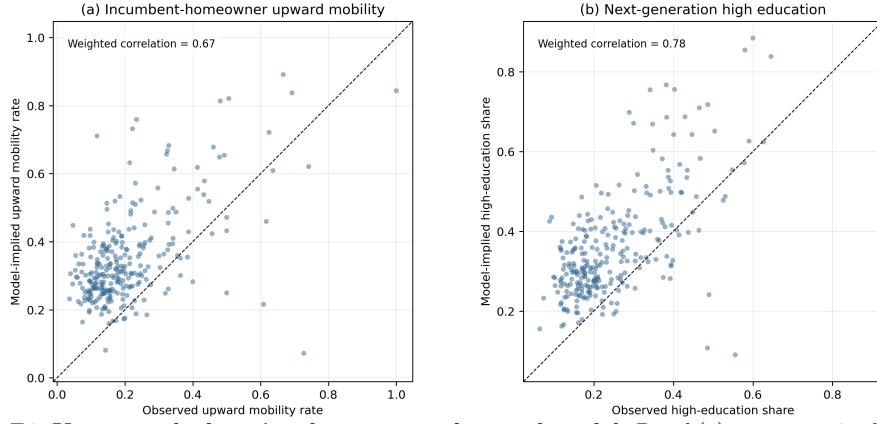


Figure E4: **Untargeted educational outcomes: data and model.** Panel (a) compares city-level upward mobility among children of low-educated incumbent homeowners. Panel (b) compares high-education shares for the 1986–1993 birth cohorts. Each point is a city; correlations are weighted by the corresponding observed child counts. Dashed lines are 45-degree lines.

E.4 UNIQUENESS

This section examines numerical uniqueness under the calibrated primitives using the fixed-point algorithm in Section E.2. I solve the baseline equilibrium from 50 randomized initial states, holding all primitives and local fundamentals fixed. In each run, the initial guesses for each city’s housing price, high- and low-education wages, and public-education resources are jointly perturbed by independent lognormal multipliers.

All 50 runs converge. Figure E5 shows that, despite dispersed initial values, the maximum log distance between each run’s current state and the baseline equilibrium declines toward zero. Across the converged runs, the maximum pairwise difference in log housing prices, wages, and public-education resources is 2.0×10^{-8} . These results provide numerical evidence for uniqueness under the calibrated primitives.

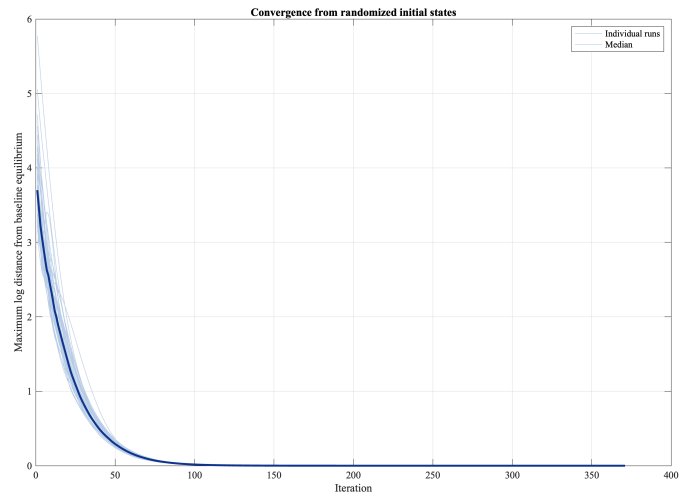


Figure E5: **Convergence from randomized initial states.** Each thin line reports the maximum log distance from the independently solved baseline equilibrium for one of 50 randomized initial states. The thick line is the cross-run median.

F SUMMARY OF VARIABLES

Table F1: Primitive Parameters

Category	Parameter	Value	Description	Source / Target
<i>Housing preferences</i>				
	$\alpha^{H\varnothing}$	0.07	H, no dependent child	UHS, 2013–2014
	α^{H0}	0.05	H, non-co-residing child	UHS, 2013–2014
	α^{H1}	0.05	H, co-residing child	UHS, 2013–2014
	$\alpha^{L\varnothing}$	0.06	L, no dependent child	UHS, 2013–2014
	α^{L0}	0.04	L, non-co-residing child	UHS, 2013–2014
	α^{L1}	0.02	L, co-residing child	UHS, 2013–2014
	$\bar{h}^{\varnothing}$	25.77	Minimum housing requirement, no dependent child	UHS, 2013–2014
	$\bar{h}^0$	25.27	Minimum housing requirement, non-co-residing child	UHS, 2013–2014
	$\bar{h}^1$	30.05	Minimum housing requirement, co-residing child	UHS, 2013–2014
	h^O	26.09	Effective per-capita housing demand, residual population	2005 Census
<i>Child education production</i>				
	ζ_H	0.818	H baseline intergenerational transmission	CMDS 2016
	ζ_L	0.490	L baseline intergenerational transmission	CMDS 2016
	$\tilde{\nu}$	0.08	Adjusted private education parameter	Calibrated, Chi and Qian (2016)
	ν	0.19	Private education elasticity	Attanasio et al. (2020)
	ι^H	0.42	H parental altruism	Implied by $\tilde{\nu}/\nu$
	ι^L	0.42	L parental altruism	Implied by $\tilde{\nu}/\nu$
	q	1.2	Public education elasticity	Hyman (2017)
<i>Nested choice parameters</i>				
	ϵ^H	4.27	H location Fréchet shape	Fan (2019)
	ϵ^L	3.91	L location Fréchet shape	Fan (2019)
	ρ^H	3.25	H family-arrangement Fréchet shape	Imbert et al. (2026)
	ρ^L	3.25	L family-arrangement Fréchet shape	Imbert et al. (2026)
<i>Migration costs and family arrangement</i>				
	T_1^e	See Table 5	Relative scale of co-residence preference	MLE, 2005 Census
	$\{\mu^{ef}, \gamma_P^{ef}, \gamma_R^{ef}, \kappa^{ef}\}$	See Table 5	Family-arrangement-specific migration costs	MLE, 2005 Census
<i>Production parameters</i>				
	ϑ	1.67	Elasticity of substitution between education types	Krusell et al. (2000)
<i>Population scale and normalization</i>				
	$\bar{H}$	721,675,277	Total adult population	2005 Census
	φ	0.658	Dependent-child equivalents per parental unit	Calibrated, 2005 Census

Notes: This table reports location-invariant parameters used in the current baseline simulation code. H and L denote high- and low-education adults; $f = \varnothing$ denotes childless adults, $f = 0$ non-co-residing parents, and $f = 1$ co-residing parents.

Table F2: Calibrated Local Characteristics and Fundamentals in the Baseline Simulation

Object	Interpretation	Data counterpart
<i>Initial populations</i>		
$H_o^{e\delta\emptyset}$	Childless adults by origin, education, and initial housing status	2005 Census
$H_o^{e\delta}$	Parents by origin, education, and initial housing status	2005 Census
O_j	Immobile residual population	2005 Census
<i>Production fundamentals</i>		
B_j^H	High-skilled productivity	Inverted, 2005 Census
B_j^L	Low-skilled productivity	Inverted, 2005 Census
<i>Amenities</i>		
A_j^{ef}	Education- and family-arrangement-specific destination amenity	Inverted jointly with migration costs, 2005 Census
<i>Housing, education access, and fiscal fundamentals</i>		
$E_j^{H\mathcal{B}}$	Housing endowment of high-skilled homeowners	Calibrated, 2000 Census
$E_j^{L\mathcal{B}}$	Housing endowment of low-skilled homeowners	Calibrated, 2000 Census
η_j	Housing-supply elasticity	Estimated, 2005 and 2015 Census
l_j	Construction-cost shifter	Calibrated, 2005 Census and CSYRE
τ_j	Effective fiscal education ratio	Calibrated, CSY
Δ_j	Migrant education-access wedge	Calibrated, CMDS

Notes: This table reports the local primitives used in the baseline equilibrium.